

Group 12: Honeywell

“Distributed Wireless” Home Security System

Vivian Nguyen, James Burch, Garrett Arcand, Collin Yuen, Mohammed Abdul-Haadi
Ansari, Abinaya Ramesh, Hailey McElroy

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Executive Summary

Project Proposal:

Objective:

To design and outline a supply chain for Honeywell's Distributed Wireless Home Security System by defining the product architecture, identifying subsystems, and mapping supplier, manufacturing, and distribution relationships.

Key Results:

Developed a clear product concept with subsystems, created use case and FAST diagrams, and built both stage and network supply chain representations linking product design to sourcing and distribution.

Project Phase I:

Objective:

To develop the supply chain business model and strategy, estimate initial demand, and perform basic demand forecasting.

Key Results:

Created a supply chain business model aligned with competitive strategy, estimated demand for the first 8 quarters, and applied static forecasting to project future demand.

Project Phase II:

Objective:

To refine demand forecasting, analyze inventory decisions, and evaluate the financial feasibility of the supply chain.

Key Results:

Applied advanced forecasting (including Winter's), optimized cycle inventory policies, and conducted a 5-year NPV analysis to assess profitability.

Project Phase III:

Objective:

To optimize supply chain operations through safety inventory planning, network design, and overall system integration.

Key Results:

Developed safety inventory strategies, improved supply chain network and transportation design, and integrated all components into a cohesive and optimized supply chain system

Group 12: Project Proposal

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Objective

To design and manage the supply chain for Distributed Wireless in Honeywell, by means of structuring a stage and network representation of said supply chain around the below items. Our main approach will be to tie in our existing solution principles, sub-systems, and primary use case to formulate a preliminary supply chain design.

Overview

To execute this supply chain design systematically, the team has developed a structured timeline that allocates specific responsibilities across a ten-week period. This time-phased approach ensures that each component of the supply chain—from initial strategy through final integration—receives dedicated analysis and development.

Following the time phase plan, in the phase of the Project Proposal, the team put together an Updated Final Concept and Preliminary Supply Chain Design. The Updated Final Concept features a visual concept design to lay out the system with a use case and FAST diagram to break down how the system works. These dissections laid out the subsystems of the security system for proper sourcing to take place. The Preliminary Supply Chain Design broke down the stages in Honeywell's supply chain connecting it back to suppliers needed for sub-systems in a stage representation and was further developed in the network representation.

Time-Phase Project Plan (Step 3)

Week	Phase	Member Assigned	Meeting Schedule
1 (1/6)	Project Proposal for SCM of Distributed Wireless Security System	<i>Garrett Arcand</i> : Time-Phase plan <i>Hailey McElroy</i> : Conceptual design review/FAST diagram <i>James Burch</i> : Preliminary infrastructure (Stage rep) <i>Vivian Nguyen</i> : Preliminary infrastructure (Network rep) <i>Abdul-Haadi</i> : Post-processing <i>Collin Yuen</i> : Network representation <i>Abinaya Ramesh</i> : Post-processing	Thursday 1/15 post-lecture Sunday/Monday (1/18-19) 4:00PM-5:00PM
2-3 (1/13)	Phase 1: SCM Strategy/Design	TBA	Sunday/Monday 3:00PM
3 (1/20)	Phase 1 Review/Report Due	“”	Sunday/Monday 3:00PM

4-5 (1/27)	Phase 2: SC Modeling/Planning (demand forecasting)	“”	Sunday/Monday 3:00PM
5 (2/3)	Phase 2 Review/Report Due	“”	Sunday/Monday 3:00PM
6-7 (2/10)	Phase 3: SC Cycle/Safety Inventory Mgmt.	“”	Sunday/Monday 3:00PM
7 (2/17)	Phase 3 Review/Report Due	“”	Sunday/Monday 3:00PM
8-9 (2/24)	Phase 4: SC Facilities/Transportation Network Design	“”	Sunday/Monday 3:00PM
9 (3/3)	Phase 4 Review/Report Due	“”	Sunday/Monday 3:00PM
10 (3/10)	Integration using SCM and Network Design from 172B	“”	Sunday/Monday 3:00PM

**Each segment of the week-by-week project plan is subject to change*

This table outlines a ten-week project timeline for the SCM of the proposed Distributed Wireless security system. The project will be structured into distinct phases, beginning with a proposal, building into the four core SCM phases, and ending with final integration.

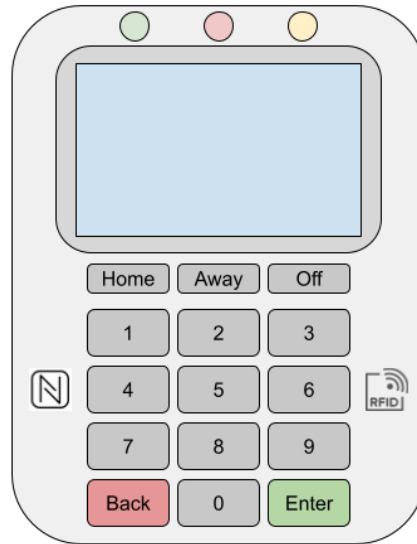
Before structuring the supply chain stages and network, it is essential to understand the product architecture and functional requirements. The following section details the *Distributed Wireless* (i.e. our selected concept) system's design, subsystems, as well as operational flow, which directly informs our supplier selection and manufacturing decisions.

Updated Final Concept (Step 4)

The final concept selection is the Distributed Wireless Home Security System. This security system requires connection to Wi-Fi and the mobile application for connectivity between the keypad, remote key fob, siren, floor vibration sensors, door sensors, and window sensors to be established. Upon proper installation with connectivity and power established the following Use-Case diagram outlines the functionality and sub-systems Distributed Wireless utilizes.

Keypad and Keyfob

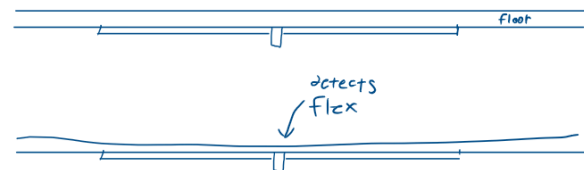
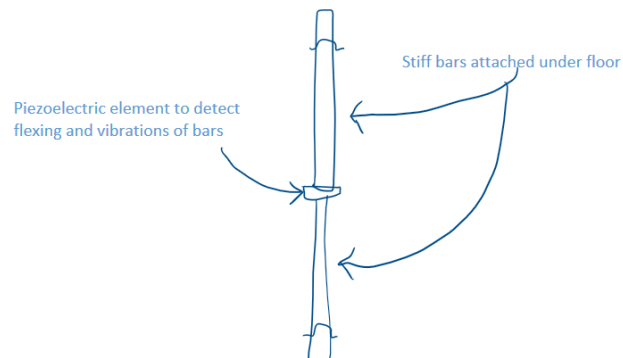
- Contains Wi-Fi and mesh radio for communication with other devices on the system
- Large display communicates system status and other information to the user
- Contains NFC and RFID readers to allow for authentication with smartphones, access cards, and keyfobs



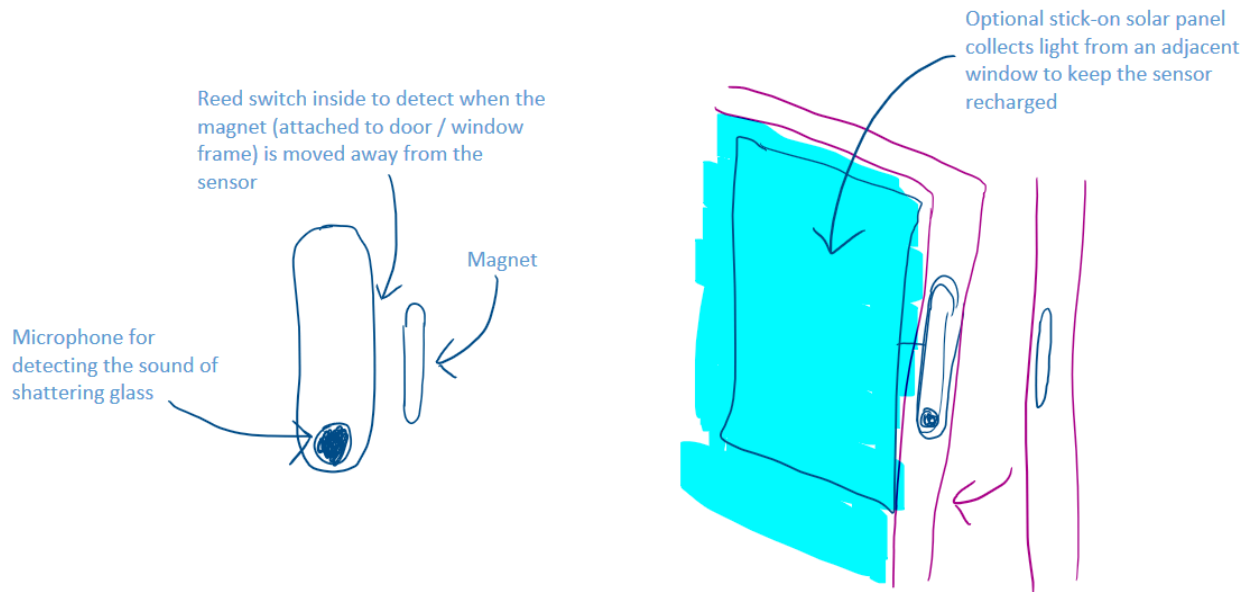
- Keyfob allows for remote control and / or user authentication at the keypad



Floor Vibration Sensor



Door / Window Sensor with Solar Panel



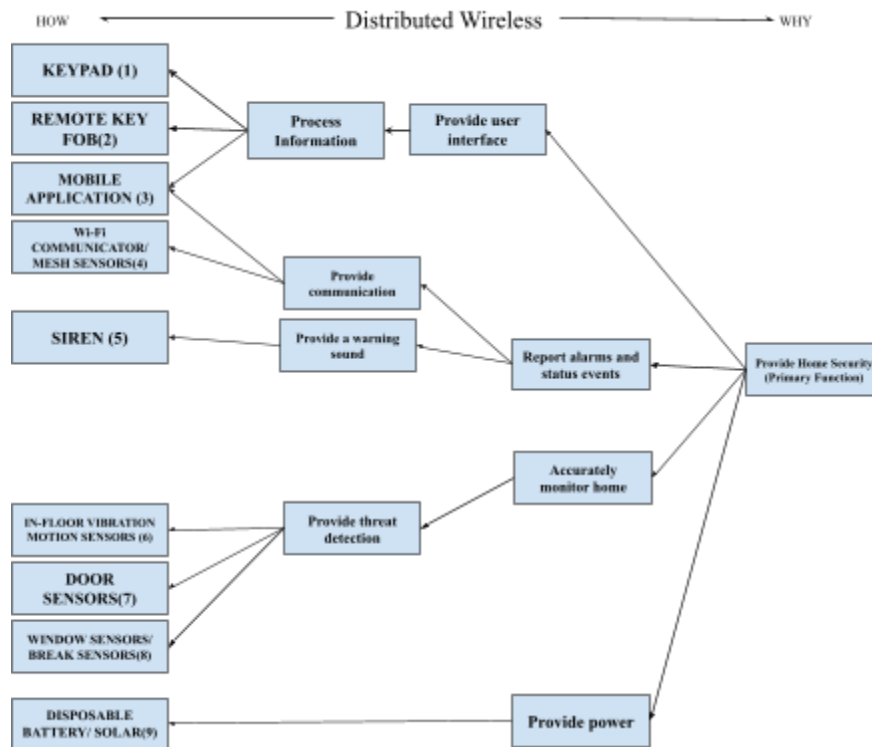
Siren

- Contains a loud piezoelectric noisemaker to alert homeowner and intruders



With the system layout above, we will further break down Distributed Wireless using a use case diagram and fast diagram to put together the “How;s?” and “Why’s?”. We do this to understand the different subsystems and products that will need to be sourced from different suppliers that will produce these subsystems Honeywell does not.

Updated FAST Diagram



The FAST diagram development breaks down the subsystems and subfunctions that define the overall structure of Honeywell's Distributed Wireless home security system. This dissection lays out the recurring functionalities that need to be physically realized within the final product. It serves as a visual representation of the product's structure, showing how the home-security system should operate, its key components, and how these elements will connect to the overall supply chain.

To translate the functional requirements into an implementable architecture, the identified functions are broken down into specifics of the physical components and technologies based on the shared responsibilities and interactions. This process outlines the specific components and technologies that create the basis of the subsystems presented in the following section.

These subsystems, sub-functions, and solution principles were realized during the first quarter of development of this home security system, and were sourced from real market data and analysis.

Sub-systems

Key sub-systems:

1. Keypad
2. Remote key fob
3. Mobile Application
4. Wi-Fi communicator / mesh sensors
5. Siren
6. Floor vibration sensors
7. Door sensors
8. Window sensors / break sensors
9. Disposable Battery / solar

Key sub-functions:

1. Provide a user interface
2. Provide warning sound
3. Provide threat detection
4. Process information
5. Provide communicator link between devices
6. Provide power

Given this set of sub-systems, the following set of solution principles were realized in product development and will need to be sourced to develop the supply chain design.

Solution Principles

- In-floor piezoelectric sensors for detecting motion
 - Monitor vibrations and is controlled with the adjustment of sensitivity for an entry point.
- Magnetic Reed switch door sensors / Acoustic Glass break sensors
 - Sensors combined into one monitor for more accurate wireless monitoring.
- RFID / NFC + Mobile app + Remote Keyfob
 - Allows users simple access to the armed household given they have the application or keyfob on hand to identify their access.
- Distributed processing
 - Provides greater resilience with continuous monitoring despite possible points of system failure
- Piezoelectric siren + app notifications
 - Current Honeywell product sirens are loud and compact allowing for simple set up.
- Disposable lithium CR2450 batteries + solar power panels
 - Provide power to the system.

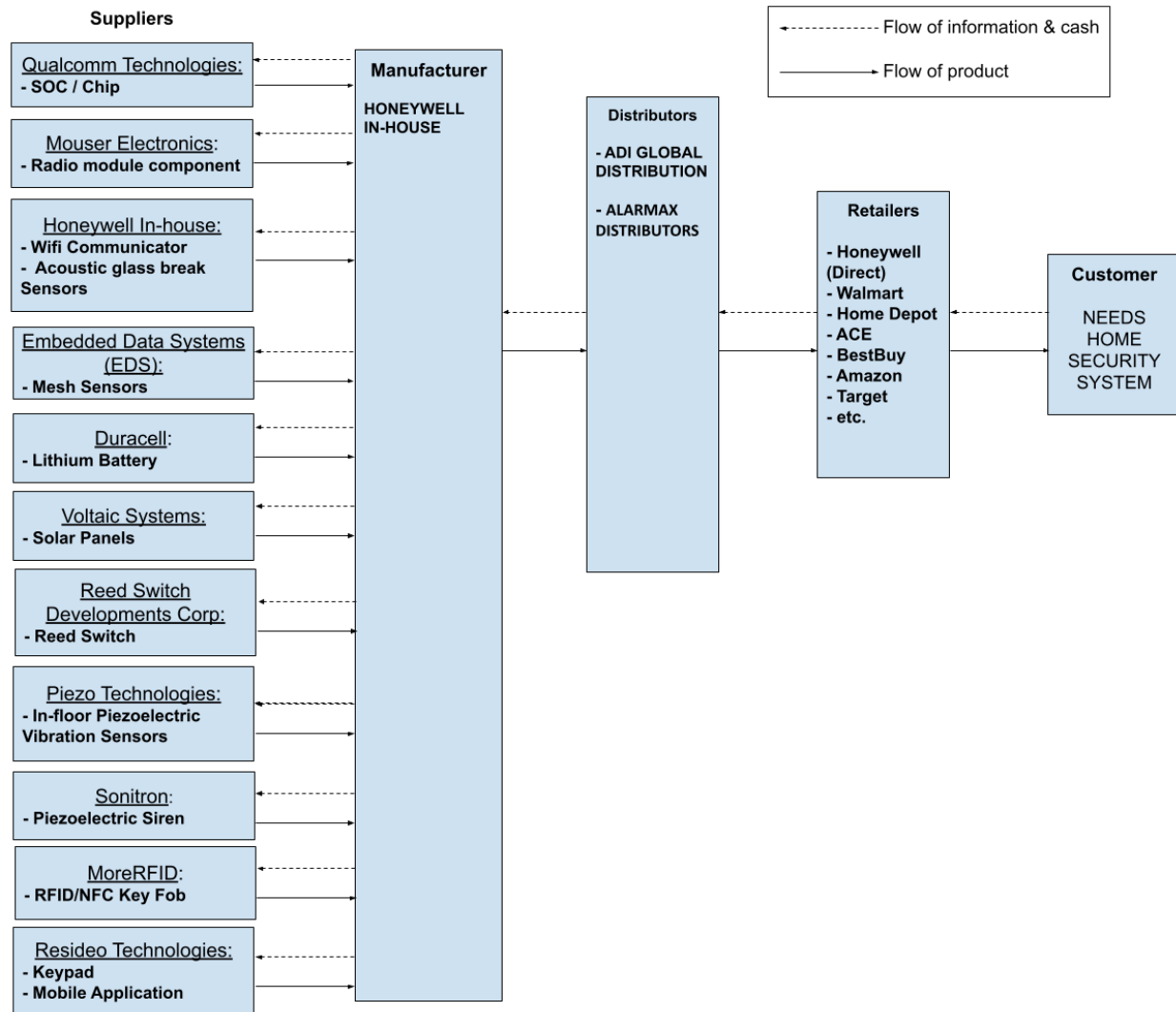
- Wi-Fi communicator + mesh sensors
 - Provide connection between sub-systems.

With the aforementioned product architecture, sub-systems, and solution principles clearly defined, the team can now map out a supply chain design. A stage and network representations allows us to demonstrate how Honeywell sources, manufactures, distributes, and delivers the *Distributed Wireless* system from B2C/B2B.

Preliminary Supply Chain Design (Step 5)

Supply Chain Stage Representation

A supply chain is a complex system involving various stages that work together in order to supply high quality products on time and at the lowest possible cost. This trade off of responsiveness and efficiency is important to note as the ultimate goal is to maximize the supply chain profitability given a supply chain is only as strong as its weakest link. The following diagram puts together the stages involved in the supply chain for Honeywell to manufacture and sell their home security systems.



It is important to note that Honeywell is involved in more than one stage manufacturing, distribution (through partnerships), and retail (direct sales). Honeywell utilizes a differentiated competitive strategy leveraging its expertise across various industries including the building automation industry. This vertical integration supports their differentiated strategy by enabling tighter quality control over critical security components, faster response to market feedback, and the ability to offer both DIY and professional installation channels. Their expertise in building automation allows them to manufacture proprietary sensors and communication modules in-house while partnering with distributors like ADI Global and Alarmax to reach diverse customer segments from DIY homeowners to professional installers.

Network Flows

- Product Flow
 - Physical components and the finished product move downstream starting from the suppliers, manufacturers, assembly distributors, retailers, and finally to the

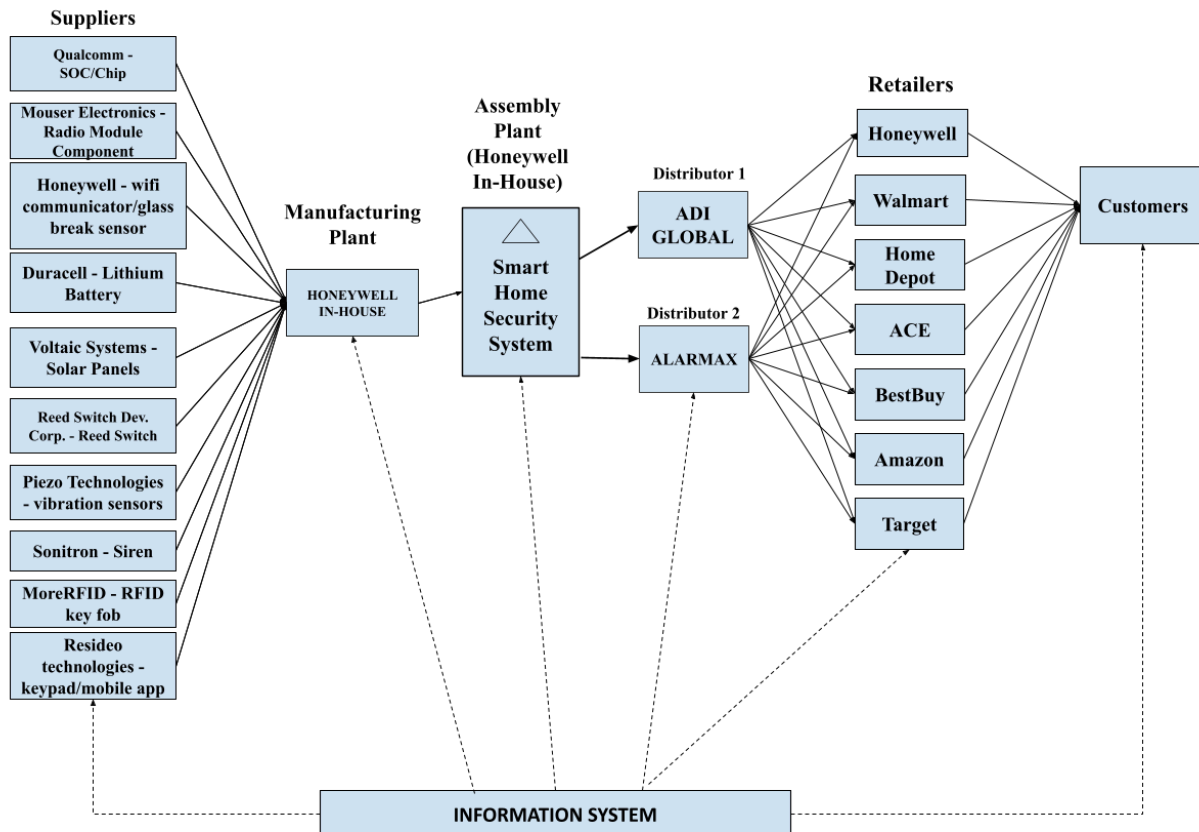
customers. Raw materials like motion sensors from an established supplier are sent to Honeywell's manufacturing for assembly into the finished security system. The products are shipped off to the distributors who deliver to one of the many retailers contracted to work with Honeywell.

- Cash Flow
 - The purchasing of the final smart home security system by customers sends the money upstream from the retailers to suppliers. When a smart home security system is purchased from one of the retailers, the payment initiates the cash flow that eventually reaches Honeywell and its suppliers. This makes sure that all supply chain partners are compensated on time, enabling them to reinvest in production, maintain operations, and fund further innovations.
- Information Flow
 - Information is both upstream and downstream in the supply chain.
 - For upstream, information on customer demand, inventory levels, and sales performance flows from retailers and distributors to manufacturers and suppliers. Sales data is shared with Honeywell, enabling them to adjust their production forecasts and maintain optimal stock levels at the retailer locations.
 - For downstream, the manufacturing/assembly plant and distributors communicate the product inventory availability and delivery updates to retailers and customers. Shipment tracking information details sent to the retailer are shared to the customer, allowing for real-time updates.

While the stage representation provides a linear view of supply chain progression, understanding the interconnected relationships between multiple entities at each stage requires a network perspective. The following section examines how suppliers, manufacturers, distributors, retailers, and customers interact within a web of transactions and information exchanges

Supply Chain Network Representation

The network view of the smart home security system supply chain exhibits the complex relationships between the different entities of suppliers, manufacturers, distributors, retailers, and customers. This perspective showcases the transaction and movement between the materials, money, product, and information. It shows how Honeywell's internal operations work as well as the level of reliance it has on third-party partners in order to reach customers efficiently.



The sections that follow break down the individual components of the network as outlined in the above diagram, explaining their roles and expected contribution to the making of the product.

Suppliers

The supply chain begins with third-party suppliers providing the necessary components and materials necessary for the smart home security system. These suppliers operate outside of Honeywell's organizational boundary and specialize in specific parts and technologies. Honeywell integrates these raw components through in-house manufacturing and assembly.

- System-on-Chip (SOC) components sourced from Qualcomm Technologies make the core of the system, supporting embedded processing capabilities necessary for the coordination between communication, sensing, and control within the connected devices of the smart home security system.
- Radio-Frequency (RF) modules supplied by Mouser Electronics allow for reliable short-range wireless communication between sensors and central control. All sensors within the system and any peripheral devices are able to communicate within the home environment by support of the RF modules. The mesh networking sensor module, MeshNet, supplied by Embedded Data Systems (EDS) extends the coverage of the distributed communication across all devices in the system.

- Wifi Communicator supplied by Honeywell's in-house manufacturing team allows the home security system to connect to the home's local wireless network so that data is able to be sent to the home owner's smartphone.
- The smart home security system is supported through primary and secondary sources of power. Disposable lithium CR2450 batteries produced by Duracell, offer long-term power exceeding at least a year, for low-power wireless sensors. In addition, solar panels supplied by manufacturers such as Voltaic Systems offer additional support for devices with larger energy requirements or to extend the battery life in select components.
- Reed Switches supplied by Reed Switch Developments Corporation. These switches are the primary sensors that will monitor doors and windows. The switches consist of a magnet, switch, and an electrical circuit that breaks with the open of a door or window, triggering an alarm and alerting the home owner.
- In-floor piezoelectric vibration sensors supplied by Piezo Technologies. These sensors detect vibrations caused by footsteps on a floor. When a mechanical vibration is detected, an electrical charge is sent to the piezo crystal alerting the home owner of a potential intrusion.
- Acoustic glass break sensors supplied by Honeywell's in-house manufacturing team are designed to listen for the specific sound frequency of broken glass using microphones. When the sound of glass breaking is triggered, the system detects and alerts the home owner before an intruder is able to enter.
- Piezoelectric Alarm supplied by Sonitron detects mechanical pressure and converts it into electrical energy, triggering a loud siren that will help alert the home owner and/or neighbors of a potential home intrusion.
- RFID/NFC key fob supplier supplied by MoreRFID acts as a digital key that allows the home owner to arm or disarm the home security system. The home owner simply taps the key fob against the reader allowing for a contactless and convenient process.
- Mobile application software supplied by Resideo Technologies provides remote access, real life camera feeds, and the ability to arm and disarm the home security system all conveniently on the user's smartphone or tablet.
- The keypad interface supplied by Resideo Technologies acts as the control center for the entire home security system. Typically installed near the entry points of the home, the keypad interface allows the user to arm and disarm and view sensors, cameras and notifications throughout the home with a pin.

These suppliers ship components to Honeywell manufacturing facilities based on demand signals and production schedules.

Manufacturing/Assembly

Honeywell's in-house manufacturing plan uses the supplied components to create specialized parts that will come together in the assembly process. They use both third-party components as

well as in-house capabilities to produce system components. The assembly plant is where the subsystems and raw materials come together to form a fully functioning smart home security system. This stage of the supply chain network focuses on quality control, component testing, and preparation for final assembly.

Assembly Plant

The Honeywell Smart Home Systems Assembly Plant is where all the subsystems and components are integrated into a fully functional smart home system. This facility is responsible for:

- Final assembly
- System integration
- Functional testing
- packaging

Once assembly is complete, the finished products are released into the distribution network.

Distributors/Retailers

Once the product is assembled, the functional systems enter the distribution channel where distributors transport the units to regional hubs, storing and preparing to deliver the product to retailers. The products are shipped to third-party distributors that manage large-scale logistics, warehousing, and order fulfillment for retailers. Honeywell's distributors are:

- ADI Global Distribution
- Alarmax Distribution

These distributors work as intermediaries between the retailers and Honeywell, helping to improve product availability and reduce delivery lead times. As distributors manage the logistics of inventory, Honeywell's retailers make the product available to customers. Honeywell has numerous certified reliable retailers such as:

- Honeywell Website (Direct to consumer)
- Walmart
- Home Depot
- ACE Hardware
- BestBuy
- Amazon.com (online retailer)
- Target
- Other security specialty retailers

These stages act as the bridge between production and end users.

Customers

The customers who would buy the product are home owners looking for a reliable smart home security system. Their usage of the product and any feedback impact the demand of the product, which flows upstream through the network.

The Supply Chain Network representation emphasizes how suppliers, manufacturers, distributors, retailers, and customers coordinate to ensure efficient production, distribution, and high responsiveness for the Distributed Wireless system. Having established both the stage-by-stage progression and network view of this supply chain, the team can now evaluate the effectiveness of this preliminary design and identify key insights that will inform subsequent project phases.

Proposal Reflection

Assessment of Work

This project proposal successfully establishes the foundational supply chain structure for the *Distributed Wireless* system. The stage and network representations clearly identify key suppliers, manufacturing processes, and distribution channels. The FAST diagram effectively demonstrates items which allow us to clearly see the bridges between product functionality requirements and supply chain requirements. The subsequent diagrams are clear representations of how key components of the supply chain stages/network connect to one another.

Insights and Takeaways

- Honeywell's multi-stage involvement (manufacturer + retailer) → enables tighter control over quality and customer feedback loops
- Nine component supplier categories identified → subsystems mapped to supply chain dependencies through FAST diagram
- Dual-channel distribution (ADI Global/Alarmax + direct retail) → broader market reach but adds complexity

Appendix

Revisions

Issue Identified	Changes Made	Team Members Responsible
Missing real dates for weeks and group meetings, and missing tentative project coordinators throughout Project Plan	Added dates to Time Phase Project Plan and noted the projects project coordinators	Hailey McElroy
Need to provide more bridging/explanatory text between Final Concept → FAST → Subsystems → Stage Representation → Network Representation.	Reassessed explanatory/bridging text segments between the listed sections in order to make the connection between each section more seamless	Hailey McElroy Vivian Nguyen
The contribution table should clearly state the name and include the dates for the following: (1) Classroom/Lecture attendance, (2) Project review attendance, (3) Group meeting attendance	Updated member contributions table	Garrett Arcand

Member Contributions Table

Name	Classroom/Lecture Attendance		Project Review Attendance	Group Meeting Attendance		Project Report Contributions
	Tues 1/6	Thurs 1/8	Weds 1/14	1/11	1/12	
James Burch	☒	☒	☒	☒	☒	Created the Supply Chain Stage Representation diagram
Hailey McElroy	☒	☒	☒	☒	☒	- Put together a Use-Case diagram to give an overview to the products functionality - Updated final concept - Updated FAST - Addition of FAST
Vivian Nguyen	☒	☒	☒	☒	☒	- Created the Supply Chain Network Representation diagram - Worked on the explanatory content of the Supply Chain Network Representation - Worked on the network flow explanatory content
Garrett Arcand	☒	☒	☒	☒	☒	- Formulated time-phase project plan based on project proposal instructions and class layout - Coordinated meetings and assignments
Mohammed Abdul-Haadi Ansari	☒	☒	☐	☒	☒	- Phase Reflection - Major refinement of competitive strategy statement - Fixed minor typos across document
Collin Yuen	☐	☐	☐	☒	☒	- Created the Supply Chain Network Representation diagram - Worked on both SC Stage Representation diagram and SC Network Representation diagram
Abinaya Ramesh	☒	☒	☐	☐	☐	Updated the stages in the Supply chain network representation with more detailed explanations

Post-meeting Action Table

Issue	Assigned Member	Timeline
- Fix table of contents according to instructions (4 sections) - Additional sections should be converted to subsections	Collin Yuen	☒ Complete 1/27
- Relocate proposal to top “To design and manage...” - Should be on page after TOC	Garrett Arcand	☒ Complete 1/25
Add dates to time-phase plan	Garrett Arcand	☒ Complete 1/25
- Relocate and reformat overview section - Should be a setup paragraph	Hailey McElroy	☒ Complete 1/27
Final concept sketch needed on FAST!	James Burch	☒ Complete 1/27
Each section needs a tie-in (glue) paragraph	Mohammed Ansari	☒ Complete 1/27
- Name suppliers in our SC diagrams - Research suppliers for this type of product	Vivian Nguyen/Collin Yuen	☒ Complete 1/27
- Improve network stages explanation - Introduce boundaries, relabel duplicates as necessary	Garrett Arcand	☒ Complete 1/27
Class attendance in contribution appendix	Garrett Arcand	☒ Complete 1/25

Group 12: Phase I

“Distributed Wireless” Home Security System

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Executive Summary

Purpose:

After careful review of our Project Proposal, in this Phase I we will develop a business model, develop the overall supply chain strategy, create/implement a process for estimating the first 8 quarters of demand, and perform the Static Forecasting method for years 3 and 4.

Key Findings:

Our demand estimation shows Year 1 sales of approximately 174,800 units, increasing to about 181,300 units in Year 2. Demand varies by quarter, with higher demand in Q3 due to seasonality factors such as increased home sales and property crime rates. Using the first 8 quarters of data, a total of 356,100 units, we calculated a static forecast of about 44,513 units per quarter for years 3 and 4.

Conclusion:

The supply chain should focus on being more responsive during the early stages of the product due to uncertain demand, and then gradually shift toward efficiency as demand stabilizes. The current supply chain design supports this balance, but the static forecasting method has limitations since it doesn't account for seasonality or growth, which can lead to less accurate predictions.

Objective

To design and manage the supply chain for Distributed Wireless in Honeywell. Building off of our revised Final Product Concept from our Project Proposal, in this phase we seek to develop a business model, the overall supply chain strategy, create and implement a process to estimate the first eight quarters of demand data, and finally perform demand forecasting on the data obtained.

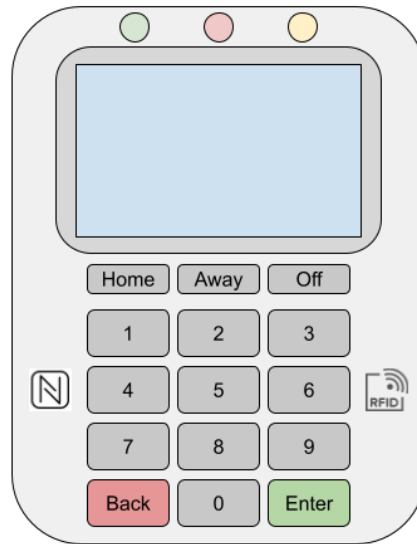
Product Concept (1)

We will be developing the supply chain business model, strategy, and demand forecasting for our product concept, the Distributed Wireless Home Security System. In order to do so, we must review our product concept and its supply chain.

This security system requires connection to Wi-Fi and the mobile application for connectivity between the keypad, remote key fob, siren, floor vibration sensors, door sensors, and window sensors to be established. Our concept is detailed in the following Use Case diagram, FAST diagram, and sub-system explanation.

Keypad and Keyfob

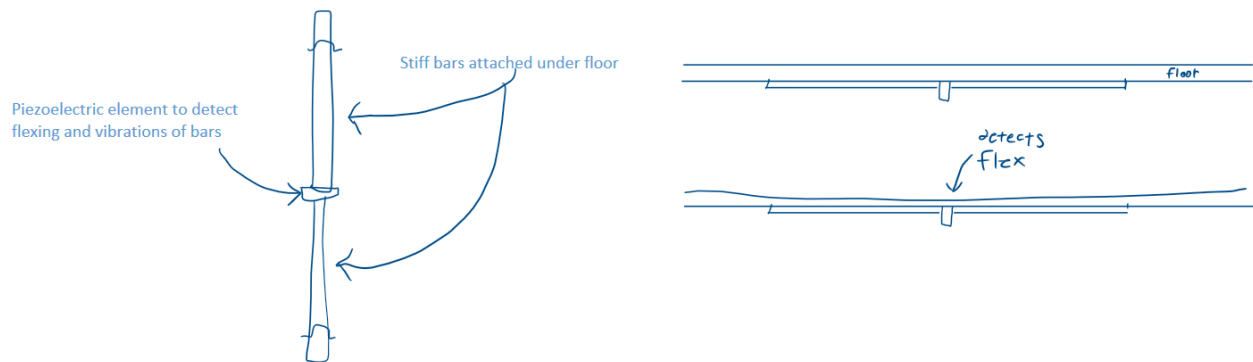
- Contains Wi-Fi and mesh radio for communication with other devices on the system
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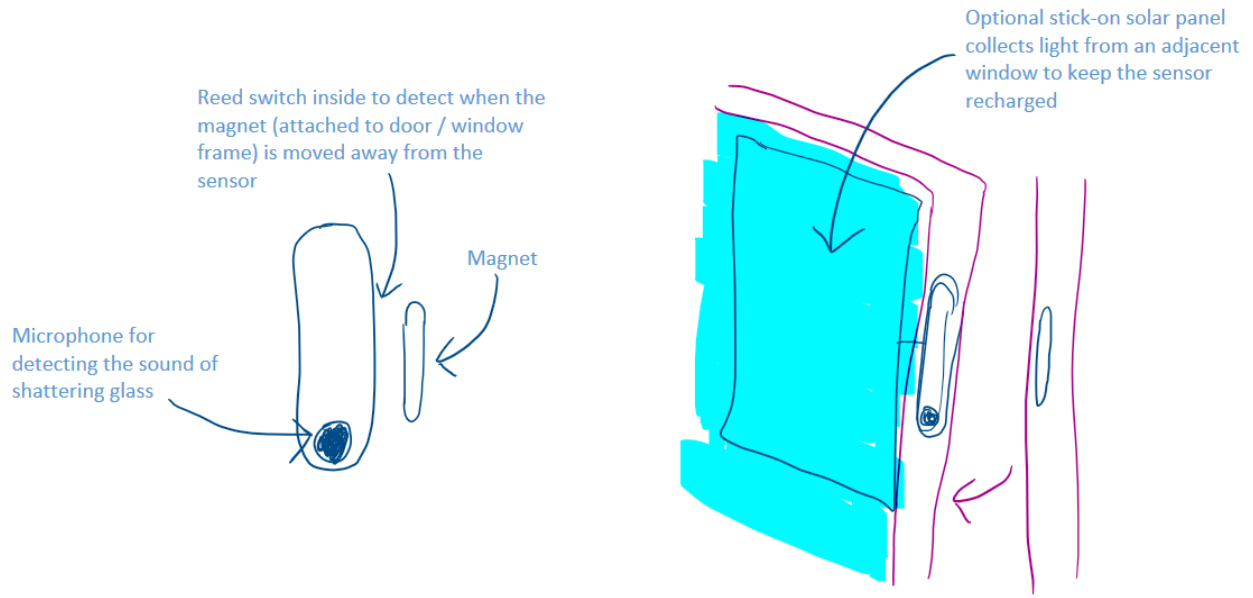
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Floor Vibration Sensor



Door / Window Sensor with Solar Panel

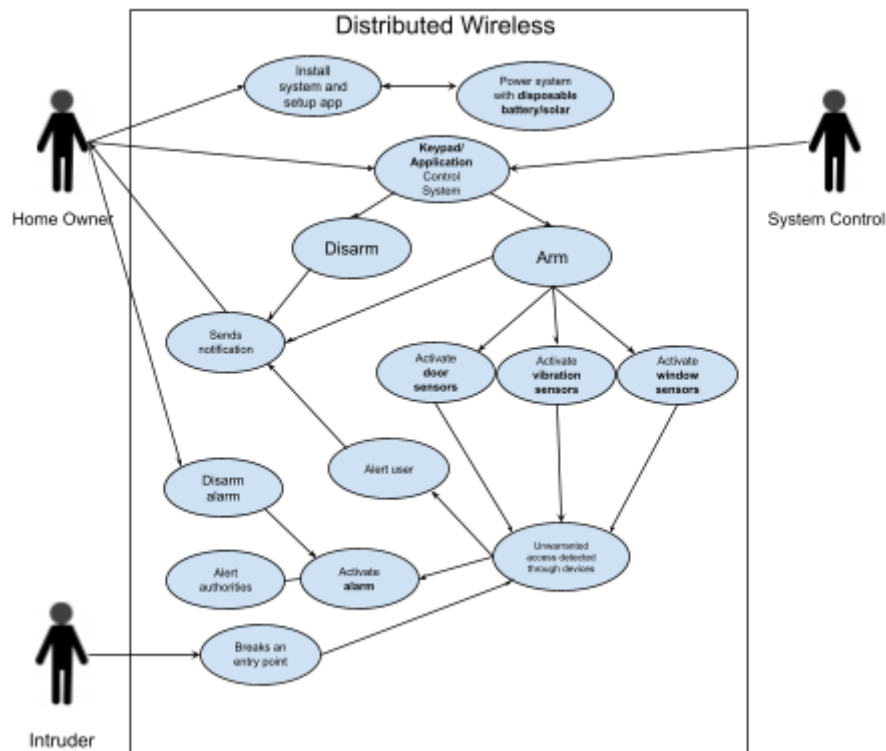


Siren

- Contains a loud piezoelectric noisemaker to alert homeowner and intruders



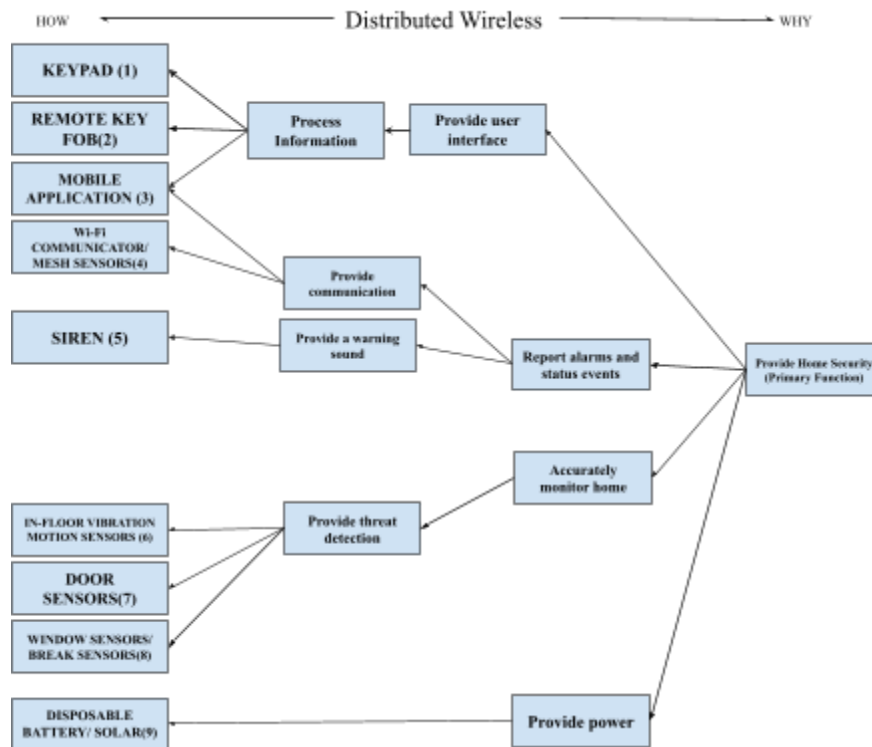
Use Case Diagram



The user is able to control the security of their household using the keypad, keyfob, or application. Upon arming or disarming the system with any one of these system controls the user is notified by the application and the status of the system is updated. All sensors activated will then monitor the household access. If unwarranted access is detected the alarm is activated and a notification is sent to the user allowing them to deactivate the alarm, disarm the system, or alert the authorities.

Given this functional overview, the following FAST diagram is utilized to break down the home security system's primary purpose into its primary function and supporting sub-functions. The diagram describes the necessary functions to provide the intended home security performance. Each overarching function is achieved through identified supporting actions. The FAST diagram intends to clearly outline the logical functioning foundation of Honeywell's Distributed Wireless home security system to guide the subsequent architecture design implementation.

Updated FAST Diagram



The FAST diagram development breaks down the subsystems and subfunctions that define the overall structure of Honeywell's Distributed Wireless home security system. This dissection lays out the recurring functionalities that need to be physically realized within the final product. It serves as a visual representation of the product's structure, showing how the home-security system should operate, its key components, and how these elements will connect to the overall supply chain.

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Sub-systems

Key sub-systems:

1. Keypad
2. Remote key fob
3. Mobile Application
4. Wi-Fi communicator / mesh sensors
5. Siren
6. Floor vibration sensors
7. Door sensors
8. Window sensors / break sensors
9. Disposable Battery / solar

Key sub-functions:

1. Provide a user interface
2. Provide warning sound
3. Provide threat detection
4. Process information
5. Provide communicator link between devices
6. Provide power

Given this set of sub-systems, the following set of solution principles were realized in product development and will need to be sourced to develop the supply chain design.

Solution Principles

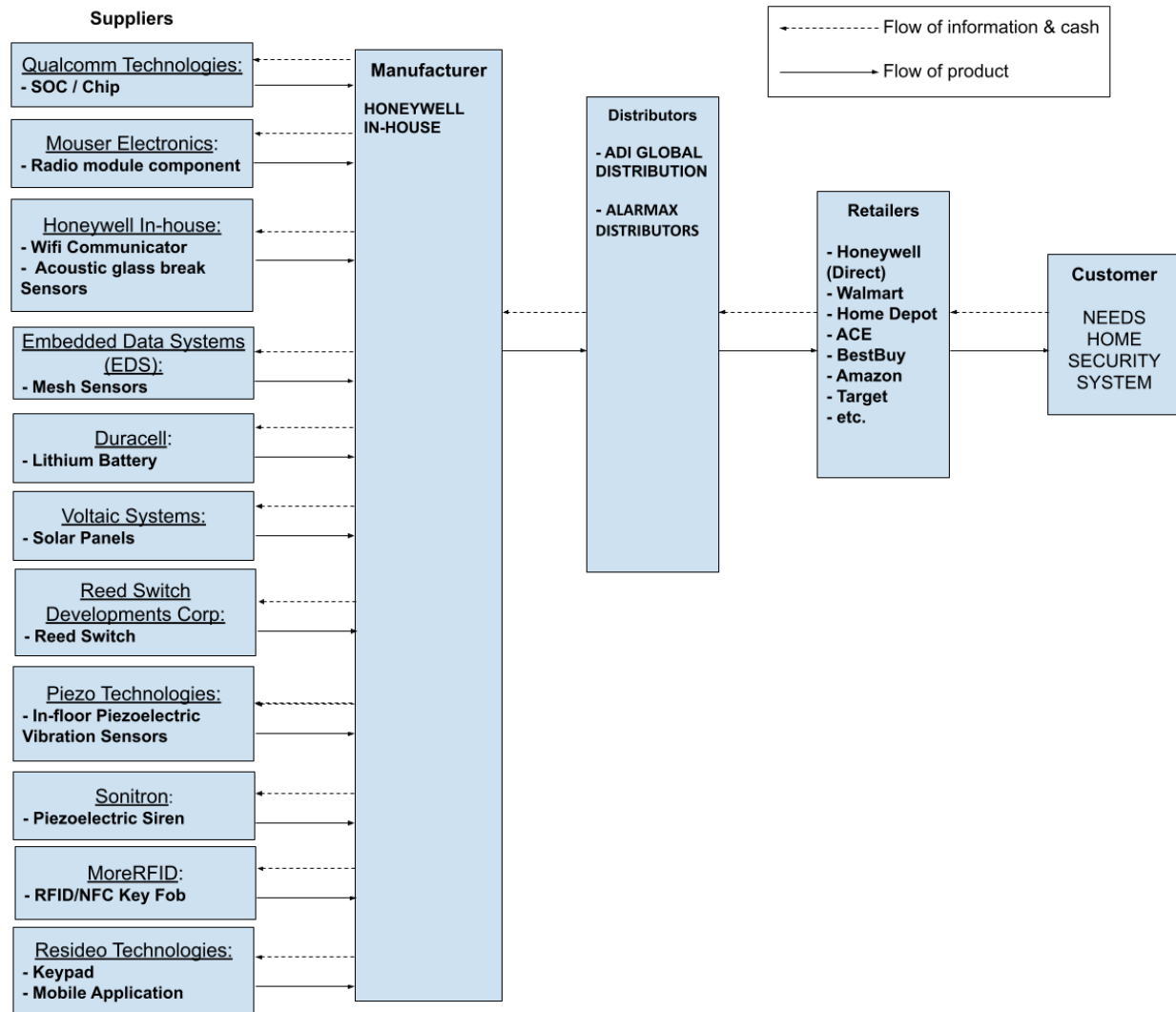
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- Wi-Fi communicator + mesh sensors
 - Provide connection between sub-systems.

With the aforementioned product architecture, sub-systems, and solution principles clearly defined, the team can now map out a supply chain design. A stage and network representations allows us to demonstrate how Honeywell sources, manufactures, distributes, and delivers the *Distributed Wireless* system from B2C/B2B.

Supply Chain Stage Representation

A supply chain is a complex system involving various stages that work together in order to supply high quality products on time and at the lowest possible cost. This trade off of responsiveness and efficiency is important to note as the ultimate goal is to maximize the supply chain profitability given a supply chain is only as strong as its weakest link. The following diagram puts together the stages involved in the supply chain for Honeywell to manufacture and sell their home security systems.



It is important to note that Honeywell is involved in more than one stage manufacturing, distribution (through partnerships), and retail (direct sales). Honeywell utilizes a differentiated competitive strategy leveraging its expertise across various industries including the building automation industry. This vertical integration supports their differentiated strategy by enabling tighter quality control over critical security components, faster response to market feedback, and the ability to offer both DIY and professional installation channels. Their expertise in building automation allows them to manufacture proprietary sensors and communication modules in-house while partnering with distributors like ADI Global and Alarmax to reach diverse customer segments from DIY homeowners to professional installers.

Network Flows

- Product Flow
 - Physical components and the finished product move downstream starting from the suppliers, manufacturers, assembly distributors, retailers, and finally to the customers. Raw materials like motion sensors from an established supplier are

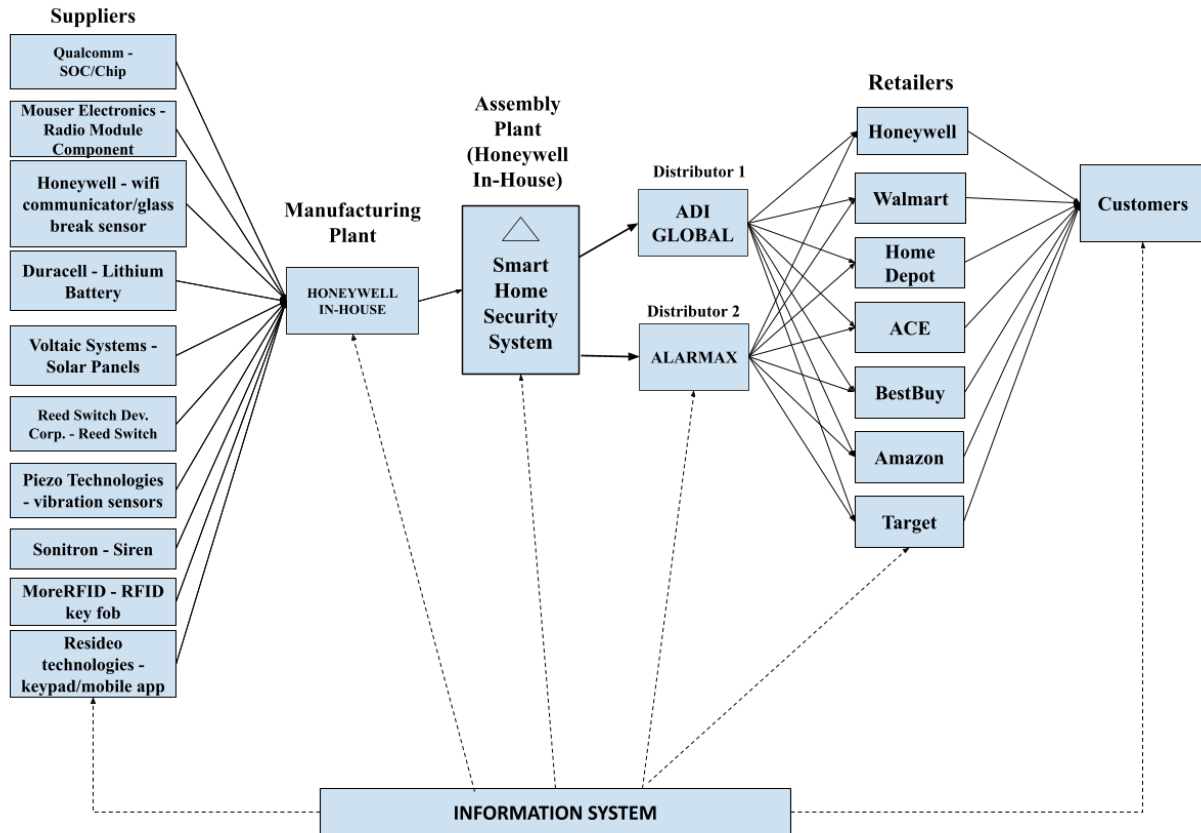
sent to Honeywell's manufacturing for assembly into the finished security system. The products are shipped off to the distributors who deliver to one of the many retailers contracted to work with Honeywell.

- **Cash Flow**
 - The purchasing of the final smart home security system by customers sends the money upstream from the retailers to suppliers. When a smart home security system is purchased from one of the retailers, the payment initiates the cash flow that eventually reaches Honeywell and its suppliers. This makes sure that all supply chain partners are compensated on time, enabling them to reinvest in production, maintain operations, and fund further innovations.
- **Information Flow**
 - Information is both upstream and downstream in the supply chain.
 - For upstream, information on customer demand, inventory levels, and sales performance flows from retailers and distributors to manufacturers and suppliers. Sales data is shared with Honeywell, enabling them to adjust their production forecasts and maintain optimal stock levels at the retailer locations.
 - For downstream, the manufacturing/assembly plant and distributors communicate the product inventory availability and delivery updates to retailers and customers. Shipment tracking information details sent to the retailer are shared to the customer, allowing for real-time updates.

While the stage representation provides a linear view of supply chain progression, understanding the interconnected relationships between multiple entities at each stage requires a network perspective. The following section examines how suppliers, manufacturers, distributors, retailers, and customers interact within a web of transactions and information exchanges.

Supply Chain Network Representation

The network view of the smart home security system supply chain exhibits the complex relationships between the different entities of suppliers, manufacturers, distributors, retailers, and customers. This perspective showcases the transaction and movement between the materials, money, product, and information. It shows how Honeywell's internal operations work as well as the level of reliance it has on third-party partners in order to reach customers efficiently.



The sections that follow break down the individual components of the network as outlined in the above diagram, explaining their roles and expected contribution to the making of the product.

Suppliers

The supply chain begins with third-party suppliers providing the necessary components and materials necessary for the smart home security system. These suppliers operate outside of Honeywell's organizational boundary and specialize in specific parts and technologies. Honeywell integrates these raw components through in-house manufacturing and assembly.

- System-on-Chip (SOC) components sourced from Qualcomm Technologies make the core of the system, supporting embedded processing capabilities necessary for the coordination between communication, sensing, and control within the connected devices of the smart home security system.
- Radio-Frequency (RF) modules supplied by Mouser Electronics allow for reliable short-range wireless communication between sensors and central control. All sensors within the system and any peripheral devices are able to communicate within the home environment by support of the RF modules. The mesh networking sensor module, MeshNet, supplied by Embedded Data Systems (EDS) extends the coverage of the distributed communication across all devices in the system.

- Wifi Communicator supplied by **Honeywell's in-house manufacturing team** allows the home security system to connect to the home's local wireless network so that data is able to be sent to the home owner's smartphone.
- The smart home security system is supported through primary and secondary sources of power. Disposable lithium CR2450 batteries produced by **Duracell**, offer long-term power exceeding at least a year, for low-power wireless sensors. In addition, solar panels supplied by manufacturers such as Voltaic Systems offer additional support for devices with larger energy requirements or to extend the battery life in select components.
- Reed Switches supplied by **Reed Switch Developments Corporation**. These switches are the primary sensors that will monitor doors and windows. The switches consist of a magnet, switch, and an electrical circuit that breaks with the open of a door or window, triggering an alarm and alerting the home owner.
- In-floor piezoelectric vibration sensors supplied by **Piezo Technologies**. These sensors detect vibrations caused by footsteps on a floor. When a mechanical vibration is detected, an electrical charge is sent to the piezo crystal alerting the home owner of a potential intrusion.
- Acoustic glass break sensors supplied by **Honeywell's in-house manufacturing team** are designed to listen for the specific sound frequency of broken glass using microphones. When the sound of glass breaking is triggered, the system detects and alerts the home owner before an intruder is able to enter.
- Piezoelectric Alarm supplied by **Sonitron** detects mechanical pressure and converts it into electrical energy, triggering a loud siren that will help alert the home owner and/or neighbors of a potential home intrusion.
- RFID/NFC key fob supplier supplied by **MoreRFID** acts as a digital key that allows the home owner to arm or disarm the home security system. The home owner simply taps the key fob against the reader allowing for a contactless and convenient process.
- Mobile application software supplied by **Resideo Technologies** provides remote access, real life camera feeds, and the ability to arm and disarm the home security system all conveniently on the user's smartphone or tablet.
- The keypad interface supplied by **Resideo Technologies** acts as the control center for the entire home security system. Typically installed near the entry points of the home, the keypad interface allows the user to arm and disarm and view sensors, cameras and notifications throughout the home with a pin.

These suppliers ship components to Honeywell manufacturing facilities based on demand signals and production schedules.

Manufacturing/Assembly

Honeywell's in-house manufacturing plan uses the supplied components to create specialized parts that will come together in the assembly process. They use both third-party components as

well as in-house capabilities to produce system components. The assembly plant is where the subsystems and raw materials come together to form a fully functioning smart home security system. This stage of the supply chain network focuses on quality control, component testing, and preparation for final assembly.

Assembly Plant

The Honeywell Smart Home Systems Assembly Plant is where all the subsystems and components are integrated into a fully functional smart home system. This facility is responsible for:

- Final assembly
- System integration
- Functional testing
- packaging

Once assembly is complete, the finished products are released into the distribution network.

Distributors/Retailers

Once the product is assembled, the functional systems enter the distribution channel where distributors transport the units to regional hubs, storing and preparing to deliver the product to retailers. The products are shipped to third-party distributors that manage large-scale logistics, warehousing, and order fulfillment for retailers. Honeywell's distributors are:

- ADI Global Distribution
- Alarmax Distribution

These distributors work as intermediaries between the retailers and Honeywell, helping to improve product availability and reduce delivery lead times. As distributors manage the logistics of inventory, Honeywell's retailers make the product available to customers. Honeywell has numerous certified reliable retailers such as:

- Honeywell Website (Direct to consumer)
- Walmart
- Home Depot
- ACE Hardware
- BestBuy
- Amazon.com (online retailer)
- Target
- Other security specialty retailers

These stages act as the bridge between production and end users.

Customers

The customers who would buy the product are home owners looking for a reliable smart home security system. Their usage of the product and any feedback impact the demand of the product, which flows upstream through the network.

The Supply Chain Network representation emphasizes how suppliers, manufacturers, distributors, retailers, and customers coordinate to ensure efficient production, distribution, and high responsiveness for the Distributed Wireless system. Having established both the stage-by-stage progression and network view of this supply chain, the team can now evaluate the effectiveness of this preliminary design and identify key insights that will inform subsequent project phases.

Supply Chain Business Model (2)

Having put together the supply chain network representation in section 1 we shall develop a business model to ensure the flows of inventory and The following section develops a supply chain business model for Honeywell's Distributed Wireless home security system.

Process to Develop the Supply Chain Business Model

Focusing on the Home Security Market of Honeywell's Building Automation market we will develop a supply chain business model by taking the following steps:

Part 1: Competitive Strategy

- Identify customers motivators for purchasing security systems
- Map the competitive strategies of the market players

Part 2: Develop Honeywells SC strategy

- Define the processes of the supply chain
 - o Create a cycle view
- Define and understand the flows of the supply chain

Part 3:

- Put together Honeywell's supply chain model

Execution

Part 1: Competitive Strategy

Customer motivators for purchasing a home security system:

The following list ranks the highest to lowest motivators for purchasing a system identified by the [SafeHome.org Home Security Market Report](https://www.safehome.org/home-security-market-report/):

1. Professional Monitoring Options

2. Price
3. Integration with smart devices/apps
4. Ease of installation
5. Brand
6. Month to month contract options
7. Portability

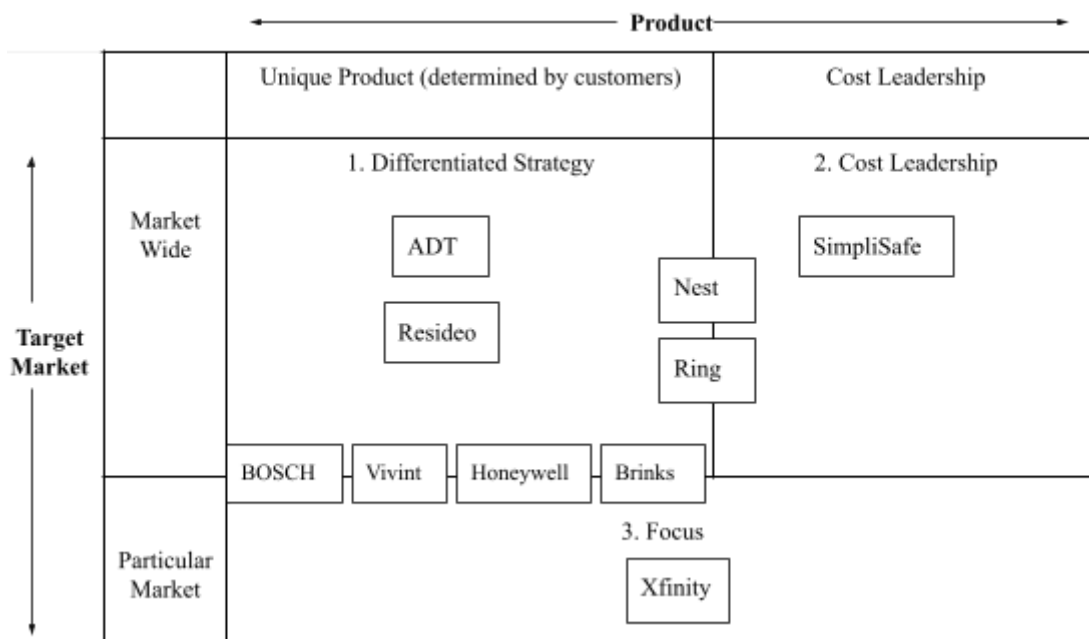
With the motivators listed above we have derived the following list of needs for the supply chain outside of the products customer needs explained by the product development team.

1. Responsive customer support
2. System that is cheap (cost has been minimized with efficiency)

Having laid out these customer motivators and needs, when developing the supply chain model we can identify how these needs and demands for the product to be supplied will influence the model. The following Competitive Map in Figure 2 identifies the different competitors and their strategies for meeting customer needs in the security system market.

Market's Competitive Strategy Map

Figure 2 - Porter's Competitive Strategy Map



**Companies in the Home Security market can take on a combination of competitive strategies given the customer needs, and technologies that are continually evolving allowing for high tech customizable systems that fit a variety of markets. The companies established in Figure 2 reflect the market snapshot and analytics developed by [Home Security Systems Market Analysis](#)*

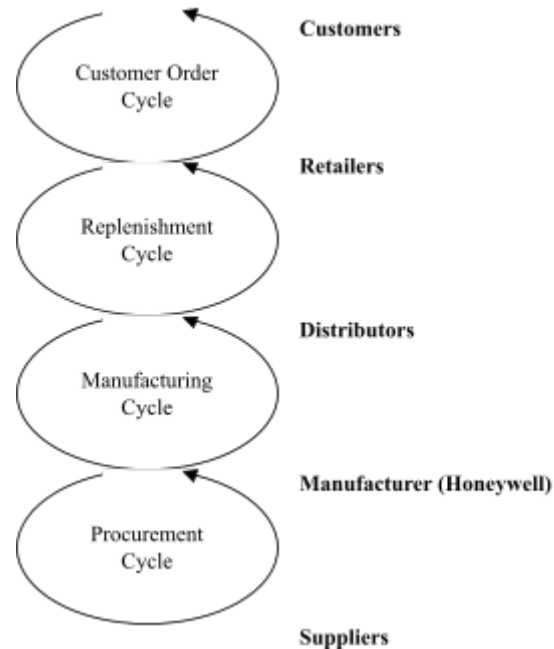
In the scope of Honeywell in the Home Security market with Distributed Wireless, as the product development team and supply chain management team has positioned itself to have a more focused strategy

Competitive Strategy	Company	Explanation
Differentiated Strategy	ADT	Valued for their monitored systems as a well established brand offering bundles.
	Resideo	Specialized security products that are valued for their performance and offered market wide.
Differentiated/Cost Leadership	Nest	Competitive pricing offered market wide given the advantage of Google's reach
	Ring	Low cost hardware and plan subscriptions making the system accessible and customizable.
Cost Leadership	SimpliSafe	Affordable plans offered market wide made flexible to fit customer needs.
Differentiated Strategy/ Focus	Honeywell	As a company Honeywell maintains a differentiated strategy for their wide variety of highly valued products available to a wide market. While Honeywell as a whole maintains a differentiated competitive strategy with a focus in their priority markets which include Building Automation.
	Vivint	Targets a higher end customer market requiring a professional installation and valued responsiveness.
	Brinks	Targets customers in need of professionally monitored systems while prioritizing the responsiveness and reliability of their systems making them highly valued.
	BOSCH	Offering products market wide with a focus on tailoring the systems for their sectors in industries and households.
Focus	Xfinity	Targets Xfinity customers to bundle their home security system with their other services.

With this overview and analysis of the market itself we can see the supply chain stages of suppliers, manufacturers (like Honeywell), and buyers realized. As home security systems on the market are currently sold primarily through major retailers, storefronts or online platforms, the inclusion of a distributor and retailer in the supply chain stages is expected by customers. Upon manufacturing their products Honeywell sells through both distributors and retailers as well as through their own online platform [Honeywell.com](#). With this in mind the supply chain processes that are key to monitoring and stocking inventory come into play.

Supply Chain Processes

There are operational cycles that occur between adjacent stages in a supply chain. Given that Honeywell's supply chain stages that Honeywell is involved in the following cycle view can be put together to give their operations overview:

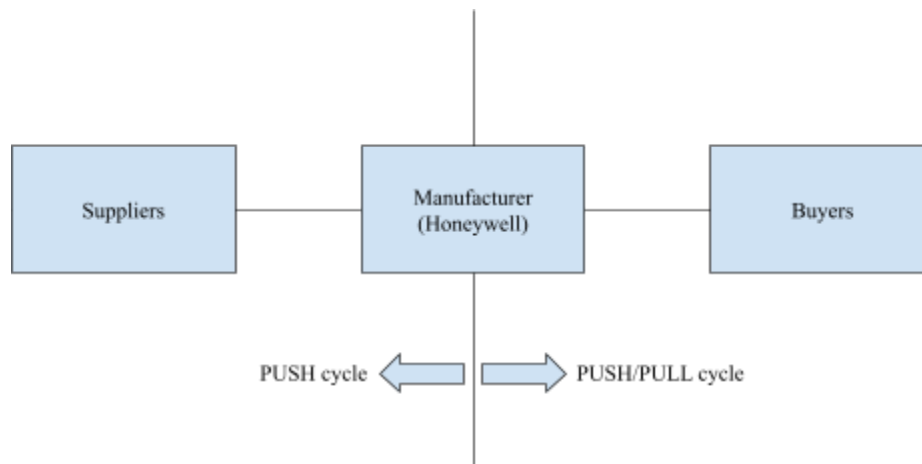


Given that each cycle has a set of operations associated with it the following table breaks down each.

Customer Order Cycle	Replenishment Cycle	Manufacturing Cycle	Procurement Cycle
- customer places an order - retailer receives orders - retailer processes order - retailer ships order to customer - customer receives the order	- retailer places an order to replenish stock - distributor receives orders - distributor processes order - distributor ships order to retailer - retailer receives the order	- distributor places an order to replenish stock - manufacturer receives order - manufacturer processes order - manufacturer ships order to distributor - distributor receives the order	- manufacturer places an order to replenish stock - suppliers receives order - suppliers processes order - suppliers ships order to manufacturer - manufacturer receives the order

Supply Chain Business Model

Honeywell's supply chain business model will rely on their involvement in a variety of stages in the supply chain. As a manufacturer and assembler of their products (with Honeywell in-house) as well as selling directly as a retailer Honeywell has more insight into the network flows between cycles. These cycles are separated as push or pull cycles, where push cycles are initiated in anticipation of demand and pull cycles are initiated by a request from a customer. In anticipation of demand Honeywell will manufacture Distributed Wireless to be supplied to their partnered distributors, to the retailer, then customer (PUSH cycle), or built directly to order upon receiving a request from a customer (PULL cycle). In anticipation of this demand Honeywell would stock the supplies needed to make their home security systems in anticipation of customer demand (PUSH cycle). Putting this all together the following supply chain model is realized:

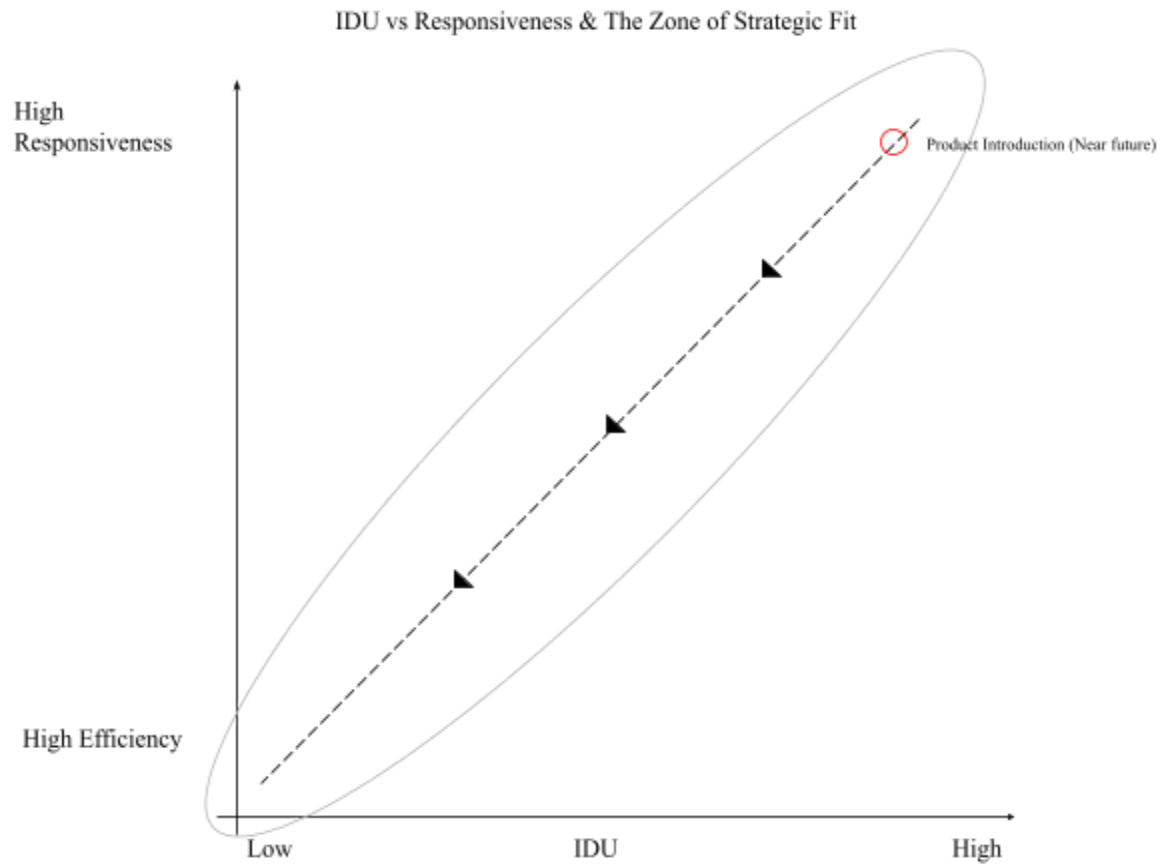


Overall Supply Chain Strategy (3)

Supply Chain Performance

As we are using a differentiated, semi-market-wide competitive strategy, our supply chain must be responsive, but also efficient. However, because our product is new to the market, the IDU of our product is high. We do not yet know how consumers will respond to our product. Due to this, at the launch time of our product, the responsiveness of our supply chain should be high. This will allow us to refine our strategy and work out ways to make it more efficient as more products are sold, while also responding to unpredictable changes in customer demand.

In the diagram below, the gray oval is the “zone of strategic fit.” This represents the region in which our supply chain strategy should remain for optimal performance. The dotted arrows are the supply chain strategy’s trajectory over the lifecycle of the product. The red circle is where our product’s supply chain currently stands.



Data Estimation (4)

The following section estimates the first 8 quarters of demand for the Distributed Wireless home security system. The estimation of demand data process aims to develop a reasonably realistic data-driven projection that aligns with the market, product lifecycle stages, and consumer patterns. This estimation approach will incorporate home purchase statistics, home intrusion statistics, and seasonality to determine the base demand and consumer behavior. Grounding the estimate with existing market data offers a credible foundation for the subsequent forecasting methods.

Process to Obtain Demand Data

1. Define Market/Target Segments
 - a. Identify the primary/expected users
 - b. Identify the main competitors
2. Product Life Cycle Analysis
 - a. Introduction Phase
 - b. Growth Phase
 - c. Maturity Phase
3. Demand based on Factors
 - a. Market Reach
 - b. Statistics related to Adjacent Security Products
 - c. Seasonality
4. Develop Realistic Adoption Estimates
 - a. Year One Adoption Estimates
 - b. Year Two Adoption Estimates
 - c. Create quarterly demand estimates based on (4a and 4b)

Execution

1. Market/Target Segments
 - a. Primary customers
 - i. Home Owners
 - b. Competitor products
 - i. SimpliSafe 21-Piece Security System:
 1. $P(\text{SimpliSafe}) = \$622.79 \text{ USD}$
 - ii. Vivint HomeProtect Pro:
 1. $P(\text{Vivint}) = \$499 \text{ USD}$
 - iii. Ring Alarm Pro Security Kit:
 1. $P(\text{Ring}) = \$595 \text{ USD}$
2. Product Life Cycle

- a. Introduction (Quarters 1-2)
 - i. The introduction phase will have limited adoption, focusing on showcasing the Distributed Wireless home security system in real home scenarios with testing trials and demonstrations.
 - b. Growth (Quarters 3-6)
 - i. The growth phase is expected to have increased adoption as positive results drive demand upwards.
 - ii. Endorsements of the Distributed Wireless by real homeowners will positively drive demand.
 - c. Maturity (Quarters 7-8)
 - i. The maturity phase expects a stabilized level of demand as the Distributed Wireless home security system is expected to achieve broader market recognition.
 - ii. Sales are driven by system upgrades, add-on component purchases, and new homeowners entering the market.
3. Demand based on Factors
- a. Home Security System Market Size⁵
 - i. The global home security system market is valued at 72.4 billion USD in 2025 and is projected to reach 109.4 USD between 2026-2030. These statistics reflect a compound annual growth rate (CAGR) of 8.6%.
 - b. Statistics related to Adjacent Security Products
 - i. Wireless security cameras¹⁸
 - 1. The market for wireless security cameras is projected to grow at 20.79% from 2025 to 2035, reaching an estimate of 128.95 billion USD.
 - ii. Smart doorbells¹⁹
 - 1. The global smart doorbell market is projected to grow at a CAGR of 33.4% from 2023 to 2030, reaching an estimated 33,428.5 million (=33.4 billion).
 - c. Influencers of Demand ^{7,8,9}
 - i. Research outlines several recurring factors on home security consumer adoption:
 - 1. The real estate market activity: home/property purchasing (long term)
 - 2. Residential crime seasonality (short term)
 - d. Home Purchasing Annual Statistics ^{10,11,12}

- i. Fig. 1 - Monthly total home sales in the United States (data collected and reported by the National Association of Realtors)¹¹.

1. The peak of home sales data appears to occur between May to August, with the lowest occurring between January and February.

Monthly total U.S. existing home sales
Without seasonal adjustment

Year	Jan	Feb	Mar	Apr	May	June	July	Aug	Sept	Oct	Nov	Dec
2016	302K	314K	421K	470K	525K	582K	513K	539K	486K	445K	418K	437K
2017	319K	315K	455K	447K	555K	600K	513K	535K	462K	458K	425K	427K
2018	313K	319K	434K	460K	535K	570K	523K	539K	421K	446K	406K	377K
2019	285K	311K	400K	456K	542K	528K	540K	532K	450K	462K	404K	434K
2020	317K	335K	416K	373K	372K	507K	597K	560K	563K	573K	493K	538K
2021	366K	366K	484K	513K	528K	615K	584K	576K	546K	526K	503K	513K
2022	352K	352K	456K	463K	499K	525K	454K	474K	428K	371K	325K	327K
2023	231K	269K	359K	337K	408K	433K	372K	401K	348K	332K	300K	297K
2024	234K	271K	325K	360K	405K	376K	390K	379K	330K	348K	315K	329K
Average	299K	315K	416K	452K	499K	540K	509K	520K	447K	440K	396K	410K

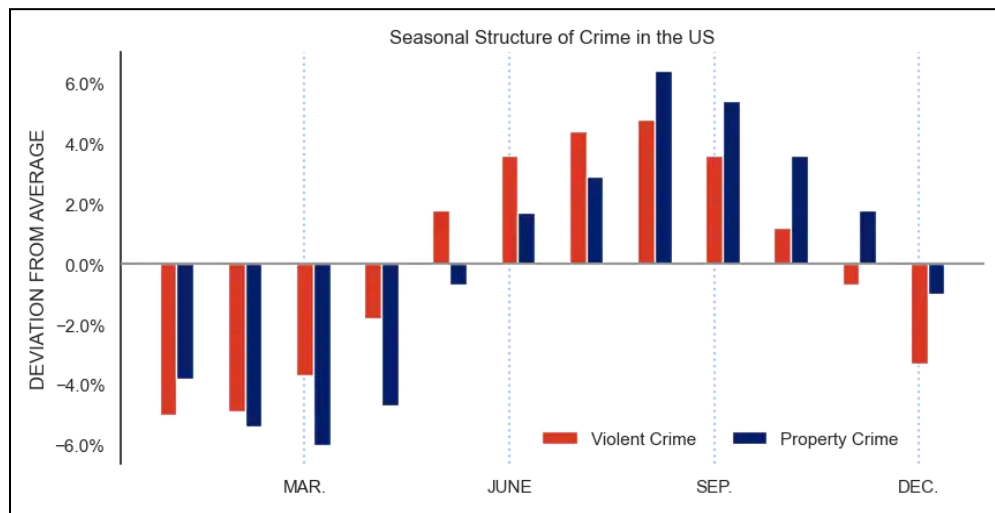
Table: Lance Lambert • Source: National Association of Realtors • Created with Datawrapper

ResiClub

- e. Home Intrusion (Property Crime) Statistics ^{7,9}

- i. Fig. 2 - Seasonal structure of crime in the U.S. (produced by the Pinkerton Crime Index data and federal agency sources)⁷

1. Research regarding the seasonality of crime in the United States reveals property related crime is much more prevalent between June to September.



4. Develop Realistic Adoption Estimates

Assumption:

1. In the first year, the demand would be lower and would gradually increase into year two as more consumers become aware of the product.

2. Home security system starting market size (2025) estimated to be \$4.37 billion¹³
3. Home security system market CAGR estimated to be 3.7%¹³
4. Honeywell's Distributed Wireless market share 2%, approximately \$87,400,000
5. Honeywell's Distributed Wireless will have an average selling price of \$500.

a. Year 1 (2026)

- i. Estimated Year 1 sales volume:

$$\$87,400,000 / \$500 = 174,800 \text{ units}$$

- ii. Estimated Year 2 sales volume:

$$174,800 \text{ units} \times (1.037) = \text{approximately } 181,300 \text{ units.}$$

b. Quarterly demand estimates based on (4a and 4b)

- i.

Year	Quarter	Period (t)	Demand (units)
2026	1	1	31,220
	2	2	47,200
	3	3	61,180
	4	4	35,200
TOTAL			174,800
2027	1	5	33,600
	2	6	50,200
	3	7	65,320
	4	8	32,180
TOTAL			181,300

- ii. Using the home security system starting market size in 2025 as a baseline, and a target of 2% of the market share for Honeywell in its first year entering the market, we built a sales demand and volume estimation for years one and two.

Demand Forecasting (5)

There are five different methods used to forecast demand data: Static, Moving Average, Simple Exponential Smoothing, Holt's, and Winter's methods. With the historical demand data for home security systems, developed in the Demand Estimation section and laid out below, we have performed demand forecasting using the Static method for years 3 and 4, 2028 and 2029 respectively.

Historical Demand Data

Quarter	Demand(units)
Q1	31,220
Q2	47,200
Q3	61,180
Q4	35,200
Q5	33,600
Q6	50,200
Q7	65,320
Q8	32,180

- Total Demand for quarters 1-8: 356,100 units
- # of Periods: 8 quarters

Static Demand Forecast Calculation(Years 3 and 4)

- Static Forecast = Total Historic Demand / # of periods
 $\rightarrow 356,100 / 8 = 44,512.5 = 44,513$ units per quarter

Quarter	Demand(units)
Q9	44,513
Q10	44,513
Q11	44,513
Q12	44,513
Q13	44,513

Q14	44,513
Q15	44,513
Q16	44,513

- Total Forecasted Demand for years 3 and 4 = $44,513 * 8 = 356,104$ units

**see worked spreadsheet calculations below*

Demand Forecasting - Distributed Wireless Home Security System								
Historical Demand Data (Quarters 1-8)								
Year	Quarter	Period (t)	Demand (Dt)	Notes				
2026	Q1	1	31220	Introduction Phase				
2026	Q2	2	47200	Introduction Phase				
2026	Q3	3	61180	Growth Phase				
2026	Q4	4	35200	Growth Phase				
2027	Q1	5	33600	Growth Phase				
2027	Q2	6	50200	Growth Phase				
2027	Q3	7	65320	Maturity Phase				
2027	Q4	8	32180	Maturity Phase				
Total Demand (Q1-Q8)			356100					
Average Demand per Quarter			44512.5					
Market Assumptions								
Market Size 2025			4370000000	USD				
Market CAGR			0.037	3.7%				
Target Market Share			0.02	2%				
Average Selling Price			500	USD/unit				
Periodicity (p)			4	Quarters per year				

Static Forecasting Method					
Formula: Forecast = Average of Historical Demand					
Static Forecast = Sum(D1-D8) / 8					
Year	Quarter	Period (t)	Forecast (Ft)	Method	Notes
2028	Q1	9	44513	Static	Years 3-4 forecast
2028	Q2	10	44513	Static	Years 3-4 forecast
2028	Q3	11	44513	Static	Years 3-4 forecast
2028	Q4	12	44513	Static	Years 3-4 forecast
2029	Q1	13	44513	Static	Years 3-4 forecast
2029	Q2	14	44513	Static	Years 3-4 forecast
2029	Q3	15	44513	Static	Years 3-4 forecast
2029	Q4	16	44513	Static	Years 3-4 forecast
Total Forecasted Demand (Q9-Q16)			356104		
Calculation Details					
Sum of Historical Demand			356100	Q1-Q8	
Number of Periods			8		
Static Forecast Value			44512.5	Rounded to 44513	
Limitations of Static Method					
- Does not account for seasonality					
- Does not account for growth trend					
- High-demand quarters underestimated					
- Low-demand quarters overestimated					

Deseasonalized Demand Calculation			
Periodicity (p) = 4 quarters			
Period (t)	Actual Demand (Dt)	Deseasonalized Demand	Calculation Method
1	31220		Not enough data
2	47200		Not enough data
3	61180	43700	Avg of periods 1-4 and 2-5
4	35200	44675	Avg of periods 2-5 and 3-6
5	33600	46825	Avg of periods 3-6 and 4-7
6	50200	47000	Avg of periods 4-7 and 5-8
7	65320	45325	Avg of periods 5-8
8	32180		Not enough data
Linear Regression for Level and Trend			
Using deseasonalized demand (periods 3-7)			
Regression Results			
Initial Level (L0)		18439	Intercept
Trend (T)		524	Slope
Deseasonalized Demand Formula			
$\bar{D}(t) = L0 + T \cdot t$			
$\bar{D}(t) = 18439 + 524 \cdot t$			

Seasonal Factor Calculation					
Period (t)	Actual (Dt)	Deseasonalized	Seasonal Factor (St)	Quarter	Avg Seasonal Factor
1	31220	18963	1.65	Q1	
2	47200	19487	2.42	Q2	
3	61180	20011	3.06	Q3	
4	35200	20535	1.71	Q4	
5	33600	21059	1.60	Q1	
6	50200	21583	2.33	Q2	
7	65320	22107	2.95	Q3	
8	32180	22631	1.42	Q4	
Average Seasonal Factors by Quarter					
Quarter	Periods Used	Calculation	Avg Factor	Description	
Q1	1, 5	$(1.65 + 1.60) / 2$	1.63	Low demand quarter	
Q2	2, 6	$(2.42 + 2.33) / 2$	2.38	Medium demand quarter	
Q3	3, 7	$(3.06 + 2.95) / 2$	3.01	Peak demand quarter	
Q4	4, 8	$(1.71 + 1.42) / 2$	1.57	Low demand quarter	
Note: Seasonal factors show strong seasonality					
Q3 has highest demand (summer installation peak)					
Q1 and Q4 have lowest demand (winter months)					

Similar Product Analysis			
Used to validate our product demand estimates			
Product Category	Market Size	CAGR	Purpose
Wireless Security Cameras	128.95B by 2035	20.79%	Validate growth expectations
Smart Doorbells (Ring, etc)	33.4B by 2030	33.4%	Entry-level security adoption
Full Home Security Systems	109.4B by 2030	8.6%	Primary market

See full export:

<https://drive.google.com/file/d/1MZsRpzJnvjao8SSUN7RuVw9J4pFt5K5Z/view?usp=sharing>

Conclusions

Using the static forecasting method, demand for years 3 and 4 is projected to stay stable at around 44,500 units per quarter. This forecast provides a baseline estimate that is useful for long-term capacity planning and supplier coordination. The static method does not account for seasonality or growth, both of which are present in the historical demand data. As a result, high-demand quarters may be underestimated and low-demand quarters may be overestimated.

Appendix

Revisions

Issue Identified	Changes Made	Team Members Responsible
Missing a meaningful executive summary that includes real results		Abinaya Ramesh
Your SC business model includes too much filler, and you need to more explicitly align your SC business model (section 2) with your SC stage and network representations (section 1).		Hailey McElroy
Incorrect Competitive Strategy and incomplete SC Strategy; you must clearly show: Competitive strategy → SC strategy → SC Structure.	Corrected competitive strategy to fit this project. Added more bridging text to clearly show the development from competitive strategy to SC strategy to SC structure.	Hailey McElroy
Too much filler within Demand Estimation (pages 18-23), and your demand estimation process needs to be mathematicized with symbols and equations.	Reduced explanatory filler, Revised estimation process to be more clearly mathematicized.	Vivian Nguyen Collin Yuen

Member Contributions Table

Name	Class 2/3	Meeting 2/8	Meeting 2/11	Class 2/5	Project Report Contributions
James Burch	☒	☒	☒	☒	- Added 1st section containing the final concept, FAST diagram, sub-systems, stage representation of the product's supply chain, and network representation of the product's supply chain - Redeveloped the overall supply chain strategy
Hailey McElroy	☒	☒	☒	☐	- Added explanatory text to connect each section in 1 & 2 - Redeveloped the Business Model - Removed the details I included explaining supply chain flows
Vivian Nguyen	☒	☒	☒	☒	Found similar products demand data to improve the demand estimation section
Garrett Arcand	☒	☒	☒	☒	- Adjusted workload distribution to make things more "fair" - Added real dates to time-phase project plan and improved formatting - Created post-meeting action table to manage assignments and explanations
Mohammed Abdul-Haadi Ansari	☒	☐	☐	☒	Forecasted demand data for distributed wireless system using various methods from TIM172B
Collin Yuen	☒	☐	☐	☐	Found similar products demand data to improve the demand estimation section
Abinaya Ramesh	☒	☒	☐	☒	Forecasted demand data for distributed wireless system using various methods from TIM172B

Post-meeting Action Table

Issue	Assigned Member	Timeline
• Create post-meeting action table • Issue assignments based on feedback	Garrett Arcand	☒ Complete 2/8-2/11
• Add project proposal to (1)	James Burch	☒ Complete 2/8-2/11
• Revised supply chain strategy (3)	James Burch	☒ Complete 2/8-2/11
• Business Model Improvements ◦ Implement competitive strategy map ◦ SC implementations should start with who should and shouldn't be included	Hailey McElroy	☒ Complete 2/8-2/11
• Improve flow of (1) → (2) → (3)	Garrett Arcand	☒ Complete 2/8-2/11
• Data Estimation Improvements ◦ Observe similar products' demand data ◦ Explain how price was found ◦ Check work by using sales data	Vivian Nguyen Collin Yuen	☒ Complete 2/8-2/11
• Forecasting Improvements ◦ Needs to match complexity of HWs ◦ Explain in detail	Mohammed Ansari Abinaya Ramesh	☒ Complete 2/8-2/11

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Group 12: Phase II

“Distributed Wireless” Home Security System

Vivian Nguyen, James Burch, Garrett Arcand, Collin Yuen, Mohammed Abdul-Haadi Ansari,
Abinaya Ramesh, Hailey McElroy

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Executive Summary

This phase outlines the strategic plan for Distributed Wireless to capture a 2% share of the global home security market. Our financial analysis establishes a competitive selling price of \$500 per unit against a production cost of \$270.37 per unit, which includes a \$243.97 base component sum plus a 25% buffer for auxiliary materials such as housings and packaging. To support this rollout, the project accounts for \$8.74M in development costs and \$4.37M in initial ramp-up expenses.

While initial estimates placed first-year demand at 174,800 units, our adaptive forecasting analysis identifies Winter's method as the most accurate model due to its ability to account for level, trend, and seasonality. According to this method, the projected demand for 2028 is 213,927 units, with a peak of 71,454 units occurring in the third quarter. This aligns with our research into U.S. property crime and housing trends, which show significantly higher activity between June and September.

To manage this volume, the supply chain focuses on aggressive cycle inventory for high-volume components. Duracell batteries and magnetic reed switches will be managed with frequent 2-day replenishment cycles, maintaining a lean average flow time of just one day in inventory. Specifically, the model dictates optimal lot sizes of 3,720 batteries and 4,624 switches per order. In contrast, the finished security systems, which carry the full \$270.37 unit cost, will be managed with larger lot sizes of 5,407 units on an 11-day cycle to optimize holding costs while ensuring the assembly line remains continuously fed.

Our initial financial model is promising. Our nominal (base case) profit based on a 5-year NPV analysis with a discount factor of 10% is \$1,730,000. While this profit is relatively low for a 5-year period, it indicates that our project will likely be profitable.

Objective

To design and manage the supply chain for Distributed Wireless in Honeywell.

1. Estimating Demand

Definition of Problem

The following section estimates the first 8 quarters of demand for the Distributed Wireless home security system. The estimating demand process aims to develop a data-driven numerical demand projection which aligns with the market size, expected market share, product lifecycle stages, seasonal demand fluctuations, and predicted consumer patterns. The approach will incorporate related product and industry statistics to inform seasonality/consumer behaviors to determine the estimated base demand. Grounding the estimate with existing market data offers a credible foundation for the subsequent forecasting and supply chain decisions.

Process Plan to Obtain Demand Data

1. Identify Related Products
 - a. Prices of competitor products
 - b. Similar Markets
2. Upper Limit Definition
3. Home Security Market Size
4. Determine revenue, sales volume, price
5. Product Life Cycle
 - a. Analysis
 - i. Introduction Phase
 - ii. Growth Phase
 - iii. Maturity Phase
 - b. Factor in products life cycle
6. Demand based on Influencers
 - i. Home Purchase Statistics
 - ii. Home/Property Crime Statistics
7. Seasonality
8. Quarterly Demand Estimates
 - a. Year One Adoption Estimates
 - b. Year Two Adoption Estimates
9. Visualization(s)

Execution

Variable	Definition	Value(s)
H	Total homes owned in the US	97,703,784
$D_n^{\max}$	Maximum possible annual demand	
P_H^1	Probability of homeowners adopting home security system (Year 1)	0.18%
P_H^2	Probability of homeowners adopting home security system (Year 2)	0.19%
M_0	Total home security system market value at the base year	\$4.37 billion

M_n	total home security system market value in year n	
g	Annual market growth rate	3.7%
s	Honeywell market share	2%
P	Average selling point	\$500
V_n	Estimated unit sales volume in year n	TBD
W_q	Seasonal Weight for quarter q	TBD

1. Identify Related Products

Let:

P = price range

a. Prices of competitor products

i. SimpliSafe 21-Piece Security System:

1. $P(\text{SimpliSafe}) = \$622.79 \text{ USD}$

ii. Vivint HomeProtect Pro:

1. $P(\text{Vivint}) = \$499 \text{ USD}$

iii. Ring Alarm Pro Security Kit:

1. $P(\text{Ring}) = \$595 \text{ USD}$

b. Determining a price for Distributed Wireless

- i. Appropriate product price is between \$500 to \$700 per unit for the Distributed Wireless according to the prices from competitors:
 $\Rightarrow P = \$500.$

c. Similar Markets

Let:

g = annual market growth rate

a. Wireless security cameras¹⁸

- i. The market for wireless security cameras is projected to grow at 20.79% from 2025 to 2035:

1. $g(\text{wireless security cameras}) = 20.79\%$
 $\Rightarrow g > 0$

b. Smart doorbells¹⁹

- i. The global smart doorbell market is projected to grow at a CAGR of 33.4% from 2023 to 2030:
 - 1. $g(\text{smart doorbells}) = 33.4\%$
 $\Rightarrow g > 0$
- c. Both markets for the related products to home security are statistically expecting a positive growth rate.

2. Upper Limit Definition

Let:

- H = Total homes owned in the US
- $D_n^{\max}$ = Maximum possible annual demand
- V_n = Estimated annual unit demand in year n

- a. Number of homeowners in the United States:^{21,22}
 - i. Homeownership rate in the United States = 65.7% (US Census, 2026)
 - ii. Recorded number of existing US home properties is 148,712,000. (US Census, 2026)

$$\begin{aligned}
 H &= (\text{US home properties}) * (\text{2025 US homeownership rate}) \\
 &= (148,712,000) * (65.7\%) \\
 H &= 97,703,784 \text{ homes}
 \end{aligned}$$

- b. Maximum Demand

$$D_n^{\max} = H$$

$$V_n \leq D_n^{\max}$$

- c. Interpretation:
 - i. H serves as the upper limit of potential adopters of the Distributed Wireless home security system.
 - ii. Annual unit demand can't exceed the total number of homeowners.

3. Home Security Market Size

Let:

- M_0 = total home security system market value in the base year (2025)
- M_n = total home security system market value in year n
- g = annual market growth rate

- a. 2025 Market Values ¹³

- i. The global home security system market is valued at 4.37 billion USD in 2025 and has a compound annual growth rate (CAGR) of 3.7%.

$$M_0 = \$4.37 \text{ billion USD}$$

$$g = 3.7\% \\ \Rightarrow 0.037$$

where $g > 0$

ii. $M_n = M_0 * (1+g)^n$

- b. Interpretation from values:

- i. The market will continue to increase over time due to the positive growth rate.

$$\Rightarrow M_n > M_{n-1}$$

4. Determine revenue, sales volume, price

Let:

H = Total homes owned in the US

P_H^n = Probability of homeowners adopting home security system

S = Honeywell market share

P = Average selling point

- a. Year 1 Revenue, Sales Volume, Price

1. Sales Volume

$$V_1 = H * P_H^1 * S = \\ 97,703,784 * 0.0018 * 0.0033 = 580 \text{ units}$$

2. Price

$$P = \$500$$

3. Revenue

$$R_1 = V_1 * P \\ 3517 * 500 = \$1758500 \\ \$1758500 \times .1 \times 1$$

- b. Year 2 Revenue, Sales Volume, Price

1. Sales Volume

$$V_1 = H * P_H^1 * S =$$

$$97,703,784 * 0.0019 * 0.02 = 3713 \text{ units}$$

2. Price

$$P = \$500$$

3. Revenue

$$R_1 = V_1 * P$$

$$3713 * 500 = \$1856500$$

c. Interpretation from values:

- i. The revenue and sales volume show positive growth and will continue to increase from Year 1 to Year 2 with the probability of adoption increasing each year that the product is in the market.

5. Product Life Cycle

a. Analysis

i. Introduction (Quarters 1-2)

1. The introduction phase will have limited adoption, focusing on showcasing the Distributed Wireless home security system in real home scenarios with testing trials and demonstrations.
2. Customer feedback during this phase will refine subsequent adjustments to the system.
3. Marketing during this phase will emphasize the variety of features and advantages.

ii. Growth (Quarters 3-6)

1. The growth phase is expected to have increased adoption as positive results drive demand upwards.
2. Endorsements of the Distributed Wireless by real homeowners will positively drive demand.

iii. Maturity (Quarters 7-8)

1. The maturity phase expects a stabilized level of demand as the Distributed Wireless home security system is expected to achieve broader market recognition.
2. Sales are driven by system upgrades, add-on component purchases, and new homeowners entering the market.

b. Factor in products life cycle

i.

Quarter	Phase	Weight (W_n)	Sales	Revenue
Q1	Introduction	10%	352	\$176,000
Q2	Introduction	15%	528	\$264,000
Q3	Growth	35%	1,231	\$615,500
Q4	Growth	40%	1,406	\$703,000
TOTAL		100%	3,517 units	\$1,758,000
Q5	Growth	30%	1,114	\$557,000
Q6	Growth	30%	1,114	\$557,000
Q7	Maturity	20%	743	\$371,500
Q8	Maturity	20%	742	\$371,500
TOTAL		100%	3713 units	\$1,856,500

6. Demand based on Influencers

Let:

a. Influencers

i. Home Purchase Statistics^{10,11,12}

1. Fig. 1 - Monthly total home sales in the United States (data collected and reported by the National Association of Realtors)¹¹.

- a. The peak of home sales data is evident between **May to August**, with the lowest occurring between January and February.

Year	Jan	Feb	Mar	Apr	May	June	July	Aug	Sept	Oct	Nov	Dec
2016	302K	314K	421K	470K	525K	582K	513K	539K	486K	445K	418K	437K
2017	319K	315K	455K	447K	555K	600K	513K	535K	462K	458K	425K	427K
2018	313K	319K	434K	460K	535K	570K	523K	539K	421K	446K	406K	377K
2019	283K	311K	400K	456K	542K	528K	540K	532K	450K	462K	404K	434K
2020	317K	335K	416K	373K	372K	507K	597K	560K	563K	573K	493K	538K
2021	366K	366K	484K	513K	528K	615K	584K	576K	546K	526K	503K	513K
2022	352K	352K	456K	463K	499K	525K	454K	474K	428K	371K	325K	327K
2023	231K	269K	359K	337K	408K	433K	372K	401K	348K	332K	300K	297K
2024	234K	271K	325K	360K	405K	376K	390K	379K	330K	348K	315K	329K
Average	299K	315K	416K	452K	499K	540K	509K	520K	447K	440K	396K	410K

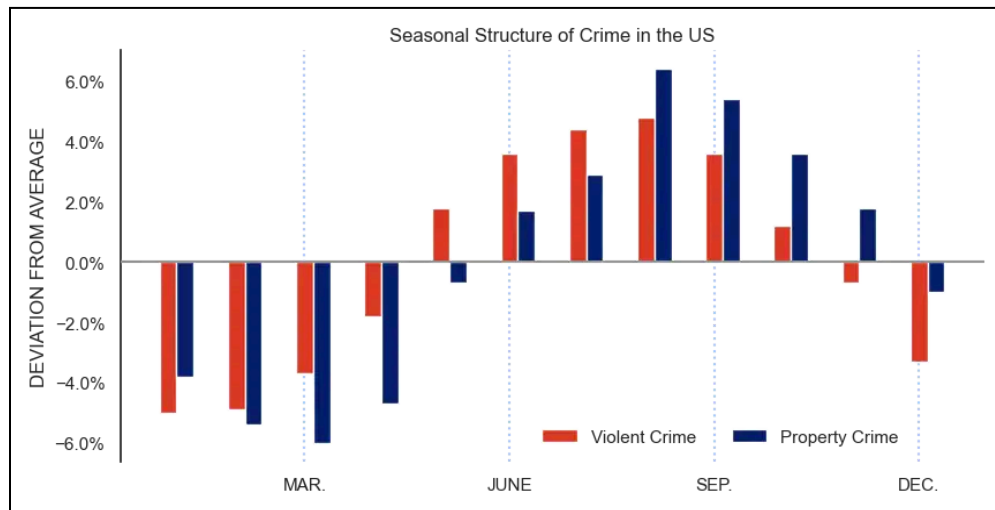
Table: Lance Lambert • Source: National Association of Realtors • Created with Datawrapper

ResiClub

ii. Home/Property Crime Statistics^{7,9}

1. Fig. 2 - Seasonal structure of crime in the U.S. (produced by the Pinkerton Crime Index data and federal agency sources)⁷

- a. Research regarding the seasonality of crime in the United States reveals property related crime is much more prevalent between **June to September**.



7. Seasonality

Let:

W_q = Seasonal Weight for quarter q

i. Seasonality Equation : $W_1 + W_2 + W_3 + W_4 = 1$

a. Where:

ii. W_1 = Winter Q1

1. Lowest demand due to lowest statistics in home property sales and property crime
 - iii. $W_2 = \text{Spring Q2}$
 1. Growing demand due to increase in home property sales and rise in property crime
 - iv. $W_3 = \text{Summer Q3}$
 1. Peak demand due to peak home property sales and property crime
 - v. $W_4 = \text{Fall Q4}$
 1. Moderate demand due to lower home property sales and property crime
- b. Assumption: Highest seasonal weights will be assigned to Q2 and Q3 due to expected peak demand through the seasons. In year 2, we will assume that due to consumer interest, demand isn't as concentrated during Q3.

Year	Quarter	Season	Weight (W_n)
1	Q1	Winter	0.10
	Q2	Spring	0.30
	Q3	Summer	0.35
	Q4	Fall	0.25
2	Q1	Winter	0.11
	Q2	Spring	0.30
	Q3	Summer	0.33
	Q4	Fall	0.26

- Total estimated demand equation:

$$\circ D_n = \text{annual demand} = V_n = \frac{M_0(1+g)^n(s)}{P}$$

■ Where:

- M_0 = Total home security system market value at the base year
- g = Annual market growth rate
- n = year
- s = Honeywell Distributed Wireless market share
Honeywell's Distributed Wireless market share 2%, approximately \$87,400,000

- P = average selling price
- $D_n * W_q$ = Quarterly Demand
 - Where:
 - D_n = demand of year n
 - W_q = seasonal weight for quarter

8. Quarterly Demand Estimates

a. Assumption:

1. In the first year, the demand would be lower and would gradually increase into year two as more consumers become aware of the product.
2. Home security system starting market size (2025) estimated to be \$4.37 billion¹³
3. Home security system market CAGR estimated to be 3.7%¹³
4. Honeywell's Distributed Wireless will have an average selling price of \$500.

$$Vn = \frac{M0(1+g)^n(s)}{P} = Demand$$

a. Year 1 (2026)

- i. Estimated Year 1 sales volume: \$87,400,000/\$500 = 174,800 units

$$V1 = \frac{\$4,370,000,000 (1.037^1)(0.02)}{\$500} = 174,800$$

b. Year 2 (2027)

- i. Estimated Year 2 sales volume: 174,800 units X (1.037) = approximately 181,300 units.

$$V2 = 174,800 (1.037)^1 = 181,268 \rightarrow 181,300$$

c. Quarterly demand estimates based on (7a and 7b)

Year	Quarter	Season	Weight (W_n)
1	Q1	Winter	0.10
	Q2	Spring	0.30
	Q3	Summer	0.35
	Q4	Fall	0.25
2	Q1	Winter	0.11
	Q2	Spring	0.30

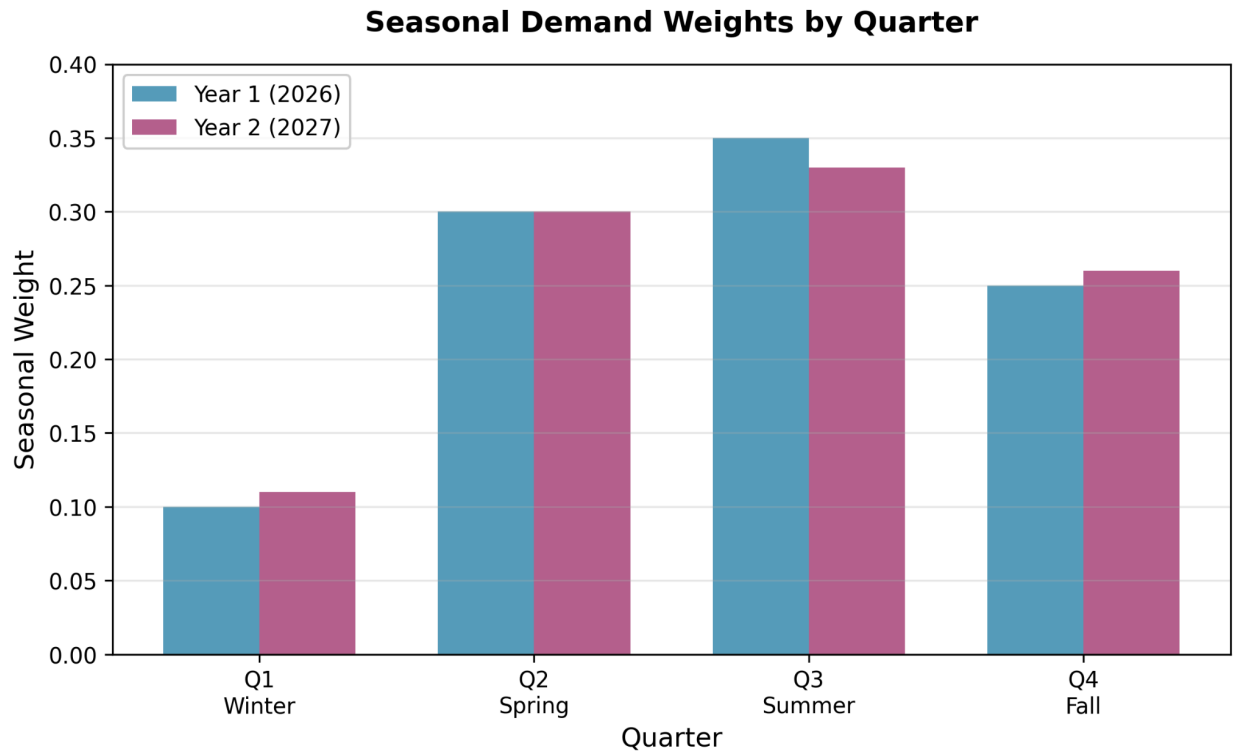
	Q3	Summer	0.33
	Q4	Fall	0.26

Year	Quarter	Period (t)	Demand (units)
2026	1	1	17,480
	2	2	52,440
	3	3	61,180
	4	4	43,700
TOTAL			174,800
2027	1	5	19,943
	2	6	54,390
	3	7	59,829
	4	8	47,138
TOTAL			181,300

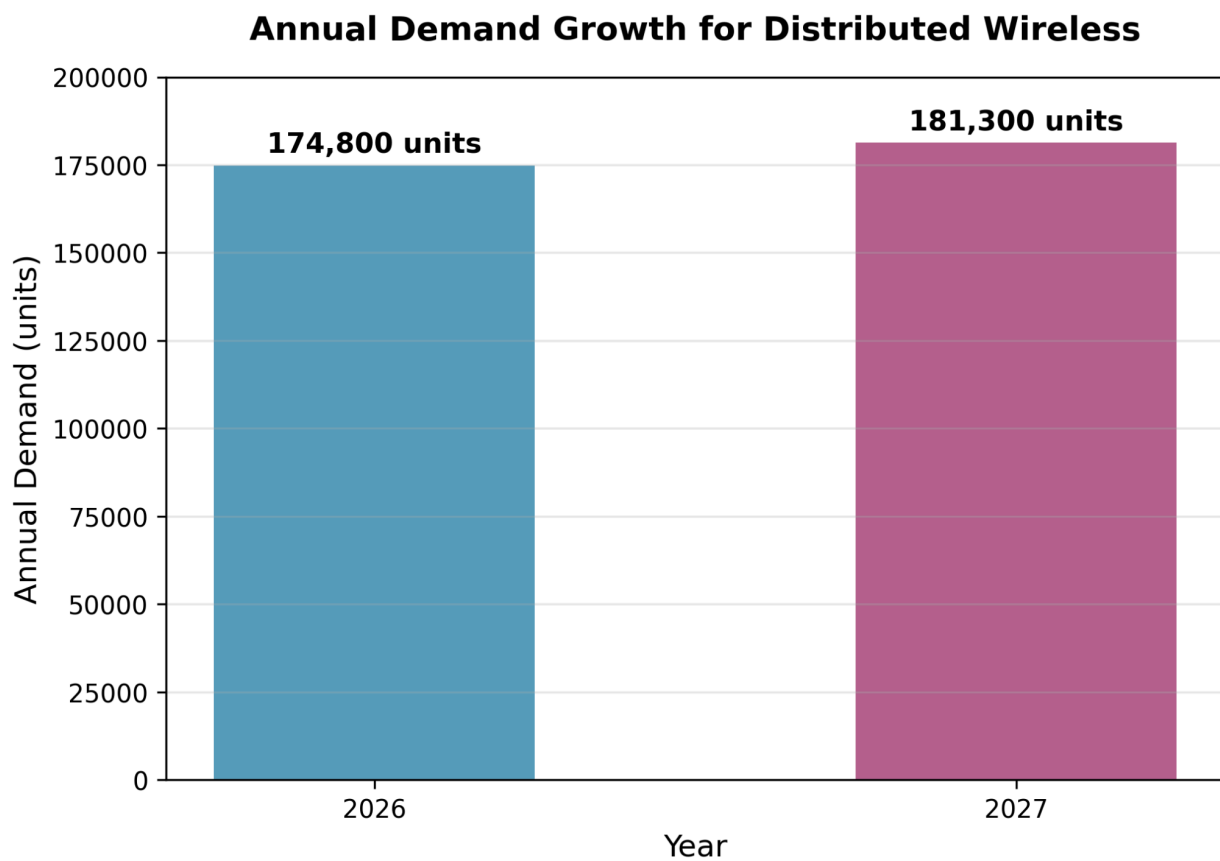
- i. Using the home security system starting market size in 2025 as a baseline, and a target of 2% of the market share for Honeywell in its first year entering the market, we built a sales demand and volume estimation for years one and two.

9. Visualizations

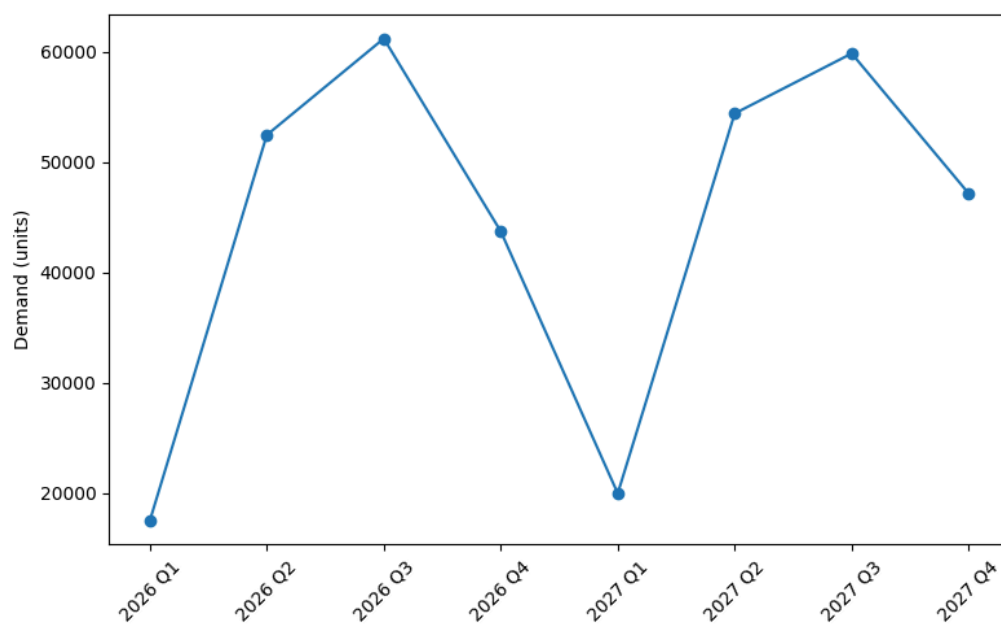
A. Seasonality Graph



B. Annual demand growth ($y_{sub 1} \rightarrow y_{sub 2}$)



C. Quarterly demand



2. Demand Forecasting

In order to plan every aspect of Honeywell's supply chain, we must determine the forecast of the demand of Distributed Wireless accurately. There are five forecasting methods that we will use: Static, Moving Average, Simple Exponential Smoothing, Holt's, and Winter's. Performing each of these methods allows us to then determine the best method to use by performing an error analysis based on the following error metrics and their formulas:

1. Forecast Error

$$E_i \triangleq F_i - D_i \quad (i = 0, 1, 2, 3, \dots)$$

2. Mean Square Error

$$(MSE)_n \triangleq 1/n \left[\sum_{i=1}^n (E_i)^2 \right]$$

3. Absolute Error (Deviation)

$$A_i \triangleq |E_i| \quad (i = 0, 1, 2, 3, \dots)$$

4. Mean Absolute Deviation

$$(MAD)_n \triangleq 1/n \left[\sum_{i=1}^n A_i \right]$$

5. Percent Error

$$(PE)_i \triangleq (E_i / D_i) * 100 \quad (i = 0, 1, 2, 3, \dots)$$

6. Absolute Percent Error

$$(APE)_i \triangleq (A_i / D_i) * 100 \quad (i = 0, 1, 2, 3, \dots)$$

7. Mean Absolute Percent Error

$$(MAPE)_n \triangleq 1/n \left[\sum_{i=1}^n (A_i / D_i) \right] * 100$$

8. Error Bias

$$[Bias]_n \triangleq \sum_{i=1}^n E_i$$

9. Tracking Signal

$$(TS)_n \triangleq (Bias_n / MAD_n)$$

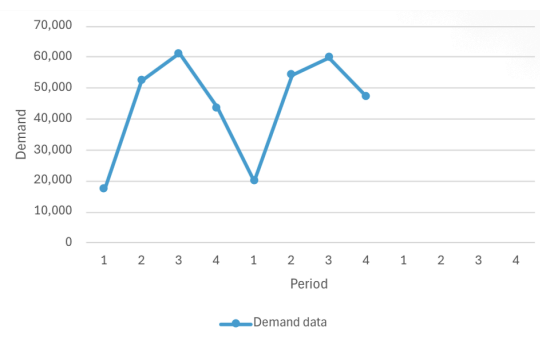
Actual Demand Data

Figure 1 lays out the given data in the 1st four columns of each method's spreadsheet while Figure 2 puts together the visual representation of the demand data for each quarter.

Figure 1

Year	Quarter	Period, t	Demand, D_t
		0	
2026	1	1	17,480
	2	2	52,440
	3	3	61,180
	4	4	43,700
2027	1	5	19,943
	2	6	54,390
	3	7	59,829
	4	8	47,138
2028	1	9	
	2	10	
	3	11	
	4	12	

Figure 2



Static Forecasting [Distributed Wireless Static Method .xlsx](#)

Static forecasting assumes that the level, trend, and seasonality of the demand data does not vary with time.

Level (L): It represents your theoretical starting point for demand at period zero before anything else influences it, $\bar{D}_i = L + iT$.

Trend (T): Long-term, underlying trajectory of your demand (is demand upward, shrinking, or staying completely flat?).

Seasonality (S_i): These are the predictable, repeating, short-term fluctuations in demand that happen at the same time every year (i.e., cycle). For example, home purchases and property crimes peak in the Summer (Q_3) and dip in the Winter (Q_1). Seasonality is the multiplier that accounts for these expected, calendar-driven spikes and dips.

Once again, we are to **assume that these values are non-changing when doing static forecasting**.

One might ask, if the aforementioned is the case, then why even bother with static forecasting?

- In Data Science, the “baseline” is our starting point – i.e., the most straightforward model you can build
- Adaptive models (discussed later in this document) are quite more complex than static forecasting – If your complex Winter's Method doesn't yield a lower Mean Absolute Deviation (MAD) or Mean Absolute Percent Error (MAPE) than your basic Static Method, then the complex model isn't worth the extra effort. You always need a simple baseline to prove your advanced models are actually doing a better job.
- Steps 1-5 is a repeat of what we will do later anyway

With this demand data we can take the following steps to perform the Static Method:

1. De-seasonalize the demand data given the periodicity of the data is 4 (quarterly)

Formula:

$$\bar{D}'_3 = \frac{1}{2}[(D_1 + D_5 + 2(D_2 + D_4))/8]$$

$$\bar{D}'_4 = \frac{1}{2}[(D_2 + D_6 + 2(D_3 + D_5))/8]$$

$$\bar{D}'_5 = \frac{1}{2}[(D_3 + D_7 + 2(D_4 + D_6))/8]$$

$$\bar{D}'_6 = \frac{1}{2}[(D_4 + D_8 + 2(D_5 + D_7))/8]$$

Using the formulas above we put together the following de-seasonalized demand data in the 5th column shown in Figure A.

Figure A

Year	Quarter	Period, t	Demand, D _t	Deseasonalized Demand
		0		
2026	1	1	17,480	
	2	2	52,440	
	3	3	61,180	44,008
	4	4	43,700	44,560
2027	1	5	19,943	44,634
	2	6	54,390	44,895
	3	7	59,829	
	4	8	47,138	
2028	1	9		
	2	10		
	3	11		
	4	12		

2. Regress the de-seasonalized demand data

Formula:

$$\bar{D}_i = L + iT \quad (i = 0, 1, 2, \dots, t)$$

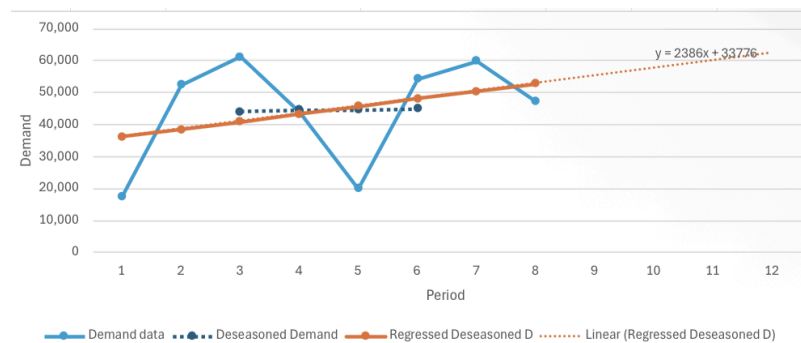
L = Level(y-intercept)
 T = Trend(slope)

Using the formula above the regressed deseasonalized data was then placed in column 6 in Figure B and visualized with Figure C as follows.

Figure B

Year	Quarter	Period, t	Demand, D_t	Deseasonalized Demand	Regressed Deseasonalized Demand
2026	1	1	17,480		36,162
	2	2	52,440		38,548
	3	3	61,180	44,008	40,934
	4	4	43,700	44,560	43,320
2027	1	5	19,943	44,634	45,706
	2	6	54,390	44,895	48,092
	3	7	59,829		50,478
	4	8	47,138		52,864

Figure C



3. Compute the seasonal factors of the demand data

Formula:

$$S_i = D_i / \bar{D}_i \quad (i = 1, 2, \dots, p)$$

Figure D

Year	Quarter	Period, t	Demand, D_t	Deseasonalized Demand	Regressed Deseasonalized Demand	Seasonal Factor
2026	1	1	17,480		36,162	0.48
	2	2	52,440		38,548	1.36
	3	3	61,180	44,008	40,934	1.49
	4	4	43,700	44,560	43,320	1.01
2027	1	5	19,943	44,634	45,706	0.44
	2	6	54,390	44,895	48,092	1.13
	3	7	59,829		50,478	1.19
	4	8	47,138		52,864	0.89

4. Compute the average seasonal factors for the demand data

As we have two years of actual demand data to use we have 2 cycles of data available where the periodicity p is equal to 4.

Formula:

$$\overline{S_1} = (S_1 + S_5) / 2$$

$$\overline{S_2} = (S_2 + S_6) / 2$$

$$\overline{S_3} = (S_3 + S_7) / 2$$

$$\overline{S_4} = (S_4 + S_8) / 2$$

Figure E

Year	Quarter	Period, t	Demand, D _t	Regressed			
				Deseasonalized Demand	Deseasonalized Demand	Seasonal Factor	Averaged Seasonal Factor
2026	1	1	17,480		36,162	0.48	0.46
	2	2	52,440		38,548	1.36	1.25
	3	3	61,180	44,008	40,934	1.49	1.34
	4	4	43,700	44,560	43,320	1.01	0.95
2027	1	5	19,943	44,634	45,706	0.44	
	2	6	54,390	44,895	48,092	1.13	
	3	7	59,829		50,478	1.19	
	4	8	47,138		52,864	0.89	

5. Re-seasonalize the regressed de-seasonalized demand data

By re-seasonalizing this data we obtain the forecast of the demand shown in Figure F and visualized as a forecast for 2028 & 2029 as shown in Figure G

Formula:

$$F_{t+l} = (\overline{S_{t+l}})(\overline{D_{t+l}})$$

$$F_{t+l} = (\overline{S_{t+l}})[L + (t+l) * T]$$

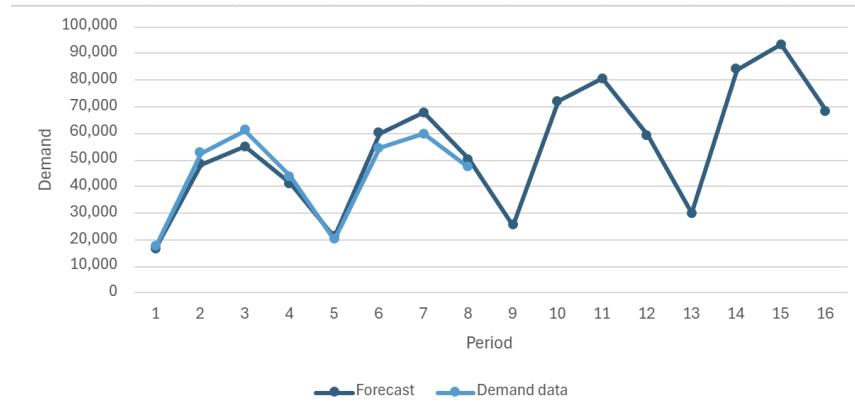
$$(i = t+l)$$

$$(l = 0, 1, 2, \dots)$$

Figure F

Year	Quarter	Period, t	Demand, D _t	Deseasonalized Demand	Regressed			
					Deseasonalized Demand	Seasonal Factor	Averaged Seasonal Factor	Forecast
2026	1	1	17,480		36,162	0.48	0.46	16,629
	2	2	52,440		38,548	1.36	1.25	48,018
	3	3	61,180	44,008	40,934	1.49	1.34	54,848
	4	4	43,700	44,560	43,320	1.01	0.95	41,164
2027	1	5	19,943	44,634	45,706	0.44		21,018
	2	6	54,390	44,895	48,092	1.13		59,907
	3	7	59,829		50,478	1.19		67,637
	4	8	47,138		52,864	0.89		50,233
2028	1	9						25,407
	2	10						71,795
	3	11						80,425
	4	12						59,302
2029	1	13						29,796
	2	14						83,684
	3	15						93,213
	4	16						68,371

Figure G



Having put together the Static Method to forecast demand we can use error analysis to quantify how reliable and good of a forecast it is. After completing the error metric steps Figure H displays the complete Static Method with error analysis.

Figure H

Year	Quarter	Period, t	Demand, D _t	Deseasonalized Demand	Regressed Deseasonalized Demand	Seasonal Factor	Averaged Seasonal Factor	Forecast	Forecast Error	MSE	AE (Deviation)	MAD	PE	APE	MAPE	Bias	TS
2026	1	1	17,480			36,162	0.48	16,629	-851	723651.8809	850.6773071	31634.76	4.866575	4.866575	8.075158	3,354	0.11
	2	2	52,440			38,548	1.36	48,018	-4,422	19553450.66	4421.928387	15392.04	8.432358	8.432358	8.533527	4,205	0.27
	3	3	61,180	44,008		40,934	1.49	54,848	-6,332	40087997.61	6331.50832	8787.385	10.34898	10.34898	8.550388	8,627	0.98
	4	4	43,700	44,560		43,320	1.01	41,164	-2,536	6431892.404	2536.117585	5007.662	5.803473	5.803473	8.190669	14,958	2.99
2027	1	5	19,943	44,634		45,706	0.44	21,018	1,075	1156035.69	1075.191002	3498.906	5.39132	5.39132	8.787468	17,495	5.00
	2	6	54,390	44,895		48,092	1.13	59,907	5,517	30434444.98	5516.742243	2736.556	10.14293	10.14293	9.919517	16,419	6.00
	3	7	59,829			50,478	1.19	67,637	7,808	60960745.86	7807.736283	1557.514	13.05009	13.05009	9.807808	10,903	7.00
	4	8	47,138			52,864	0.89	50,233	3,095	9578156.248	3094.859649	3094.86	6.56553	6.56553	6.56553	3,095	1.00
2028	1	9						25,407									
	2	10						71,795									
	3	11						80,425									
	4	12						59,302									
2029	1	13						29,796									
	2	14						83,684									
	3	15						93,213									
	4	16						68,371									

Adaptive Forecasting

Adaptive (or dynamic) forecasting assumes that the level, trend, and seasonality of the demand data varies with time; which in turn, needs updated estimates as new demand data is made available. While the static method provides a solid baseline, real-world demand is rarely perfectly constant. Therefore, we transition to Adaptive (or dynamic) forecasting, which assumes that the level, trend, and seasonality of the demand data *do* vary with time and require continuous updating as new data becomes available.

To achieve this ‘dynamic updating’, these models utilize smoothing values to weigh recent data more heavily than past data. They are α , β , & γ ... and their specificities are as follows:

Parameter	Symbol	Purpose	Range
Alpha	α	Smoothing constant for Level	$0 \leq \alpha \leq 10$

Beta	β	Smoothing constant for Trend	$0 \leq \beta \leq 10$
Gamma	γ	Smoothing constant for Seasonality	$0 \leq \gamma \leq 10$

There are four methods of *adaptive forecasting* which are explored: Moving Average, Simple Exponential Smoothing, Holt's Method, and Winter's Method.

Moving Average [Distributed Wireless Moving Average.xlsx](#)

This method assumes that the demand data only had level, L , ignoring trend and seasonality. With this assumption the key idea is that the level calculated for any time will provide the best forecast of demand.

With the same demand data laid out for the Static Method, we can take the following steps to perform the Moving Average Method:

1. Calculate level

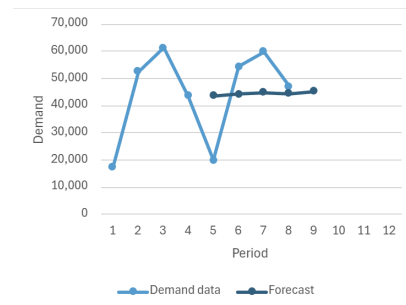
Formula:

$$L_i \triangleq \frac{D_i + D_{i-1} + D_{i-2} + D_{i-3}}{4}, (i = 4, 5, 6, 7, 8)$$

2. Forecast demand

$$F_i = L_{i-1}$$

Figure I



Having put together the Moving Average Method to forecast demand we can use error analysis to quantify how reliable and good of a forecast it is. After completing the error metric steps, Figure J displays the complete Moving Average Method with error analysis.

Figure J

Year	Week	Period, t	Demand, Dt	Deseasonalized Demand	Regressed Deseasonalized Demand	Seasonal Factor	Level Lt	Forecast Ft	Error Et	Mean Squared Error MSET	Absolute Error AET (Deviation)	Mean Absolute Deviation MADt	Percent Error Pet	Abs. Percent Errors APET	Mean Abs. Percent Error MAPET	Bias	Tracking Signal TSt
2026	1	1	17,480		36,162	0.48											
	2	2	52,440		38,548	1.36											
	3	3	61,180	44,008	40,934	1.49											
	4	4	43,700	44,560	43,320	1.01	43,700										
	5	5	19,943	44,634	45,706	0.44	44,316	43,700	23,757	564,395,049	23,757	23,757	119	119	119	23,757	1.00
2027	2	6	54,390	44,895	48,092	1.13	44,803	44,316	-10,074	332,942,781	10,074	16,916	-19	19	69	13,683	0.81
	3	7	59,829		50,478	1.19	44,466	44,803	-15,026	297,219,575	15,026	16,286	-25	25	54	-1,343	-0.08
	4	8	47,138		52,864	0.89	45,325	44,466	-2,673	224,700,245	2,673	12,882	-6	6	42	-4,016	-0.31
	1	9						45,325									
	2	10															
2028	3	11															
	4	12															

Moving Average is the "rearview mirror" approach. It assumes that recent history is the only thing that matters, smoothing out all the random noise to give you a trailing baseline.

- **What it means:** "To guess what we will sell next quarter, let's just look at the average of what we sold over the last year (the last 4 quarters)."
- **The massive blind spot:** As your report correctly notes, this method completely ignores Trend and Seasonality.

Why it yielded a massive 71% Error Rate

If you look at your Method Comparisons table on Page 23, the Moving Average resulted in an average MAPE of ~71% and a MAD of ~17,460. That means your forecasts were missing the mark by an average of 17,460 units every single quarter!

- Because Distributed Wireless system has a strong upward growth trend (sales are increasing over time), a Moving Average will *always* forecast too low. It anchors your future prediction to older, lower sales numbers. It is perpetually playing catch-up.
- Your product has a massive seasonal spike in Q3 due to peak home sales and property crime. The Moving Average completely flattens that spike out. When Q3 hits, your Moving Average forecast will severely under-predict demand, leading to massive stockouts and lost revenue.

Simple Exponential Smoothing [Distributed Wireless Simple Exponential Smoothing.xlsx](#)

This method assumes that the demand data only had level, L, ignoring trend and seasonality. With this assumption the key idea is that by using a smoothing constant, α , to smooth the level this method will best forecast demand.

With the same demand data laid out for the Static Method, we can take the following steps to perform the Simple Exponential Smoothing Method:

1. Initialize the Level
Formula:

$$L_0 = \frac{1}{n} \sum_{i=1}^n D_i$$

Using the formula above to average the demand data we get an initial level of 44,513.

Essentially, if we strip away all the summer sales spikes, the winter slumps, and our overall market growth, the Distributed Wireless system naturally drives a baseline of about 44,513 unit sales per quarter. From an SCM perspective, **this is the baseline volume the manufacturing plants/suppliers need to be capable to producing** (on a quarterly basis – so around 500 or so units/day incl. weekends and near 700 not incl. weekends).

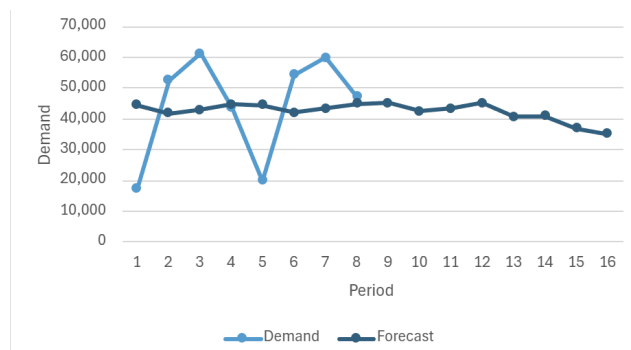
2. Determine the Levels for every period of data

Now that we have the initial level we can calculate the levels for the rest of the demand data using

Formula:

$$L_i \triangleq \alpha * D_i + (1 - \alpha) * [L_{i-1}], (i = 1, 2, 3, \dots)$$

Figure K



Actual demand (the light blue) is highly volatile—i.e., it spikes in the summer and drops in the winter while trending upward. Forecast (dark blue) is practically a flat horizontal line. *Why?* Because Simple Exponential Smoothing assumes that business has zero trend and zero seasonality. It is mathematically blind to your summer spikes.

Having put together the Simple Exponential Smoothing Method to forecast demand we can use error analysis to quantify how reliable and good of a forecast it is. After completing the error metric steps Figure L displays the complete Simple Exponential Smoothing Method with error analysis.

Figure L

Year	Quarter	Period, t	Demand, D _t	Deseasonalized Demand	Regressed Deseasonalized Demand	Seasonal Factor	Level Lt	Forecast Ft	Error Et	Mean Squared Error MSEt	Absolute Error AEt (Deviation)	Mean Absolute Deviation MADt	Percent Error Pet	Abs. Percent		Bias	Tracking Signal Tst	
														Erros APEt	Mean Abs. Percent Error MAPEt			
2026		0					44,513											
		1	1	17,480		36,162	0.48	41,809	44,513	27,033	843,768,902	27,033	37,663	155	155	87	16,402	0.44
		2	2	52,440		38,548	1.36	42,872	41,809	-10,631	56,506,423	10,631	5,315	-20	20	20	-10,631	-2.00
		3	3	61,180	44,008	40,934	1.49	44,703	42,872	-18,308	149,394,603	18,308	9,646	-30	30	25	-28,938	-3.00
		4	4	43,700	44,560	43,320	1.01	44,603	44,703	1,003	112,297,501	1,003	7,485	2	2	17	-27,935	-3.73
		1	5	19,943	44,634	45,706	0.44	42,137	44,603	24,660	211,458,983	24,660	10,920	124	124	44	-3,276	-0.30
		2	6	54,390	44,895	48,092	1.13	43,362	42,137	-12,253	201,239,284	12,253	11,142	-23	23	40	-15,529	-1.39
		3	7	59,829		50,478	1.19	45,009	43,362	-16,467	211,227,671	16,467	11,903	-28	28	38	-31,996	-2.69
2027		4	8	47,138		52,864	0.89	45,222	45,009	-2,129	185,390,892	2,129	10,681	-5	5	33	-34,125	-3.19
2028		1	9			55,250		42,448	45,222									
		2	10			57,636		43,447	42,448									
		3	11			60,022		45,220	43,447									
		4	12			62,408		40,698	45,220									
		1	13			64,794		40,998	40,698									
		2	14			67,180		36,898	40,998									
		3	15			69,566		35,203	36,898									
	2029		4	16		71,952		37,122	35,203									

The Error Table above (i.e., Figure L) proves it is chronically under-forecasting: Look at "Bias" column in. By Period 8, Bias is at -34,125. A negative bias means forecast is consistently lower than actual demand. Because this method ignores your product's underlying growth trend, it is constantly lagging behind reality.

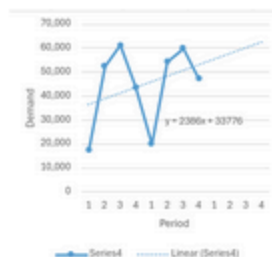
While Simple Exponential Smoothing is a slight step up from a basic Moving Average, because it reacts faster to recent data via our smoothing constant,

With the same demand data laid out for the Static Method, we can take the following steps to perform Holt's Method:

1. Determine the initial values of level and trend

To determine the initial values of level and trend we graphed the demand data and added a trend line that gave us an initial level of 33,776 and trend of 2,386 as shown in Figure M.

Figure M



2. Adapt using two smoothing constants α & β for level and trend respectively

Formula:

$$L_i \triangleq \alpha D_i + (1 - \alpha)[L_{i-1} + T_{i-1}] (i = 1, 2, 3, \dots)$$

$$T_i \triangleq \beta (L_i - L_{i-1}) + (1 - \beta)[T_{i-1}] (i = 1, 2, 3, \dots)$$

Figure N

27

Year	Quarter	Period, t	Demand, D _t	Level L _t	Trend T _t
		0		33776	2386
	1	1	17,480	34,294	2,012
	2	2	52,440	37,920	2,335
	3	3	61,180	42,347	2,754
2026	4	4	43,700	44,961	2,726
	1	5	19,943	44,912	2,171
	2	6	54,390	47,813	2,317
	3	7	59,829	51,100	2,511
2027	4	8	47,138	52,963	2,381
	1	9		49,810	1,274
	2	10		45,978	253
	3	11		41,606	-672
2028	4	12		36,841	-1,491
	1	13		31,815	-2,168

is still

inadequate for Distributed Wireless. The model's inability to account for our aggressive market

growth and summer seasonality resulted in an average MAPE of ~38%, meaning it would still cause severe stockouts during peak seasons.

Holt's Method [Distributed Wireless Holt's Model.xlsx](#)

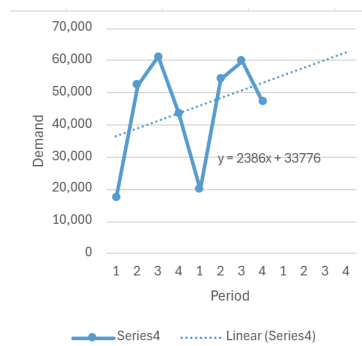
Holt's Method, also referred to as Level & Trend Exponential Smoothing, assumes that level and trend are given by the demand data and seasonality is ignored.

With the same demand data laid out for the Static Method , we can take the following steps to perform Holt's Method:

1. Determine the initial values of level and trend

To determine the initial values of level and trend we graphed the demand data and added a trend line that gave us an initial level of 33,776 and trend of 2,386 as shown in Figure M.

Figure M



2. Adapt using two smoothing constants α & β for level and trend respectively

Formula:

$$L_i \triangleq \alpha D_i + (1 - \alpha)[L_{i-1} + T_{i-1}](i = 1, 2, 3, \dots)$$

$$T_i \triangleq \beta(L_i - L_{i-1}) + (1 - \beta)[T_{i-1}](i = 1, 2, 3, \dots)$$

Figure N

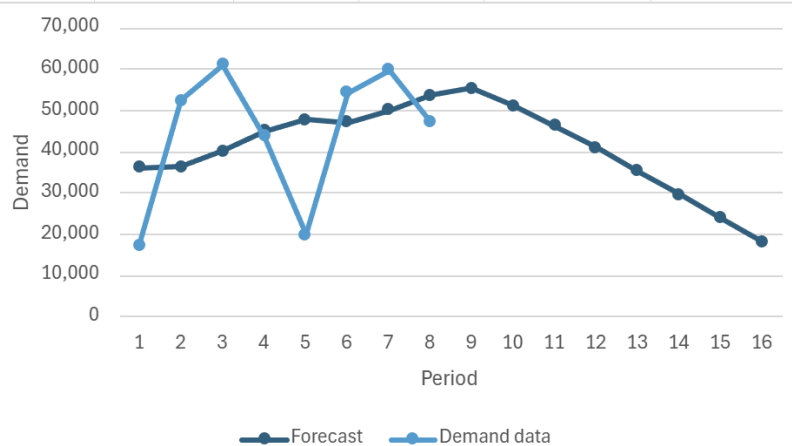
Year	Quarter	Period, t	Demand, D _t	Level L _t	Trend T _t
2026		0		33,776	2,386
	1	1	17,480	34,294	2,012
	2	2	52,440	37,920	2,335
	3	3	61,180	42,347	2,754
2027	4	4	43,700	44,961	2,726
	1	5	19,943	44,912	2,171
	2	6	54,390	47,813	2,317
	3	7	59,829	51,100	2,511
2028	4	8	47,138	52,963	2,381
	1	9		49,810	1,274
	2	10		45,976	253
	3	11		41,606	-672
2029	4	12		36,841	-1,491
	1	13		31,815	-2,198
	2	14		26,656	-2,790
	3	15		21,480	-3,267
	4	16		16,391	-3,631

3. Forecast the data points

Formula:

$$F_i \triangleq (L_{i-1} + T_{i-1}), \quad (i = 0, 1, 2, 3, \dots)$$

Figure O



Having put together the Holt's Method to forecast demand we can use error analysis to quantify how reliable and good of a forecast it is. After completing the error metric steps Figure P displays the complete Holt's Method with error analysis.

Figure P

Year	Quarter	Period, t	Demand, D _t	Level L _t	Trend T _t	Forecast F _t	Error E _t	Mean Squared Error MSE _t	Absolute Error AEt (Deviation)	Mean Absolute Deviation MAD _t	Percent Error PE _t	Abs. Percent Error APET	Mean Abs. Percent Error MAPET	Bias	Tracking Signal TSt
		0		33776	2386										
	1	1	17,480	34,294	2,012	36,162	18,682	609,317,917	18,682	34,816	107	107	68.82	2,548	0.07
	2	2	52,440	37,920	2,335	36,306	-16,134	130,150,397	16,134	8,067	-31	31	30.77	-16,134	-2.00
	3	3	61,180	42,347	2,754	40,255	-20,925	232,724,654	20,925	12,353	-34	34	32.48	-37,059	-3.00
2026	4	4	43,700	44,961	2,726	45,101	1,401	175,033,958	1,401	9,615	3	3	22.72	-35,659	-3.71
	1	5	19,943	44,912	2,171	47,686	27,743	293,963,452	27,743	13,241	139	139	51.82	-7,915	-0.60
	2	6	54,390	47,813	2,317	47,082	-7,308	253,869,496	7,308	12,252	-13	13	44.14	-15,223	-1.24
	3	7	59,829	51,100	2,511	50,130	-9,699	231,040,918	9,699	11,887	-16	16	39.49	-24,922	-2.10
2027	4	8	47,138	52,963	2,381	53,611	6,473	207,397,868	6,473	11,210	14	14	35.81	-18,449	-1.65
	1	9		49,810	1,274	55,345									
	2	10		45,976	253	51,085									
	3	11		41,606	-672	46,229									
2028	4	12		36,841	-1,491	40,934									
	1	13		31,815	-2,198	35,350									
	2	14		26,656	-2,790	29,618									
	3	15		21,480	-3,267	23,866									
2029	4	16		16,391	-3,631	18,212									

Holt's Method is great because it finally recognizes that your business is growing over time (that's the Trend part). But because it ignores seasonality, it draws a straight line right through the middle of the year, completely missing the reality that the customers buy way more security systems in the summer (Q3) and way fewer in the winter (Q1).

Winter's Method [Distributed Wireless Winter's Model .xlsx](#)

Winter's Method (also referred to as Level, Trend, & Seasonality Exponential Smoothing) assumes that level, trend, and seasonality are *given by the demand data*. With this in mind, this method uses the same initial steps of the Static Method's Forecasting however uses smoothing constants for the data.

With the same demand data laid out for the Static Method, we can take the following steps to perform Winter's Method:

1. De-seasonalize the demand data given the periodicity of the data is 4 (quarterly)

Formula:

$$\begin{aligned}\bar{D}_3 &= \frac{1}{2}[(D_1 + D_5 + 2(D_2 + D_4))/8] \\ \bar{D}_4 &= \frac{1}{2}[(D_2 + D_6 + 2(D_3 + D_5))/8] \\ \bar{D}_5 &= \frac{1}{2}[(D_3 + D_7 + 2(D_4 + D_6))/8] \\ \bar{D}_6 &= \frac{1}{2}[(D_4 + D_8 + 2(D_5 + D_7))/8]\end{aligned}$$

2. Regress the de-seasonalized demand data

Formula:

$$\begin{aligned}\bar{D}_i &= L + iT \quad (i = 0, 1, 2, \dots, t) \\ L &= \text{Level (y-intercept)} \\ T &= \text{Trend (slope)}\end{aligned}$$

3. Compute the average seasonal factors for the demand data

As we have two years of actual demand data to use we have 2 cycles of data available where the periodicity p is equal to 4.

Formula:

$$\begin{aligned}\overline{S_1} &= (S_1 + S_5) / 2 \\ \overline{S_2} &= (S_2 + S_6) / 2 \\ \overline{S_3} &= (S_3 + S_7) / 2 \\ \overline{S_4} &= (S_4 + S_8) / 2\end{aligned}$$

4. Re-seasonalize the regressed de-seasonalized demand data

By re-seasonalizing this data we obtain the forecast of the demand shown in Figure 1F and visualized as a forecast for 2028 & 2029 as shown in Figure 1G

$$\begin{aligned}F_{t+l} &= (\overline{S}_{t+l})(\overline{D}_{t+l}) \\ F_{t+l} &= (\overline{S}_{t+l})[L + (t+l) * T] \\ (i &= t+l) \\ (l &= 0, 1, 2, \dots)\end{aligned}$$

Figure Q puts together the demand information from the four steps laid out above.

Figure Q

Year	Quarter	Period, t	Demand, D _t	Deseasonalized Demand	Regressed Deseasonalized Demand
		0			
2026	1	1	17,480		36,162
	2	2	52,440		38,548
	3	3	61,180	44,008	40,934
	4	4	43,700	44,560	43,320
2027	1	5	19,943	44,634	45,706
	2	6	54,390	44,895	48,092
	3	7	59,829		50,478
	4	8	47,138		52,864
2028	1	9			55,250
	2	10			57,636
	3	11			60,022
	4	12			62,408
2029	1	13			64,794
	2	14			67,180
	3	15			69,566
	4	16			71,952

5. Compute the initial values of level and trend, and the averaged seasonal factors.

The initial values for the level, trend, and seasonality can be taken from the same trend line laid out in Figure M and put together in Figure R.

Figure R

Year	Quarter	Period, t	Demand, D _t	Deseasonalized Demand	Regressed Deseasonalized Demand	Seasonal Factor	Seasonality S _t	Level L _t	Trend T _t
		0						33776	2386
2026	1	1	17,480		36,162	0.48	0.46		
	2	2	52,440		38,548	1.36	1.25		
	3	3	61,180	44,008	40,934	1.49	1.34		
	4	4	43,700	44,560	43,320	1.01	0.95		
2027	1	5	19,943	44,634	45,706	0.44			
	2	6	54,390	44,895	48,092	1.13			
	3	7	59,829		50,478	1.19			
	4	8	47,138		52,864	0.89			
2028	1	9							
	2	10							
	3	11							
	4	12							
2029	1	13							
	2	14							
	3	15							
	4	16							

6. Adapt using the smoothing constants for level, trend, and seasonality respectively

Formula:

$$L_i \triangleq \alpha (D_i / S_{i-p}) + (1 - \alpha)[L_{i-1} + T_{i-1}], (i = 1, 2, 3, \dots)$$

$$T_i \triangleq \beta(L_i - L_{i-1}) + (1 - \beta)[T_{i-1}], (i = 1, 2, 3, \dots)$$

Note about eqn for L_i : the demand data must need to be divided by deseasonalized data

To adapt for seasonality use the following:

$$S_i \triangleq \gamma * (D_i / L_i) + (1 - \gamma) * S_{i-4}, (i = 5, 6, 7, \dots)$$

Figure S

Year	Quarter	Period, t	Demand, D _t	Deseasonalized Demand	Regressed Deseasonalized Demand	Seasonal Factor	Seasonality S _t	Level L _t	Trend T _t
		0						33,776	2,386
2026	1	1	17,480		36,162	0.48	0.46	36,254	2,395
	2	2	52,440		38,548	1.36	1.25	38,822	2,412
	3	3	61,180	44,008	40,934	1.49	1.34	41,456	2,435
	4	4	43,700	44,560	43,320	1.01	0.95	43,995	2,445
2027	1	5	19,943	44,634	45,706	0.44	0.46	46,276	2,429
	2	6	54,390	44,895	48,092	1.13	1.26	48,435	2,402
	3	7	59,829		50,478	1.19	1.35	50,505	2,368
	4	8	47,138		52,864	0.89	0.95	52,699	2,351
2028	1	9			55,250		0.46	52,297	2,076
	2	10			57,636		1.24	51,654	1,804
	3	11			60,022		1.34	50,785	1,537
	4	12			62,408		0.95	49,706	1,275
2029	1	13			64,794		0.41	48,432	1,020
	2	14			67,180		1.12	46,979	773
	3	15			69,566		1.20	45,365	534
	4	16			71,952		0.85	43,604	305

7. Forecast demand

Using the seasonality, level, and trends from step 6, laid out in Figure S, the following formula computes the forecasts of the demand data laid out in Figure T and visualized in Figure U.

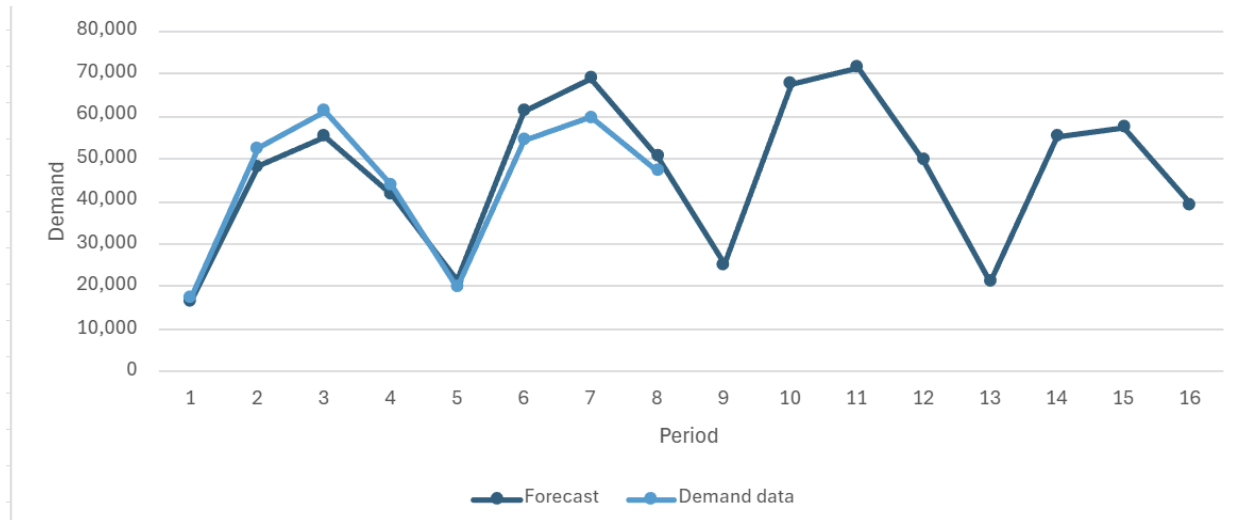
Formula:

$$F_i \triangleq (L_{i-1} + T_{i-1}) * S_i \quad (i = 0, 1, 2, 3, \dots)$$

Figure T

Year	Quarter	Period, t	Demand, D _t	Deseasonalized	Regressed Demand	Seasonal Factor	Seasonality S _t	Level L _t	Trend T _t
		0						33,776	2,386
2026	1	1	17,480		36,162	0.48	0.46	36,254	2,395
	2	2	52,440		38,548	1.36	1.25	38,822	2,412
	3	3	61,180	44,008	40,934	1.49	1.34	41,456	2,435
	4	4	43,700	44,560	43,320	1.01	0.95	43,995	2,445
2027	1	5	19,943	44,634	45,706	0.44	0.46	46,276	2,429
	2	6	54,390	44,895	48,092	1.13	1.26	48,435	2,402
	3	7	59,829		50,478	1.19	1.35	50,505	2,368
	4	8	47,138		52,864	0.89	0.95	52,699	2,351
2028	1	9			55,250		0.46	52,297	2,076
	2	10			57,636		1.24	51,654	1,804
	3	11			60,022		1.34	50,785	1,537
	4	12			62,408		0.95	49,706	1,275
2029	1	13			64,794		0.41	48,432	1,020
	2	14			67,180		1.12	46,979	773
	3	15			69,566		1.20	45,365	534
	4	16			71,952		0.85	43,604	305

Figure U



Having put together the Winter's Method to forecast demand we can use error analysis to quantify how reliable and good of a forecast it is. After the error metric steps and their associated formulas listed below Figure V displays the complete Winter's Method with error analysis.

Figure V

Year	Quarter	Period, t	Demand, Dt	Regressed			Seasonality St	Level Lt	Trend Tt	Forecast Ft	Error Et	Mean Squared Error MSEt	Absolute Error AEt (Deviation)	Mean Absolute Deviation MADt	Percent Error PET	Abs. Percent Error APET	Mean Abs. Percent Error MAPEt	Bias	Tracking Signal TS
				Deseasonalized Demand	Deseasonalized Demand	Seasonal Factor													
		0						33,776	2,386										
2026	1	1	17,480		36,162	0.48	0.46	36,254	2,395	16,629	-851	723,652	851	851	-5	5	4.87	-851	-1.00
	2	2	52,440		38,548	1.36	1.25	38,822	2,412	48,145	-4,295	9,586,154	4,295	2,573	-8	8	6.53	-5,146	-2.00
	3	3	61,180	44,008	40,934	1.49	1.34	41,456	2,435	55,251	-5,929	18,107,169	5,929	3,692	-10	10	7.58	-11,075	-3.00
	4	4	43,700	44,560	43,320	1.01	0.95	43,995	2,445	41,706	-1,994	14,574,415	1,994	3,267	-5	5	6.83	-13,069	-4.00
	1	5	19,943	44,634	45,706	0.44	0.46	46,276	2,429	21,459	1,516	12,119,475	1,516	2,917	8	8	6.98	-11,552	-3.96
	2	6	54,390	44,895	48,092	1.13	1.26	48,435	2,402	61,182	6,792	17,789,017	6,792	3,563	12	12	7.90	-4,760	-1.34
	3	7	59,829		50,478	1.19	1.35	50,505	2,368	68,808	8,979	26,764,345	8,979	4,337	15	15	8.92	4,219	0.97
	4	8	47,138		52,864	0.89	0.95	52,699	2,351	50,469	3,331	24,805,988	3,331	4,211	7	7	8.68	7,550	1.79
2027	1	9			55,250		0.46	52,297	2,076	25,266									
	2	10			57,636		1.24	51,654	1,804	67,578									
	3	11			60,022		1.34	50,785	1,537	71,454									
	4	12			62,408		0.95	49,706	1,275	49,629									
2028	1	13			64,794		0.41	48,432	1,020	21,059									
	2	14			67,180		1.12	46,979	773	55,316									
	3	15			69,566		1.20	45,365	534	57,444									
	4	16			71,952		0.85	43,604	305	39,183									

Method Comparisons

Explanation	Forecasting Method	Avg. MAD	Avg. MAPE (%)	Greatest Bias	TS Range
TS Range does not fluctuate around zero, estimated average marginal percent error is 8.6%, and mean absolute deviation is 8,995. This notes that this method does a poor but usable job at forecasting demand and could be used given the other methods do a worse job at forecasting	Static Method	~ 8,995	~ 8.60	25,636	0.37 to 12.10
TS Range does fluctuate around zero, estimated average marginal percent error is 71%, and mean absolute deviation is 17,460. This notes that this method does a bad job at forecasting demand and should not be used.	Moving Average	~ 17,460	~ 71	23,757	-0.31 to 1.00
TS Range does not fluctuate around zero, estimated average marginal percent error is 38.09%, and mean absolute deviation is 13,095. This notes that this method does a bad job at forecasting demand and should not be used.	Simple Exponential Smoothing	~ 13,095	~ 38.09	-31,996	-3.73 to 0.44
TS Range does not fluctuate around zero, estimated average marginal percent error is 40.76%, and mean absolute deviation is 14,180. This notes that this method does a poor job at forecasting demand and should not be used.	Holt's Method	~ 14,180	~ 40.76	-37,059	-3.00 to 0.07
TS Range does fluctuate around zero, estimated average marginal percent error is 7.29%, and mean absolute deviation is 3,176. This notes that this method does a good job at forecasting	Winter's Method	~ 3,176	~ 7.29	-13,069	-4.00 to 1.79

demand and should be used.					
----------------------------	--	--	--	--	--

Selected Method: Winter's Forecasting Method

Having put together the method comparison and visualized the forecast compared to the actual demand the selected forecasting method yields the following quarterly and annual demand:

Year	Quarter	Period (t)	Demand (units)
2028	1	9	25,266
	2	10	67,578
	3	11	71,454
	4	12	49,629
TOTAL			213,927
2029	1	13	21,059
	2	14	55,316
	3	15	57,444
	4	16	39,183
TOTAL			173,002

As demonstrated by Winter's Method forecast plot (Figure U) & the MAPE, this model successfully mirrors historical reality to a high degree. Unlike the horizontal flatline produced by Simple Exponential Smoothing or the rigid diagonal trendline of Holt's Method, Winter's Method captures the distinct nuances of our market. By utilizing all three smoothing constants (alpha, beta, gamma), the model dynamically projects our baseline growth trend while accurately predicting the massive Q3 summer volume spikes.

3. Cycle Inventory Management

Cycle inventory allows a stage in the supply chain to exploit economies of scale given it is the average inventory in a supply chain stage due to the production (or purchasing) of lot sizes larger than the demand of customers. With this in mind, we will use the network representation that displays the stages of

Honeywell's Distributed Wireless supply chain, to identify the managerial inventory levels that will reduce the cycle inventory without increasing the costs for stages Honeywell will transfer inventory with.

Process

Step 1: Obtain the supply chain network

Step 2: From the supply chain network decide which subsystems to investigate and identify the stages related to them

Step 3: Layout the data on the subsystems and costs

Step 4: Perform cycle inventory calculations and analysis for years 1-4 (provide all equations)

1. Lot size per shipment minimizing total cost/Economic order quantity
2. Number of shipments per year
3. Cycle inventory holding cost
4. Replenishment cycle time
5. Average flow time

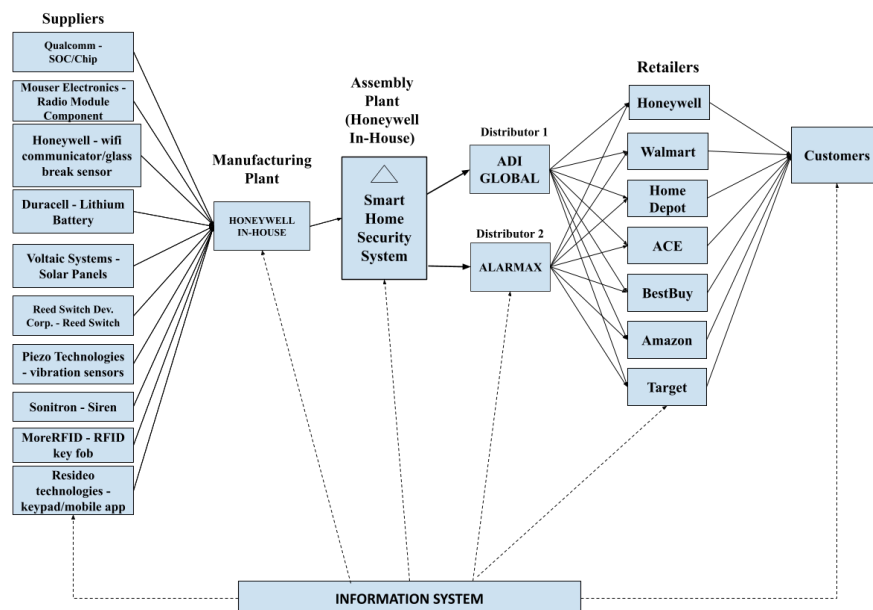
Step 5: Implement in excel

Step 6: Put together the cycle inventory of all subsystems in one diagram

Step 4: Draw conclusions

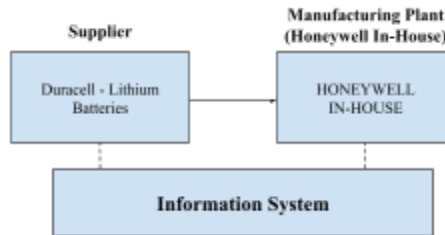
Execution

Step 1: SC Network Representation

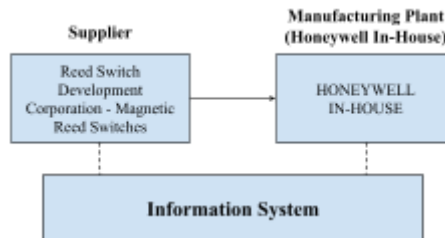


Step 2: Given the supply chain network above we decided to investigate the cycle inventory of duracell batteries, magnetic reed switches, RFID key fobs, and the entire Distributed Wireless system. The following stages zoom into the network for each associated system we are investigating.

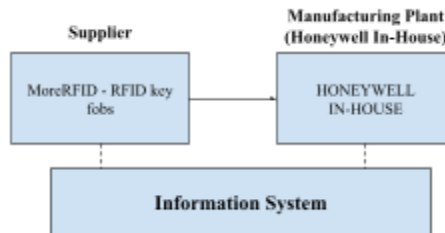
Duracell Lithium Batteries supplied to Honeywell



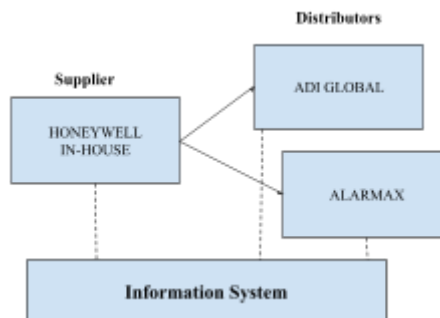
Reed Switch Development Corporation magnetic reed switches supplied to Honeywell



MoreRFID RFID key fobs supplied to Honeywell



Honeywell's Distributed Wireless supplied to ADI Global and AlarMax for distribution



Step 3: Layout the data on the subsystems and costs

To calculate both the cycle inventory and safety stock for each subsystem and the system itself the data on the unit cost, order cost, shipping cost, and demand of units per system will be identified below and displayed in a table.

Duracell Batteries (Subsystem 1)

**Assumption: Each unit uses 4 batteries, thus multiplying demand by 4*

Batteries provide power to the various components of the Distributed Wireless system, and are batched and sold in quantities of 4 per 1 system unit. Given this a single battery was calculated to cost \$2.50.

Reed Switches (Subsystem 2)

**Assumption: Each unit has five magnetic reed switches, thus multiplying demand by 5*

Magnetic reed switches use two components: a sensor and an actuator. Looking at set pricing from Radwell a unit would cost \$24.72. As no ordering cost was tacked onto either this supplier's site nor on others like DigiKey we sought to find what the fulfillment cost for any given order using current order cost averages. Using the [Speed Commerce](#) fulfillment data and noting both manufacturing and distributing is based in the U.S., the order cost has been estimated to be \$10.00 with a holding cost of 20%.

RFID key fob (Subsystem 3)

**Assumption: Each unit has 2 RFID key fobs, thus multiplying demand by 2*

As we are sourcing the RFID key fobs from MoreRFID we researched their available market products and found the average market price of this control system, used to make Distributed Wireless, may cost anywhere between \$2 to \$10 for every door, noted by [MoreRFID](#). Looking at their available key fobs, we decided that a single keyfob would cost \$5. As Honeywell would purchase these directly from MoreRFID based in China the ordering cost was estimated to be approximately \$20 with a holding cost of 20%.

Distributed Wireless (System)

As previously determined the selling price of Honeywell's Distributed Wireless is \$500. To determine the ordering cost, we looked at similar products' shipping costs and the Honeywell website which based shipment on priority thus we averaged the shipment costs to be roughly \$16.00 with a holding cost of 20%.

Subsystem	Unit Cost	Order Cost	Holding Cost	Forecasted annual demand for 2026-2029	
<i>Lithium Batteries (4 per system)</i>	C = \$2.50/battery	S = \$10/order	h = \$2.00/year	2026	D _{total} = 14,052 units/year
				2027	D _{total} = 14,613 units/year
				2028	D _{total} = 14,940 units/year
				2029	D _{total} = 15,328 units/year

<i>Magnetic Reed Switched (5 per system)</i>	C = \$24.72/sensor set	S = \$10/order	h = 20% = \$0.20	2026	D _{total} = 17,565 units/year
				2027	D _{total} = 18,265 units/year
				2028	D _{total} = 18,675 units/year
				2029	D _{total} = 19,160 units/year
<i>RFID key fob (2 per system)</i>	C = \$5/key fob	S = \$20/order	h = 10% = \$0.10	2026	D _{total} = 7,026 units/year
				2027	D _{total} = 7,306 units/year
				2028	D _{total} = 7,470 units/year
				2029	D _{total} = 7,664 units/year
<i>Distributed Wireless</i>	C = \$270.37/unit	S = \$16 per order	h = 20% = \$0.20	2026	D _{total} = 3513 units/year
				2027	D _{total} = 613 units/year
				2028	D _{total} = 746 units/year
				2029	D _{total} = 846 units/year

Step 4: Provide all equations to perform cycle inventory calculations and analysis for each year(to be implemented in an excel spreadsheet)

Annual Demand given units/system	Optimal Lot Size (lot size per shipment minimizing total cost)	Shipment Frequency (number of shipments per year)	Min. Holding Cost (Holding cost incurred)	Replenishment Cycle Time (When the inventory needs to be replenished)	Average Flow Time (Average time 1 unit spends in inventory)
= (units/system) * (systems annual demand)	$Q_L^* = \sqrt{2(DS)/hC}$	$n^* = D/Q_L^*$	$C_I^* = (Q_L^*/2)(hC)$	$T = 365/n^*$	Flow Time = T/2

Step 5: Implement in excel: [Cycle Inventory.xlsx](#)

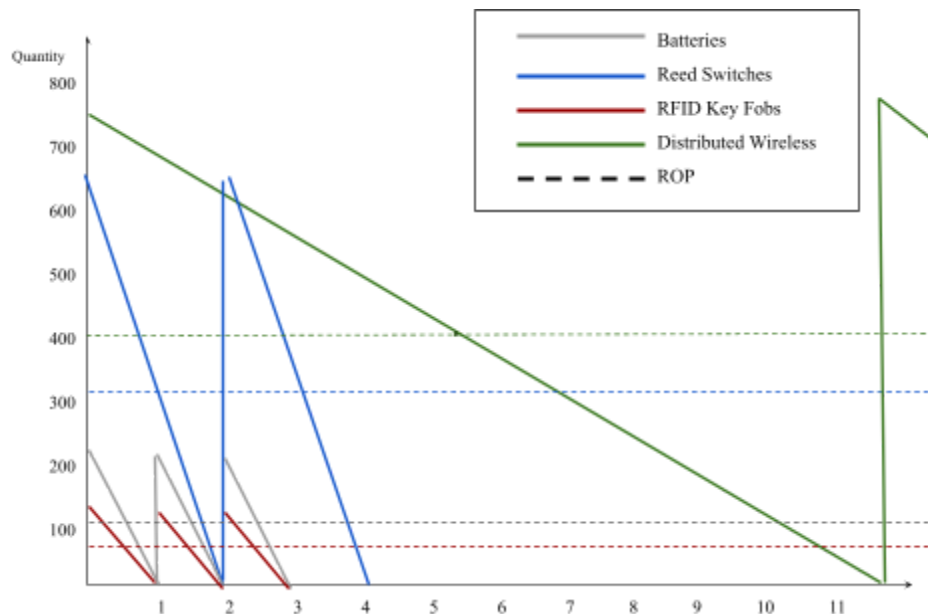
Given both the data and equations, we calculated the cycle inventory of the different subsystems and Distributed Wireless for 2026-2029 (years 1-4 respectively) as shown below.

Data					Forecasted Annual Demand					
System	Unit Cost	Order Cost	Holding Cost	Units/system		Q1	Q2	Q3	Q4	Total
Batteries	2.5	10	0.1	4	2026	351	1,054	1,230	878	3,513
Reed Switches	24.72	10	0.2	5	2027	402	1,096	1,205	950	3,653
RFID key fob	5	20	0.2	2	2028	393	1,121	1,269	952	3,735
Distributed Wireless	270.37	16	0.2	1	2029	403	1,150	1,302	977	3,832

2026 Calculations					2028 Calculations				
	Batteries	Reed Switches	RFID key fob	Distributed Wireless		Batteries	Reed Switches	RFID key fob	Distributed Wireless
Annual	14,052	17,565	7,026	3,513	Annual	14,940	18,675	7,470	3,735
Optimal lot size	265	659	133	770	Optimal lot size	273	679	137	794
Shipment frequency	53	27	53	5	Shipment frequency	55	27	55	5
Min. holding cost	13	66	13	77	Min. holding cost	14	68	14	79
Replenishment cycle time	7	14	7	80	Replenishment cycle time	7	13	7	78
Average flow time	3	7	3	40	Average flow time	3	7	3	39

2027 Calculations					2029 Calculations				
	Batteries	Reed Switches	RFID key fob	Distributed Wireless		Batteries	Reed Switches	RFID key fob	Distributed Wireless
Annual	14,612	18,265	7,306	3,653	Annual	15,328	19,160	7,664	3,832
Optimal lot size	270	672	135	786	Optimal lot size	277	688	138	805
Shipment frequency	54	27	54	5	Shipment frequency	55	28	55	5
Min. holding cost	14	67	14	79	Min. holding cost	14	69	14	80
Replenishment cycle time	7	13	7	79	Replenishment cycle time	7	13	7	77
Average flow time	3	7	3	39	Average flow time	3	7	3	38

Step 6: Put together the cycle inventory of all subsystems in one diagram



Step 7: Conclusions

System Information	Batteries	Reed Switches	RFID Key Fobs	Distributed Wireless
Cycle inventory	7 days (1 week)	14 days (2 weeks)	7 days (1 week)	77-80 days (11-12 weeks)
Avg. flow time	3 days	7 days (1 week)	3 days	38-40 days (5-6 weeks)
Is this cycle inventory reasonable?	Yes	Yes	Yes	Yes

Overall, the inventory strategy focuses on minimizing total costs by balancing component replenishment with slower finished-good cycles. For the selected high-volume subsystems (batteries and RFID key fobs), the model dictates aggregating and ordering reed switches and RFID key fobs at the same time while ordering batteries about 10 weeks sooner. In contrast, the completed security system, which has a higher unit cost of \$270.37, is managed with a larger lot size and a longer replenishment cycle. This approach ensures that while the assembly line remains continuously fed with parts, the more expensive finished inventory is turned over less frequently to optimize holding costs. The cycle inventory and flow times length makes sense given the estimated and forecasted demand is so low.

4. Profitability of Distributed Wireless

The following section determines the base case profit according to the five year NPV analysis framework. Based on the demand estimates developed in the earlier phase this analysis allows us to determine if the project offers sufficient return.

4.1 Sales & Production Volume

Year	Quarter	Period (t)	Demand (units)
2026	1	1	351
	2	2	1054

	3	3	1230
	4	4	878
TOTAL			3513
2027	1	5	402
	2	6	1096
	3	7	1205
	4	8	950
TOTAL			3653
2028	1	9	393
	2	10	1121
	3	11	1269
	4	12	952
TOTAL			3735
2029	1	13	403
	2	14	1150
	3	15	1302
	4	16	977
TOTAL			3832

Following the estimated demand data projected for years 1-4, the value of the sales and production volume is ... (2026), ... (2027), ... (2028), and ... (2029).

4.2 Base Case Input Justifications

Sales & Production Volume

$$\begin{aligned}
 \text{Sales \& Production Volume} &= \text{Unit Demand sum of years 1-4} \\
 &= Q_{2026} + Q_{2027} + Q_{2028} + Q_{2029}
 \end{aligned}$$

$$\begin{aligned}
 &= 3513 + 3653 + 3735 + 3832 \\
 &= 14,733 \\
 &\Rightarrow \text{In NPV table values are displayed as per quarter demand}
 \end{aligned}$$

Development Cost

$$\text{Development Cost (D)} = \text{Revenue}_{2026} (R) * \text{development percentage allocation (d)}$$

- Base Year Revenue(2026)

$$\begin{aligned}
 R_{2026} &= V_1 * P \\
 &= 3,513 * 500 \\
 &= 1,756,500 \\
 &\Rightarrow \text{Base year (2026) revenue is \$1,756,500}
 \end{aligned}$$
- Development percentage allocation

$$r = 10\% = 0.1$$

$\Rightarrow$ Development Cost

$$\begin{aligned}
 D &= R_{2026} * r \\
 &= 1,756,500 * 0.1 \\
 &= 175,650 \\
 &\Rightarrow \text{Development costs is \$175,650}
 \end{aligned}$$

Development cost is estimated to be \$2,900 as 10% of the projected base-year (2026) revenue as similar products do not publicize detailed development costs.

Unit price

The unit price of the Distributed Wireless (\$500) is determined by market research on similar products to ensure competitive pricing:

Leading Competitor Products	Price (\$ USD)
SimpliSafe 21-Piece Security System	\$622.79 ~ 650
Vivint HomeProtect Pro	\$499
Ring Alarm Pro Security Kit	\$595

Market Price range determination

- Minimum: \$499
- Maximum: \$650

$$\Rightarrow (499 \leq P \leq 650)$$

$\Rightarrow$ Unit Price

$$P = 500$$

- Unit Price of Distributed Wireless = \$500

Based on the pricing of competitor home security products, setting the price at \$500 keeps the product in the expected margin over production ($\$499 \leq P \leq \650) while still offering affordability to the customers.

Unit Productions Cost

Unit Production Cost (C_{unit}) = Sum of Component Costs (C) * Overhead Costs ($k * C$)

Component Cost Breakdown			
Component Type	Manufacturer	Component Name	Cost
SOC / Chip	Qualcomm Technologies	Arduino Nano R4	\$12.10
Radio module component	Silicon Labs	MGM240SA22VNA2	\$5.54
Wifi Communicator	Honeywell (in-house)	Wifi Communicator	\$6.00
Acoustic glass break Sensors	Acoustic glass break Sensors	Acoustic glass break Sensors	\$35.99
Lithium Battery	Duracell	Lithium Battery	\$10
Solar Panel	Voltaic Systems	0.4 Watt 2.5 Volt Mini Solar Panel - ETFE	\$4.29
Reed Switch	Reed Switch Developments Corp	2005-1900-400	\$24.72
In-floor Piezoelectric Vibration Sensor	TE Connectivity	MiniSense 100	\$4.26
Piezoelectric Siren	Sonitron	SMAC-25	\$15.37
RFID / NFC Keyfob	MoreRFID	RFID / NFC Keyfob	\$2.00
Keypad	Resideo Technologies	Keypad	\$127.99

Component Cost Breakdown	
Sum of component costs	\$248.26

- Sum of Component Costs (C)

$$\begin{aligned}
 C &= \sum_{i=1}^n c_i \\
 &= \sum_{i=1}^{11} c_i \\
 &= 12.10 + 5.54 + 6 + 35.99 + 10 + 4.29 + 24.72 + 4.26 + 15.37 + 2 + 127.99 \\
 C &= 248.26 \\
 &\Rightarrow \text{sum of major components of Distributed Wireless is } \$248.26
 \end{aligned}$$

- Overhead Allocation ($k * C$)
where $k = 25\% = 0.25$ (percentage)

$$\begin{aligned}
 k * C &= 0.25 * C \\
 &= 0.25 * 248.26 \\
 &= 62.065 \\
 k * C &= 62.07 \\
 &\Rightarrow \text{allocate an overhead of } \$62.07 \text{ to account for indirect manufacturing costs}
 \end{aligned}$$

$\Rightarrow$ Unit Productions Cost:

$$\begin{aligned}
 C_{\text{unit}} &= C + (k * C) \\
 &= 248.26 + 62.07 \\
 C_{\text{unit}} &= 310.33 \\
 &\Rightarrow \text{Unit Productions Cost is } \$310.33
 \end{aligned}$$

Unit production costs are estimated at \$310.33 using a bottom-up component pricing of all major components including an added 25% allocation for indirect manufacturing costs to ensure realistic cost estimates for all production expenses.

Ramp-up cost

Ramp-up Cost (RC) = Revenue₂₀₂₆ (R) * initial percentage allocation (r)

- Initial percentage allocation
 $r = 5\% = 0.05$
- Base Year Revenue(2026)

$$\begin{aligned}
 R_{2026} &= V_1 * P \\
 &= 3,153 * 500 \\
 &= 1,756,500 \\
 &\Rightarrow \text{Base year (2026) revenue is \$1,756,500}
 \end{aligned}$$

$\Rightarrow$ Ramp-Up Costs

$$\begin{aligned}
 RC &= R_{2026} * r \\
 &= 1,756,500 * 0.05 \\
 &= 87,825 \\
 &\Rightarrow \text{ramp up costs is \$87,825}
 \end{aligned}$$

Ramp-up cost is \$14,500 which was estimated as 5% of the expected revenue in the base year, representing the initial setup investments.

Marketing & support cost

Marketing Cost (M) = Revenue₂₀₂₆ (R) * marketing and support percentage allocation (m)

- Marketing and Support percentage allocation (m)

$$m = 2\% = 0.02$$

- Base Year Revenue(2026)

$$\begin{aligned}
 R_{2026} &= V_1 * P \\
 &= 3,513 * 500 \\
 &= 1,756,500 \\
 &\Rightarrow \text{Base year (2026) revenue is \$1,756,500}
 \end{aligned}$$

$\Rightarrow$ Marketing & support Cost

$$\begin{aligned}
 M &= R_{2026} * m \\
 &= 1,756,500 * 0.02 \\
 &= 35,130 \\
 &\Rightarrow \text{market and support costs is \$35,130}
 \end{aligned}$$

Market and support costs is \$35,130 which was estimated using the percentage of sales method which takes 2% of the expected revenue value in the base year.

Annual Discount Factor

$$\begin{aligned}
 r &= 10\% \\
 r &= 0.1
 \end{aligned}$$

The annual discount factor is 10% which is a common value used for NPV analysis.

Five-year NPV Analysis

The values from the tables above were used to calculate the net present value, as seen in the spreadsheet below. The years 2026 - 2027 using actual demand data, while 2028 - 2029 are using forecasted data. (Winter's method)

Final Report - Financial Modeling & Sensitivity Analysis.xlsx

BaseCase	2025				2026				2027				2028				2029			
period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
(\$ values in thousands)	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Development Cost	-43.91	-43.91	-43.91	-43.91																
Ramp-up cost				-43.91	-43.91															
Marketing & support cost				-8.7825	-8.78	-8.78	-8.78	-8.78	-8.78	-8.78	-8.78	-8.78	-8.78	-8.78	-8.78	-8.78	-8.78	-8.78	-8.78	-8.78
Production cost					-108.93	-327.09	-381.71	-272.47	-124.75	-340.12	-373.95	-294.81	-121.96	-347.88	-393.81	-295.43	-125.06	-356.88	-404.05	-303.19
Production volume					351	1,054	1,230	878	402	1,096	1,205	950	393	1,121	1,269	952	403	1,150	1,302	977
Unit production cost					-0.31	-0.31	-0.31	-0.31	-0.31	-0.31	-0.31	-0.31	-0.31	-0.31	-0.31	-0.31	-0.31	-0.31	-0.31	-0.31
Sales Revenue					175.50	527.00	615.00	439.00	201.00	548.00	602.50	475.00	196.50	560.50	634.50	476.00	201.50	575.00	651.00	488.50
Sales volume					351.00	1,054.00	1,230.00	878.00	402.00	1,096.00	1,205.00	950.00	393.00	1,121.00	1,269.00	952.00	403.00	1,150.00	1,302.00	977.00
Unit price					0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
Period Cash Flow	-43.91	-43.91	-87.83	-96.61	57.79	191.13	224.51	157.75	67.46	199.10	219.77	171.40	65.76	203.84	231.91	171.78	67.65	209.34	238.17	176.53
PV year 1, r=10%	-43.91	-42.84	-83.59	-89.71	52.36	168.93	193.60	132.71	55.37	159.42	171.68	130.63	48.89	147.87	164.13	118.61	45.57	137.58	152.71	110.42
Project NPV, \$		1,730																		
all units in thousands, \$																				

The nominal profit of the Distributed Wireless project, determined by a 5-year NPV analysis with a discount factor of 10%, is expected to be \$1,730,000. Given the current conditions, the project is expected to be profitable.

Appendix

Revisions

Issue Identified	Changes Made	Team Members Responsible
demand estimation has inconsistency with the number of units for demand (pages 10 & 14).	Corrected inconsistency by simplifying demand formula	Vivian Nguyen
In Demand Estimation and Demand Forecasting, you need to reduce filler and bulk	Reduced explanatory content filler and bulk	Vivian Nguyen
We do not see your chosen values for your smoothing constants for any of the methods. Additionally, you need to play around with adaptive forecasting smoothing parameters.		Abdul-Haadi
Need a clear mathematical, stepwise process to develop a programmable equation within your Cycle Inventory Management	Laid out step by step how the cycle inventory will be done with a process. Executed process in Excel spreadsheets	Hailey McElroy

section.		
missing a complete Network Representation of your SC in Cycle Inventory Management	Reinstated network representation diagram to its entirety, and added individual condensed diagrams	Hailey McElroy Garrett Arcand
need to show multiple replenishment cycles in your CIM graphs (page 37, 39, 40), and include a more comprehensive conclusion.	Put together multiple replenishment cycles in a graph. Added a more comprehensive conclusion <u>still need to connect to the safety inventory section to then say the graph will now need to account for the safety inventory.</u>	Hailey McElroy
Within your Financial Modeling section, you need to show a clear mathematical, stepwise process for deriving financial model inputs.	Mathematically clarified and developed the stepwise process to deriving input values	Vivian Nguyen James Burch
Your Project NPV ideally should not be negative, and your data is not properly aligned with its respective quarters within your financial model	Corrected unit conversion error in demand data	James Burch

Member Contributions Table

Name	Class 2/17	Meeting 2/16	Meeting 2/17	Class 2/19	Project Report Contributions
James Burch	☒	☒	☒	☒	- Determined the base-case costs and profit - Performed the NPV analysis
Hailey McElroy	☒	☒	☒	☐	- Document set up - Provided information for Reed Switches section of (3) - Provided information for Distributed Wireless section of (3) - Forecasting steps/process lay out for (2) - Forecasting equations laid out for (2)
Vivian Nguyen	☒	☒	☒	☒	- Reworking the Demand Estimation Section to remove filler and incorporate the explicit step wise process - Mathematicize demand data estimation with symbols, equations, and formulas. - Supported the base-case development and ramp up costs of NPV analysis
Garrett Arcand	☒	☒	☒	☒	- Updated time-phase project plan for P2 and assigned tasks evenly - Provided information for Duracell Batteries section of (3) - Added an executive summary, summarizing report findings
Mohammed Abdul-Haadi Ansari	☒	☒	☐	☒	Forecasting & Error Analysis Spreadsheet
Collin Yuen	☒	☒	☐	☒	- Summarize critical data with a formatted table - Reworking the Demand Estimation Section - Mathematicize demand data estimation with symbols, equations, and formulas. - Adjusted demand estimates with seasonality weights
Abinaya Ramesh	☒	☒	☐	☒	Forecasting & Error Analysis Spreadsheet

Member Contributions Table (Revisions)

Name	Class 2/24	Meeting 2/20	Meeting 2/22	Meeting 2/23	Project Report Contributions
James Burch	☒	☒	☒	☒	• Redid the five-year NPV analysis • Reassessed the profitability of our product, calculating corrected values for development, ramp-up, and marketing/support costs
Hailey McElroy	☐	☒	☒	☒	• Performed all forecasting methods to use • Added all graphs and screenshots for each forecasting method • Laid out the error metrics in a table • Calculated cycle inventory for the Distributed Wireless system. • Laid out purpose of each methods and simple step by step process for each
Vivian Nguyen	☒	☒	☒	☒	• Reorganized the Estimating Demand section to match expected layout • Simplified and mathematicized the Estimating Demand section • Performed added research for appropriate estimation values of product pricing • Cleared Estimating Demand filler
Garrett Arcand	☐	☒	☐	☒	• Relocated assignments table to be underneath the executive summary • Reformed/redrew SC network representation to better fit section • Calculated cycle inventory for batteries and switches • Updated the executive summary
Mohammed Abdul-Haadi Ansari	☒	☒	☒	☒	• Forecasting • Graphs (<i>Visualizations</i>) – product cycle • Demand Forecast improvement on explanations
Collin Yuen	☐	☒	☒	☒	• Reorganized the Estimating Demand section to match expected layout • Simplified and mathematicized the Estimating Demand section • Calculated yearly revenue, sales volume, price in a mathematical numerical process •
Abinaya Ramesh	☐	☐	☐	☐	•

Post-meeting Action Table

Issue	Assigned Member	Timeline
● Lengthen/broaden the executive summary to explain the “why” and results for each defined section ● Relocate assignments table underneath the executive sum. ● Relocate section numbers to the front in the TOC	Garrett Arcand	☒ Address after prof/TA review meeting independently (2/18-2/19)
● Lay out NPV as profit based on X amount of years with a discount of Y% ● Replace listed years with actual years (e.g. 2025, 2026, etc.) ● Shift tables to fit demand ● Correct demand to be reflective of (2)	James Burch Vivian Nguyen (entire group opinion may be needed here)	☒ Address in meetings (2/20-2/22)
● Add explanation for the network representation being there, and why we chose the subsystems we chose ○ Draw a simple picture that zooms in on the network rep ● We need to explain why our results make sense	Hailey McElroy Garrett Arcand	☒ Address in meetings (2/20-2/22)
● Rename “Demand Data” to “Estimating Demand” ● Section needs a “very numerical analytical process” ○ Define the variables, make more clear/simple ○ Reference hailey’s notes in the gc for more	Vivian Nguyen Collin Yuen	☒ Address both in meetings/independently
● Add plots to graph the demand data, demand vs. forecast ○ Be sure to include trend lines, etc. ● Put selected method of forecasting at the very end with a clear explanation as to why we chose it	Mohammed Ansari Abinaya Ramesh (entire group opinion needed for final selection of method)	☒ Address both in meetings/independently, notify group before deciding on a method to confirm

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Group 12: Phase III

“Distributed Wireless” Home Security System

Vivian Nguyen, James Burch, Garrett Arcand, Collin Yuen, Mohammed Abdul-Haadi Ansari,
Abinaya Ramesh, Hailey McElroy

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0. Executive Summary

Objective

To design and manage the supply chain for Distributed Wireless in Honeywell, by means of structuring a stage and network representation of said supply chain around the below items.

Key Results

Demand Estimation

Process: To estimate the demand for Distributed Wireless, we established a 9-part process that defined necessary variables, incorporated the product lifecycle, applied seasonality, and visualized/validated the results.

Conclusions: Overall, the demand total for year 1 was 580 Distributed Wireless units, and the demand total for year 2 was 612 Distributed Wireless units. In terms of general demand for home security systems in the United States, these figures make sense, showing steady growth in demand over a 2-year period.

Demand Forecasting

Process: Using the estimated demand for years 1 and 2, calculated in section 1, we forecasted demand for years 3 and 4 using five different methods of forecasting. After each method was completed, we compared each one, and selected the method that yielded the best results in terms of what is sensible for the current home security system market.

Conclusions: Winter's Method was selected despite SES yielded the lowest MAPE (31.25%) – this is because SES does not account for seasonality or trend, and the MAPE for Winter's Method was only marginally worse than that of SES anyhow.

Cycle and Safety Inventory

Process: To calculate the cycle inventory for Distributed Wireless, we first formed a general process that visualized the necessary subsystems for inventory calculations based on the subsystems that were in highest demand for our general system. We then researched the chosen subsystems and found per-unit costs, lead times, and converted these figures to match the demand of our general system. After the data was laid out, cycle inventory was calculated for years 1-4, implemented in MS Excel, and categorized into a diagram to show real results.

Conclusions: To maintain a 95% CSL, the safety inventory estimated that would be required is as follows: 251 unit safety inventory for Batteries, 443 unit safety inventory for Reed Switches, 178 unit safety inventory for RFID fobs, and a 272 unit safety inventory for the Dist. Wireless System as a whole. The ROP's corresponding with these safety inventory estimates are 1,678, 3,994, 1,607, and 6,968 units respectively. These high ROP's are a direct consequence of long replenishment lead times inherited from the cycle inventory calculations.

Cycle inventory analysis yields optimal lot sizes of 1,301 units for Duracell batteries, 3,234 units for magnetic reed switches, 1,301 units for RFID key fobs, and 6,050 units for the Distributed

Wireless system (for 2029), with corresponding replenishment cycle times of 140, 279, 281, and 2,610 days respectively. Batteries, reed switches, and RFID key fobs maintain reasonable replenishment cycles of 20–40 weeks, confirming efficient subsystem inventory turnover. While the Distributed Wireless system's replenishment cycle of approximately 372 weeks appears unreasonable on the surface, it is partly a reflection of the product's naturally low annual demand, averaging roughly 600–850 units per year across the forecast period, consistent with a specialized home security system serving a narrow market segment. That said, the EOQ lot size of ~6,000 units still substantially exceeds even several years' worth of demand, suggesting that a smaller, more frequent ordering strategy would better suit finished-goods inventory management for this product.

Financial Modeling

Process: To model the financial details of the Distributed Wireless system, we established a 4-part process, beginning with the sales and production volume, and ending with the sensitivity analyses on the calculated NPV's. Based on the real results calculated in the first 3 stages of this report, along with necessary assumptions (cited in section 4), costs for sales and production, general development, cost-per-unit, ramp-up, and marketing/support were calculated and visualized.

Conclusions: Given the results of the sensitivity analyses, the distributed wireless project is reasonably robust. Even while the input parameters were adjusted by up to 30% (50% in the case of development time), the expected profit of the project remained positive. Changes in sales volume and unit manufacturing cost had the largest impact on NPV, each decreasing it by nearly \$6.5 thousand for each percentage point increase. Thus, if either demand or manufacturing cost fall greatly short of our forecast, the project could produce a loss. This outcome, however, is unlikely, and there is a healthy margin of error.

Conclusion

Overall, the analysis demonstrates that Distributed Wireless can be supported by an efficient supply chain and a structured financial model. Our nominal (base case) profit based on a 5-year NPV analysis with a discount factor of 10% is \$357,000. By combining accurate demand forecasting, optimized inventory planning, and cost analysis, Honeywell can effectively launch and scale the product in the competitive home security market.

1. Demand Estimation

1.1 Problem Definition

The goal of the following section is to create a credible & analytically-derived quarter demand estimation for the Distributed Wireless (DW) system across its first 8 quarters of market availability (i.e., Years 1 and 2). Since the product has not yet launched, no direct historical sales data exists, demand must instead be estimated through a programmable mathematical model grounded in published market data, product life-cycle analysis, and seasonal consumer behavior.

This revision addresses two issues identified in the Phase 2 review:

(i) the absence of a single unified equation incorporating all demand factors, and (ii) the unit inconsistency between the life-cycle phase table and the final quarterly demand values.

1.2 Process

1. Defining Variables
2. Upper Limit Definition
3. Distributed Wireless Home Security Market Share
4. Annual Unit Demand
5. Product Life Cycle
6. Application of Seasonality
7. Application of Unified Demand Equation
8. Quarterly Demand Estimates
9. Validation of Results

1.3 Defining Variables

Variable	Definition	Value and/or Additional Notes
M_0	Total home security system market value in base year (i.e., 2025)	\$4.37 billion ¹ $\Rightarrow 4.37 \times 10^9$
g	Annual market growth rate (CAGR)	3.7% ¹
n	Year index ($n = 1$ for Year 1, $n = 2$ for Year 2)	1,2

¹ The Business Research Company, 2025

² Competitor benchmarking: SimpliSafe \$623, Vivint \$499, Ring \$595

S_{broad}	Honeywell's benchmark home security market share	0.2% ³
s	Honeywell Distributed Wireless market share	0.33%
P_H^1	Probability of homeowners adopting home security system (Year 1)	0.18%
P_H^2	Probability of homeowners adopting home security system (Year 2)	0.19%
R_n	Annual revenue	See below
P	Average unit selling price	500 USD ³
V_n	Estimated annual unit demand in 'n' (year)	See below
$W _{\{q=Q, n=N\}}$	Seasonal weight for quarter q in respective n	Sum of all weights is 1; Q is a specific value for q and N is a specific value for n .
$D(n, q)$	Equation with input values	

1.4 Deriving the Equation

1.4.1 Upper Limit Definition

The (theoretical) upper bound on annual demand is the total US homeowner population determined by:

$$H = 97,703,784 \text{ homes.}$$

Annual demand V_n must satisfy $V_n \leq H$. This bound is checked and satisfied by the this principle:

$$V_1 = 174,800 \ll H$$

1.4.2 Distributed Wireless Home Security Market Share

Annual demand for the Distributed Wireless system is derived from estimating the total share of the total home security market revenue and converting the revenue into units using the unit price. Market share is scaled down from the broader benchmark share using the proportion of existing Honeywell Home products that are home-security related.

³ MarketsandMarkets, 2026

$$\begin{aligned}
s &= (\text{Honeywell's benchmark market share in home security}) * (\text{frequency of similar products at Honeywell}) \\
&= s_{\text{broad}} * k \\
&= 0.02 * 0.16406 \\
&= 0.00328 \\
&\Rightarrow 0.33\%
\end{aligned}$$

1.4.3 Annual Unit Demand and Revenue

Annual unit demand in year n is derived from the upper limit of potential customers found by US homeowners, probability of adoption, and the market share of Distributed Wireless. Annual revenue is determined by taking the annual demand multiplied by the Distributed Wireless determined unit price

$$V_n = H * P_H^n * s$$

$$R_n = V_n * P$$

- Where:
 - H = Total homes owned in the US
 - P_H^n = Probability of homeowners adopting home security system
 - S = Distributed Wireless market share
 - P = unit price

Year 1 (2026):

$$\begin{aligned}
V_1 &= H * P_H^1 * s \\
&= 97,703,784 * 0.0018 * 0.0033 \\
&= 580.36 \\
&\Rightarrow 580 \text{ units of demand in Year 1} \\
R_1 &= V_1 * P \\
&= 580 * 500 \\
&\Rightarrow \$290,000 \text{ in total revenue for Year 1}
\end{aligned}$$

Year 2 (2027):

$$\begin{aligned}
V_2 &= H * P_H^2 * s \\
&= 97,703,784 * 0.0019 * 0.0033 \\
&= 612.6 \\
&\Rightarrow 613 \text{ units of demand in Year 2}
\end{aligned}$$

$$\begin{aligned}
 R_2 &= V_1 * P \\
 &= 613 * 500 \\
 &\Rightarrow \$306,500 \text{ in total revenue for Year 2}
 \end{aligned}$$

1.4.4 Incorporate Product Life-Cycle Phase Weights

The life-cycle phase structure informs the *shape* of demand within each year as the demand transitions from introduction to maturity phases.

- Q1-Q2, Y1 map to the Introduction phase (low and ramping demand).
- Q3-Q4, Y1 and Q1-Q2, Y2 map to the Growth phase.
- Q3-Q4, Y2 approach early Maturity (demand stabilizing).

In the table below, the life-cycle weights divide the estimated annual demand across quarters based on how new products typically gain adoption. In the introduction phase, demand is lower because customers are still learning about the product and distribution is limited. As the product enters the growth phase, awareness increases and demand rises more quickly, which is reflected in higher weights in the later quarters of Year 1 and early in Year 2. By the end of Year 2, we see the product move towards the maturity stage, where demand becomes more stable. Using life-cycle weights allows the model to distribute demand realistically instead of using the assumption that there will be an equal amount of sales every quarter.

Quarter	Phase	Weight (W_n)	Sales	Revenue
Q1	Introduction	10%	58	\$29,000
Q2	Introduction	15%	87	\$43,500
Q3	Growth	35%	203	\$101,500
Q4	Growth	40%	232	\$116,000
Total		100%	580 units	\$290,000
Q5	Growth	30%	184	\$92,000
Q6	Growth	30%	184	\$92,000
Q7	Maturity	20%	123	\$61,500
Q8	Maturity	20%	122	\$61,000
Total		100%	613 units	\$306,500

1.4.5 Definition of Seasonal Weights

Seasonal weights are grounded in two empirical data sources: (a) NAR monthly US home sales data, which show peak activity May - August; and (b) Pinkerton Crime Index data, showing property crime peaks June - September. Both support elevated Q2 and Q3 demand for home security systems.

Quarter	Season	Justification	$W_{q, n=1}$	$W_{q, n=2}$
1	Winter	Lowest home sales & property crime	0.10	0.11
2	Spring	Rising home sales & property crime	0.30	0.30
3	Summer	Peak home sales & property crime	0.35	0.33
4	Fall	Declining home sales & property crime	0.25	0.26
Total			1.00	1.00

Verification Note: Totals add up to 1.00

Year 2 weights reflect the expected broader consumer base demand becoming slightly less concentrated in the peak summer quarter as awareness spreads across all seasons (maturity effect).

Visualization - See 1.7.2 (#1)

1.4.6 Apply the Unified Demand Equation

Combining Steps 1.4.3 and 1.4.5 yields the single programmable equation for quarterly demand:

$$D(n,q) = V_n * W_{|q=Q, n=N\}$$

• Where:

- V_n = Estimated annual unit demand in year n
- $W_{|q=Q, n=N\}$ = Seasonal weight for quarter q in respective year n

All terms are defined in Table 1. This equation can be directly programmed in a spreadsheet by substituting $n \in \{1, 2\}$ and $q \in \{1, 2, 3, 4\}$.

1.4.8 Quarterly Demand Results (based on Unified Demand equation)

Applying the $D(n, q)$ equation results the following demand dispersion:

Year (n)	Quarter (q)	V_n	$W_{q,n}$	D(n,q) Units
1	1	580	0.10	58
	2	580	0.30	174
	3	580	0.35	203
	4	580	0.25	145
Total Demand of Year 1 (2026)				580 Units
2	1	613	0.11	67
	2	613	0.30	184
	3	613	0.33	203
	4	613	0.26	159
Total Demand of Year 2 (2027)				612 Units

Visualization - See 1.7.2 (#2)

1.4.8 Validate Results (Alternate method)

To validate the demand estimate findings, we enlist a second method that uses a top-down approach which derives market revenue factors, serving as a benchmark for estimation comparison and assessment of potential underestimation.

Alternate Demand Equation:
$$V_n = \frac{M_0(1+g)^n(s)}{P}$$

- Year 1 (2026)

- Estimated Year 1 demand = 29,909 units

$$\blacksquare V_1 = \frac{\$4,370,000,000 (1.037^1)(0.033)}{\$500} = 29,909 \text{ units}$$

- Year 2 (2027)

- Estimated Year 2 demand = 31,016 units

$$\blacksquare V_1 = \frac{\$4,370,000,000 (1.037^2)(0.033)}{\$500} = 31,016 \text{ units}$$

Rescaling of Unified Demand Equation

Using a scaling value of 10%, the original unified demand equation is adjusted to reflect the market conditions as accounted for in the alternate demand equation method. The developed equation takes the original demand estimates and adds in a fraction of the difference between both methods' demand estimates of year n.

$$V_n = D(n,q) + \alpha (V_n^{\text{alt}} - D(n,q))$$

- Year 1 (2026)

$$\begin{aligned} V_1 &= D(1,q) + \alpha (V_1^{\text{alt}} - D(1,q)) \\ &= (580) + 0.1(29909 - 580) \\ &= 580 + 0.1(29329) \\ &= 580 + 2932.9 \\ &= 3512.9 \\ &\Rightarrow \mathbf{3513 \text{ units in year 1}} \end{aligned}$$

- Year 2 (2027)

$$\begin{aligned} V_2 &= D(2,q) + \alpha (V_2^{\text{alt}} - D(2,q)) \\ &= 612 + 0.1(31016 - 612) \\ &= 612 + 0.1(30404) \\ &= 612 + 3040.4 \\ &= 3652.4 \\ &\Rightarrow \mathbf{3653 \text{ units in year 2}} \end{aligned}$$

Comparison of Methods

Year	Method 1 (Unified Demand Equation)	Method 2 (Alternate Demand Equation)	New Unified Equation Scaling Original Data Estimation
	$D(n,q) = V_n * W _{q=Q, n=N}$	$V_n = \frac{M_0(1+g)^n(s)}{P}$	$V_n = D(n,q) + \alpha (V_n^{\text{alt}} - D(n,q))$
1 (2026)	V1 = 580 Units (from 1.4.8)	$V_1 = \frac{\$4,370,000,000 (1.037^1)(0.033)}{\$500} = 29,909$	V1 = 3513 units
2 (2027)	V2 = 612 Units (from 1.4.8)	$V_2 = \frac{\$4,370,000,000 (1.037^2)(0.033)}{\$500} = 31,016$	V2 = 3653 units

After evaluating three different ways of estimating demand data, we found that the most accurate and realistic set of demand data and numbers came from the new unified equation that allowed us to scale the original values from Method 1. Method 2 set unrealistic demand numbers of approximately 30,000 units on average across the first two years. This led us to reevaluating our demand estimation and settling on a new estimation of around 3500 units across the first two years, a much more realistic set of numbers.

1.5 Final Quarterly Demand Results

After the validation of results, the original demand estimate values (1.4.7) were rescaled and adjusted.

Year (n)	Quarter (q)	V_n	$W_{q,n}$	D(n,q) Units
1	1	3513	0.10	351
	2	3513	0.30	1054
	3	3513	0.35	1230
	4	3513	0.25	878
Total Demand of Year 1 (2026)				3513 Units
2	1	3653	0.11	402
	2	3653	0.30	1096
	3	3653	0.33	1205
	4	3653	0.26	950
Total Demand of Year 2 (2027)				3653 Units

Visualization - See 1.7.3 (#3)

1.6 What was Learned, Evaluating Validity, & Additional Notes

1.6.1 Unit Consistency (Phase 2 Correction)

The Phase 2 report contained two conflicting demand values on pages 10 & 14. The p. 10 life-cycle table showed quarterly demand of numbers which hold substantial difference, computed via formula $V_n = H * P_H^n * S$ where $V_1 = 3,517$ units. The p. 14 final demand table had found $V_1 = 174,800$ (from the market-size equation). The market-size formula is dropped

due to overestimation of demand yielding unrealistic margins for a new product in the home security market.

1.6.2 Growth Check

YoY growth: $(612 - 580) / 580 = 5.52\%$. In comparison to the CAGR of the market at 3.7%, for Distributed Wireless as a new product, the rate of growth is faster than the market itself upon launch during the earlier stages of life cycle.

1.6.3 Seasonality Ratio Check

Year 1 Q3/Q1 ratio: $0.35 / 0.10 = 3.50$; Year 2 Q3/Q1 ratio: $0.33 / 0.11 = 3.00$.

Both ratios are within the expected 3.0-4.0 range for home security product launches. They both confirm higher popularity for summer demand over the winter demand which is consistent with influencing annual factors of home sales and property crime rates. The slightly lower Y2 ratio confirms the maturity-phase diffusion of demand.

1.6.4 Scaling via benchmark

A second method revealed the potential underestimation of the original demand estimates. To find security in the final demand values, a comparison was performed and confirmed the underestimation of the values and a need to scale upwards the final estimates.

1.7 Appendix

1.7.1 Equations Used

1. Distributed Wireless Market Share

$$S = S_{\text{broad}} * k$$

- Where
 - S_{broad} = benchmark market share of Honeywell in home security
 - k = frequency of similar (home security) Honeywell products available

2. Annual Unit Demand

$$V_n = H * P_H^n * s$$

- Where:
 - H = Total homes owned in the US
 - P_H^n = Probability of homeowners adopting home security system
 - s = Distributed Wireless market share

3. Annual Unit Revenue

$$R_n = V_n * P$$

- Where:
 - V_n = Estimated annual unit demand in year n
 - P = unit price

4. Unified Demand Equation

$$D(n,q) = V_n * W_{\{q=Q, n=N\}}$$

- Where:
 - V_n = Estimated annual unit demand in year n
 - $W_{\{q=Q, n=N\}}$ = Seasonal weight for quarter q in respective year n

5. Alternate Demand Equation

$$V_n = \frac{M_0(1+g)^n(s)}{P}$$

- Where:
 - M_0 = Total home security system market value at the base year
 - g = Annual market growth rate
 - n = year
 - s = Honeywell Distributed Wireless market shareHoneywell's Distributed Wireless market share 0.33%
 - P = average selling price
 - $D_n * W_q$ = Quarterly Demand

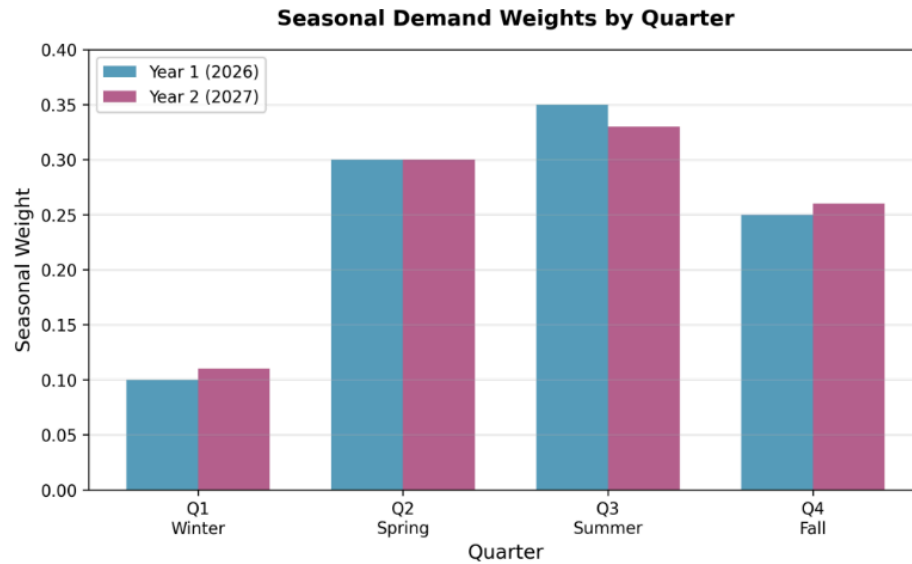
6. Rescaled Unified Demand Equation

$$V_n = D(n,q) + \alpha (V_n^{\text{alt}} - D(n,q))$$

- Where:
 - $D(n,q)$ - original Unified Demand equation
 - V_n^{alt} - Alternate Demand equation
 - Scaling value (α)
 - $\alpha = 0.1$

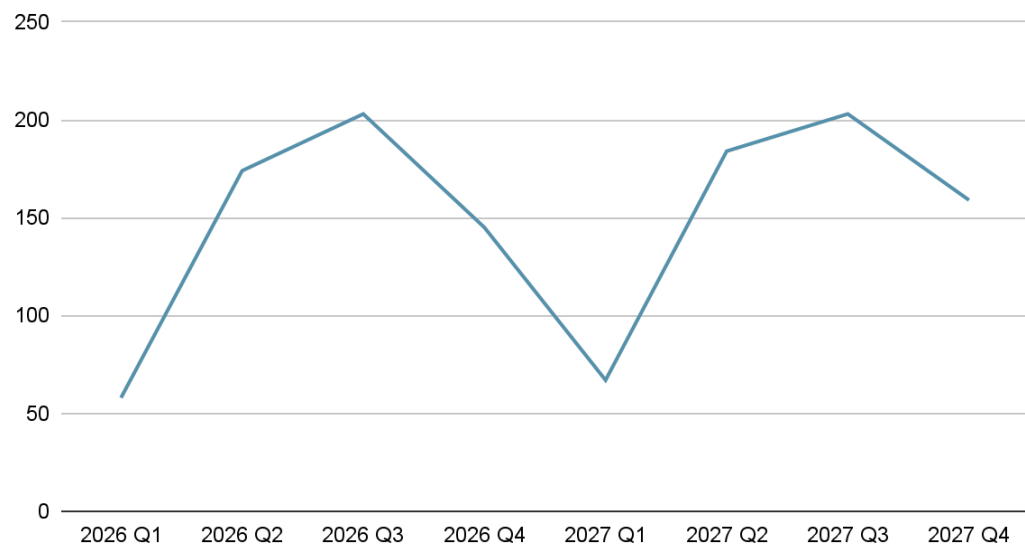
1.7.2 Visualizations

1. Seasonal Demand weights by Quarter

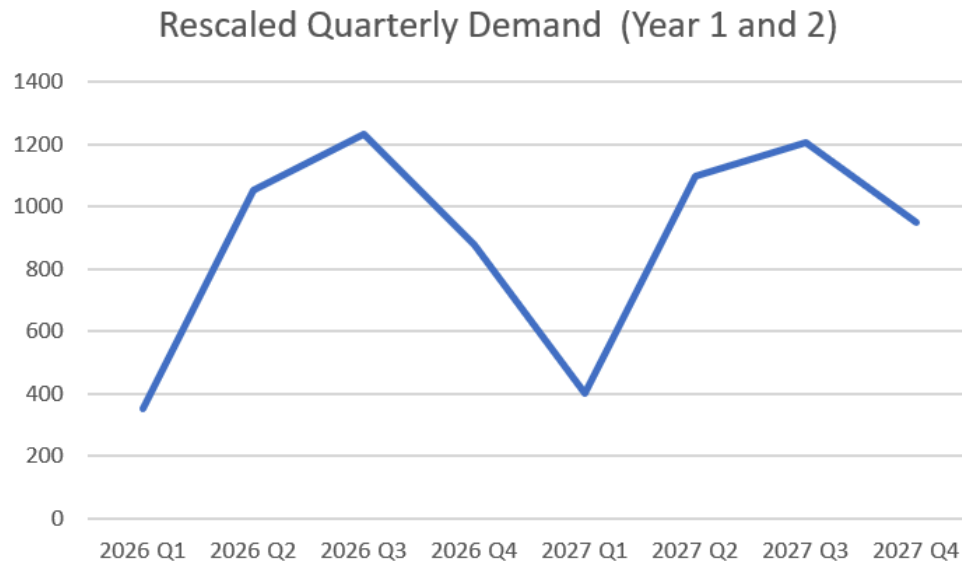


2. Quarterly Demand Graph

Quarterly Demand (Year 1 and 2)



3. Re-scaled (final) Quarterly Demand Graph



2. Demand Forecasting

2.1 Problem Definition

Using eight quarters of demand data derived from Section 1.5, this section forecasts quarterly demand for Years 3 and 4 (2028–2029) using five methods:

1. Static Forecasting
2. Moving Average (MA)
3. Simple Exponential Smoothing (SES)
4. Holt's Method
5. Winter's Method

Each method is evaluated using a complete error analysis. The best-performing method is selected to supply demand inputs for subsequent planning tasks. Smoothing constants for adaptive methods are explicitly stated, and a sensitivity analysis justifies the final selected values.

2.2 Demand Data

The 8-quarter demand series below serves as the historical input for all forecasting methods, pulled directly from the Section 1.5 quarterly demand results. All units represent complete Distributed Wireless systems sold.

Year	Quarter	Period (t)	Demand (Units)
2026	Q1	1	351
2026	Q2	2	1,054
2026	Q3	3	1,230
2026	Q4	4	878
	Year 1 Total		3,513
2027	Q1	5	402
2027	Q2	6	1,096
2027	Q3	7	1,205
2027	Q4	8	950
	Year 2 Total		3,653

2.3 Error Metrics

All five methods are evaluated using the following metrics:

Metric	Formula	Purpose
Forecast Error (E_i)	$E_i = F_i - D_i$	Signed error per period
MSE_n	$(1/n) \sum E_i^2$	Penalizes large errors
MAD_n	$(1/n) \sum E_i $	Mean absolute deviation
$MAPE_n$	$(1/n) \sum E_i/D_i $	Relative error, scale-independent
$Bias_n$	$\sum E_i$	Cumulative over/under forecast
Tracking Signal (TS_n)	$Bias_n / MAD_n$	Detects systematic bias (acceptable: ± 6)

2.4 Static Forecasting

Static forecasting assumes level (L), trend (T), and seasonality (S_i) remain constant over time. The systematic component is:

$$\bar{D}_i = L + iT$$

Step 1 — Deseasonalize ($p = 4$)

Centered moving averages (2x4 MA) are used to remove seasonality from the demand series.

Step 2 — Regression

Regression of deseasonalized demand on period t yields:

$$L = 868.43 \quad T = 6.10$$

Step 3 — Seasonal Factors

$$S_i = D_i / \bar{D}_i$$

Step 4 — Average Seasonal Factors

Averaged and normalized seasonal factors (sum = 4.00):

$$\bar{S}_1 = 0.4251 \quad \bar{S}_2 = 1.2062 \quad \bar{S}_3 = 1.3573 \quad \bar{S}_4 = 1.0114$$

Step 5 — Forecast

$$F_{t+l} = \bar{S}_{t+l} [L + (t+l)T]$$

Static Error Summary

Final MAD	Final MAPE (%)	Final Bias	TS Range
19.83	2.75%	12.69	0.14 to 2.00

The static method achieves a very low MAPE of 2.75% due to the strong, consistent seasonality in the demand series. However, since it fixes all parameters permanently, it cannot adapt to any structural shifts in the data over time.

2.5 Adaptive Forecasting

Adaptive methods update estimates each period using smoothing constants α , β , and γ . The table below summarizes all four adaptive methods.

Method	α	β	γ	Captures	Final MAPE	Assessment
Moving Average	—	—	—	Level only	~42.03%	Poor
SES	0.01	—	—	Level only	~46.67%	Poor
Holt's	0.01	0.01	—	Level + Trend	~45.70%	Poor
Winter's	0.01	0.01	0.1	Level + Trend + Seasonality	2.90%	Selected

2.5.1 Moving Average

$$L_i = (D_i + D_{i-1} + D_{i-2} + D_{i-3}) / 4 \quad F_i = L_{i-1}$$

Final MAPE $\approx 42.03\%$, MAD ≈ 260 . The complete absence of seasonal correction produces errors as large as 476 units in a single period. TS range within ± 6 : Yes.

2.5.2 Simple Exponential Smoothing (SES)

$$L_i = \alpha D_i + (1 - \alpha)L_{i-1} \quad F_i = L_{i-1}$$

Initial level: $L_0 = 895.75$ (average of all 8 historical periods)

Smoothing Constant Sensitivity Analysis

Trial	α	MAPE (%)	MAD
1	0.01	46.67	266.26
2	0.10	48.22	283.32
3	0.20	49.91	299.09
4	0.30	52.12	318.33
5	0.50	57.26	363.48

Although $\alpha = 0.01$ does not yield the lowest error across trials, all values of α produce a near-flat Y3–Y4 forecast with no quarterly variation — a structural limitation of SES regardless of α . Because the demand series exhibits a seasonal amplitude of approximately $3.2\times$ between Q1 and Q3, SES is fundamentally ill-suited to this data. $\alpha = 0.01$ is retained as the best performing value.

2.5.3 Holt's Method

$$L_i = \alpha D_i + (1 - \alpha)(L_{i-1} + T_{i-1}) \quad T_i = \beta(L_i - L_{i-1}) + (1 - \beta)T_{i-1} \quad F_i = L_{i-1} + T_{i-1}$$

Initial values: $L_0 = 868.43$, $T_0 = 6.10$

Smoothing Constant Sensitivity Analysis

Trial	α	β	MAPE (%)	MAD
1	0.10	0.10	47.50	281
2	0.05	0.05	46.48	270
3	0.05	0.10	46.54	271
4	0.01	0.01	45.70	261

Selected: $\alpha = 0.01$, $\beta = 0.01$. Adding trend over SES provides negligible improvement because seasonality—not trend—is the dominant pattern in this data. Final MAPE $\approx 45.70\%$, MAD ≈ 261 . TS range within ± 6 : Yes.

2.5.4 Winter's Method (Triple Exponential Smoothing)

$$L_i = \alpha(D_i / S_{i-4}) + (1 - \alpha)(L_{i-1} + T_{i-1})$$

$$T_i = \beta(L_i - L_{i-1}) + (1 - \beta)T_{i-1}$$

$$S_i = \gamma(D_i / L_i) + (1 - \gamma)S_{i-4}$$

$$F_i = (L_{i-1} + T_{i-1})S_i$$

Initial values: $L_0 = 868.43$, $T_0 = 6.10$, $S_0 = [0.4251, 1.2062, 1.3573, 1.0114]$

Smoothing Constant Sensitivity Analysis

Trial	α	β	γ	MAPE (%)	MAD
1	0.3	0.2	0.3	3.7	28.68
2	0.1	0.1	0.05	2.89	20.77
3	0.01	0.01	0.1	2.9	20.9
4	0.5	0.3	0.4	4.66	39.02

Selected: $\alpha = 0.01$, $\beta = 0.01$, $\gamma = 0.1$. Trials 2 and 3 produce near-identical MAPE (2.89% vs. 2.90%), confirming that results are robust to small changes in the constants. Trial 3 is retained for consistency with the other methods.

Winter's Error Summary

MAD	Final MAPE (%)	Final Bias	TS Range	TS Within ± 6
20.9	2.90%	7.6	-0.09 to 2.00	Yes

2.6 Method Comparison

Method	MAD	Final MAPE (%)	TS Range	Bias	Assessment
Static	19.83	2.75%	0.14 to 2.00	12.69	Excellent — fixed params
Moving Average	260	42.03%	-0.33 to 1.00	—	Poor — seasonal blind
SES	266	46.67%	-0.07 to 1.78	—	Poor — ignores seasonality/trend
Holt's	261	45.70%	-0.07 to 1.63	—	Poor — ignores seasonality

Winter's	20.9	2.90%	−0.09 to 2.00	7.6	Selected — best overall
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MA, SES, and Holt's all produce MAPEs of 42–47% because the demand series has a seasonal amplitude of $\sim 3.2\times$ (Q1 vs. Q3 within each year), which these methods cannot capture structurally. Static and Winter's both achieves under 3% MAPE because they explicitly model seasonal factors derived from the historical data.

Winter's is selected over Static because it is adaptive — it updates L , T , and seasonal factors each period as new data arrives, whereas Static fixes all parameters permanently. The MAPE difference between the two is negligible (0.15 percentage points), but Winter's is more appropriate for ongoing operational use as the product accumulates real sales data.

2.7 Year 3–4 Demand Forecast (Winter's Method)

Using the final trained state from Section 2.5.4 ($L = 917.17$, $T = 6.10$, $S = [0.4252, 1.2059, 1.3566, 1.0114]$) with $\alpha = 0.01$, $\beta = 0.01$, $\gamma = 0.1$:

Year	Quarter	Period	Forecast
2028	Q1	9	393
	Q2	10	1,121
	Q3	11	1,269
	Q4	12	952
Year 3 Total			3,735
2029	Q1	13	403

	Q2	14	1,150
	Q3	15	1,302
	Q4	16	977
Year 4 Total			3,832

Year 3 peaks in Q3, consistent with the established summer seasonality. Year 4 reflects continued moderate growth as the product approaches maturity. These quarterly forecasts serve as the demand inputs for subsequent inventory and financial modeling sections.

3. Cycle and Safety Inventory

A cycle inventory and safety stock allow Honeywell to maintain a plan for optimal efficiency and operational responsiveness in using the forecasted demand of their security systems to prevent discrepancies in quantifying the costs, inventory, time frames, and cycles.

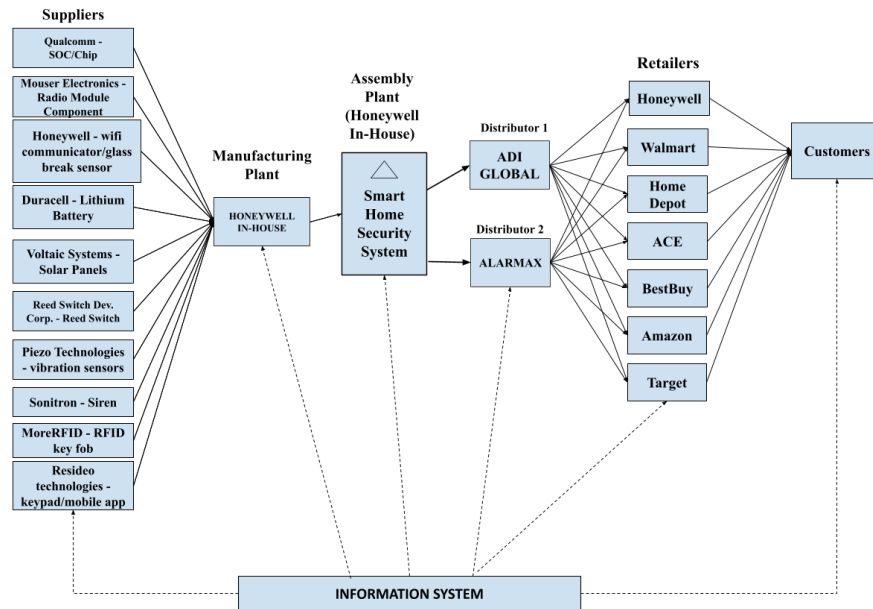
3.1 Cycle Inventory for Distributed Wireless Supply Chain

Process

- Step 1:** Obtain the supply chain network
- Step 2:** From the supply chain network decide which subsystems to investigate and identify the stages related to them
- Step 3:** Layout the data on the subsystems and costs
- Step 4:** Perform cycle inventory calculations and analysis for years 1-4 (provide all equations)
 1. Lot size per shipment minimizing total cost/Economic order quantity
 2. Number of shipments per year
 3. Cycle inventory holding cost
 4. Replenishment cycle time
 5. Average flow time
- Step 5:** Implement in excel
- Step 6:** Put together the cycle inventory of all subsystems in one diagram
- Step 4:** Draw conclusions

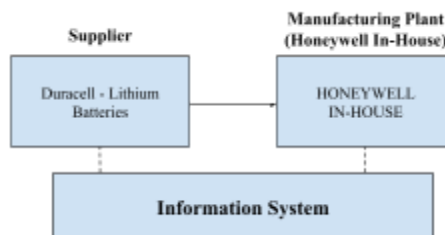
Execution

Step 1: SC Network Representation

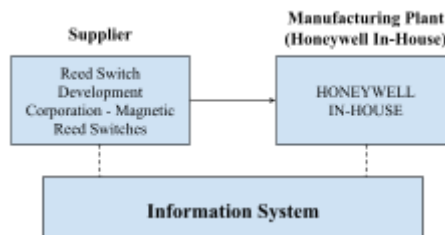


Step 2: Given the supply chain network above we decided to investigate the cycle inventory of duracell batteries, magnetic reed switches, RFID key fobs, and the entire Distributed Wireless system. The following stages zoom into the network for each associated system we are investigating.

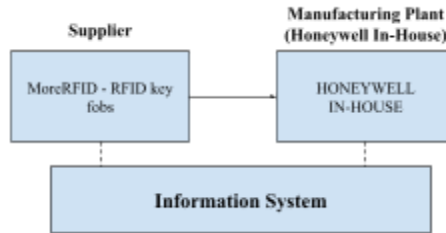
Duracell Lithium Batteries supplied to Honeywell



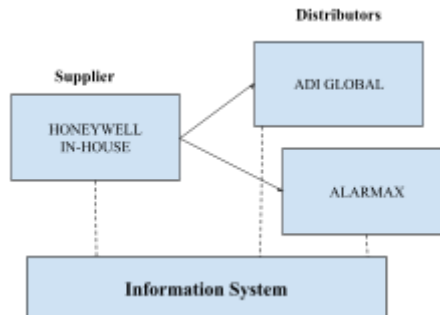
Reed Switch Development Corporation magnetic reed switches supplied to Honeywell



MoreRFID RFID key fobs supplied to Honeywell



Honeywell's Distributed Wireless supplied to ADI Global and AlarMax for distribution



Step 3: Layout the data on the subsystems and costs

To calculate both the cycle inventory and safety stock for each subsystem and the system itself the data on the unit cost, order cost, shipping cost, and demand of units per system will be identified below and displayed in a table.

Duracell Batteries (Subsystem 1)

**Assumption: Each unit uses 4 batteries, thus multiplying demand by 4*

Batteries provide power to the various components of the Distributed Wireless system, and are batched and sold in quantities of 4 per 1 system unit. Given this a single battery was calculated to cost \$2.50.

Reed Switches (Subsystem 2)

**Assumption: Each unit has five magnetic reed switches, thus multiplying demand by 5*

Magnetic reed switches use two components: a sensor and an actuator. Looking at set pricing from Radwell a unit would cost \$24.72. As no ordering cost was tacked onto either this supplier's site nor on others like DigiKey we sought to find what the fulfillment cost for any given order using current order cost averages. Using the [Speed Commerce](#) fulfillment data and noting both manufacturing and distributing is based in the U.S., the order cost has been estimated to be \$10.00 with a holding cost of 20%.

RFID fey fob (Subsystem 3)

**Assumption: Each unit has 2 RFID key fobs, thus multiplying demand by 2*

As we are sourcing the RFID key fobs from MoreRFID we researched their available market products and found the average market price of this control system, used to make Distributed Wireless, may cost anywhere between \$2 to \$10 for every door, noted by [MoreRFID](#). Looking at their available key fobs, we decided that a single keyfob would cost \$5. As Honeywell would purchase these directly from MoreRFID based in China the ordering cost was estimated to be approximately \$20 with a holding cost of 20%.

Distributed Wireless (System)

As previously determined the selling price of Honeywell's Distributed Wireless is \$500. To determine the ordering cost, we looked at similar products' shipping costs and the Honeywell website which based shipment on priority thus we averaged the shipment costs to be roughly \$16.00 with a holding cost of 20%.

Subsystem	Unit Cost	Order Cost	Holding Cost	Forecasted annual demand for 2026-2029	
<i>Lithium Batteries (4 per system)</i>	C = \$2.50/battery	S = \$10/order	h = \$2.00/ye ar	2026	D _{total} = 14,052 units/year
				2027	D _{total} = 14,613 units/year
				2028	D _{total} = 14,940 units/year
				2029	D _{total} = 15,328 units/year
<i>Magnetic Reed Switched (5 per system)</i>	C = \$24.72/sensor set	S = \$10/order	h = 20% = \$0.20	2026	D _{total} = 17,565 units/year
				2027	D _{total} = 18,265 units/year
				2028	D _{total} = 18,675 units/year
				2029	D _{total} = 19,160 units/year
<i>RFID key fob (2 per system)</i>	C = \$5/key fob	S = \$20/order	h = 10% = \$0.10	2026	D _{total} = 7,026 units/year
				2027	D _{total} = 7,306 units/year
				2028	D _{total} = 7,470 units/year
				2029	D _{total} = 7,664 units/year
<i>Distributed Wireless</i>	C = \$270.37/unit	S = \$16 per order	h = 20% = \$0.20	2026	D _{total} = 3513 units/year
				2027	D _{total} = 613 units/year
				2028	D _{total} = 746 units/year
				2029	D _{total} = 846 units/year

Step 4: Provide all equations to perform cycle inventory calculations and analysis for each year(to be implemented in an excel spreadsheet)

Annual Demand given units/system	Optimal Lot Size (lot size per shipment minimizing total cost)	Shipment Frequency (number of shipments per year)	Min. Holding Cost (Holding cost incurred)	Replenishment Cycle Time (When the inventory needs to be replenished)	Average Flow Time (Average time 1 unit spends in inventory)
$= (\text{units/system}) * (\text{systems annual demand})$	$Q_L^* = \sqrt{2(DS)/hC}$	$n^* = D/Q^*_L$	$C_I^* = (Q_L^*/2)(hC)$	$T=365/n^*$	Flow Time = T/2

Step 5: Implement in excel: [Cycle Inventory.xlsx](#)

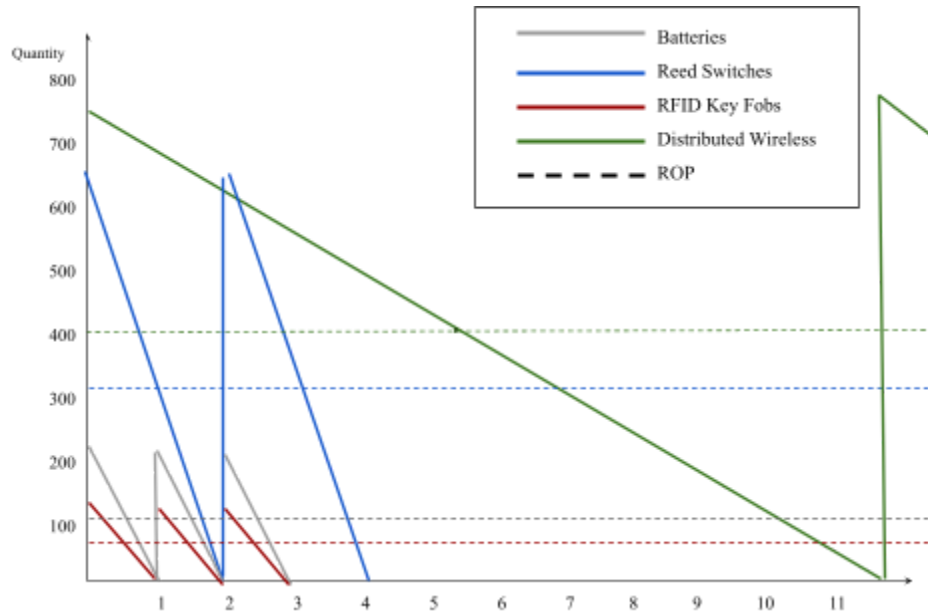
Given both the data and equations, we calculated the cycle inventory of the different subsystems and Distributed Wireless for 2026-2029 (years 1-4 respectively) as shown below.

Data					Forecasted Annual Demand					
System	Unit Cost	Order Cost	Holding Cost	Units/system		Q1	Q2	Q3	Q4	Total
Batteries	2.5	10	0.1	4	2026	351	1,054	1,230	878	3,513
Reed Switches	24.72	10	0.2	5	2027	402	1,096	1,205	950	3,653
RFID key fob	5	20	0.2	2	2028	393	1,121	1,269	952	3,735
Distributed Wireless	270.37	16	0.2	1	2029	403	1,150	1,302	977	3,832

2026 Calculations					2028 Calculations				
	Batteries	Reed Switches	RFID key fob	Distributed Wireless		Batteries	Reed Switches	RFID key fob	Distributed Wireless
Annual	14,052	17,565	7,026	3,513	Annual	14,940	18,675	7,470	3,735
Optimal lot size	265	659	133	770	Optimal lot size	273	679	137	794
Shipment frequency	53	27	53	5	Shipment frequency	55	27	55	5
Min. holding cost	13	66	13	77	Min. holding cost	14	68	14	79
Replenishment cycle time	7	14	7	80	Replenishment cycle time	7	13	7	78
Average flow time	3	7	3	40	Average flow time	3	7	3	39

2027 Calculations					2029 Calculations				
	Batteries	Reed Switches	RFID key fob	Distributed Wireless		Batteries	Reed Switches	RFID key fob	Distributed Wireless
Annual	14,612	18,265	7,306	3,653	Annual	15,328	19,160	7,664	3,832
Optimal lot size	270	672	135	786	Optimal lot size	277	688	138	805
Shipment frequency	54	27	54	5	Shipment frequency	55	28	55	5
Min. holding cost	14	67	14	79	Min. holding cost	14	69	14	80
Replenishment cycle time	7	13	7	79	Replenishment cycle time	7	13	7	77
Average flow time	3	7	3	39	Average flow time	3	7	3	38

Step 6: Put together the cycle inventory of all subsystems in one diagram



Step 7: Conclusions

System Information	Batteries	Reed Switches	RFID Key Fobs	Distributed Wireless
Cycle inventory	7 days (1 week)	14 days (2 weeks)	7 days (1 week)	77-80 days (11-12 weeks)
Avg. flow time	3 days	7 days (1 week)	3 days	38-40 days (5-6 weeks)
Is this cycle inventory reasonable?	Yes	Yes	Yes	Yes

Overall, the inventory strategy focuses on minimizing total costs by balancing component replenishment with slower finished-good cycles. For the selected high-volume subsystems (batteries and RFID key fobs), the model dictates aggregating and ordering at the same time while ordering reed switches later. In contrast, the completed security system, which has a higher unit cost of \$270.37, is managed with a larger lot size and a longer replenishment cycle. This approach ensures that while the assembly line remains continuously fed with parts, the more expensive finished inventory is turned over less frequently to optimize holding costs. Given the cycle inventory a safety inventory must account for the unforecasted excess demand developed in the section 3.2.

3.2 Safety Inventory for Distributed Wireless Supply Chain

Process/Implementation to determine SS and ROP:

Step 1: Explain why Honeywell should have a safety inventory for their Distributed Wireless system and associated components.

Step 2: Obtain weekly demand stats

1. D_w = mean weekly demand
2. σ_w = std. dev of the weekly demand

Step 3: Calculate demand stats for the supplier lead time, L

1. $(D_L)_M$ = mean value of D_L
2. σ_L = std. dev of D_L

Step 4: Calculate Safety Stock

1. $ss = z * \sigma_L$, where $z = 1.65$ (corresponding to a 95% CSL)

Step 5: Calculate ROP

1. $ROP = (D_L)_M + ss$

Calculation Assumptions:

1. Lead times are sourced from the above calculations for cycle inventory
2. All calculations will be done in an attached excel spreadsheet, with the results laid out in an appropriately structured table
3. Demand coefficient of 50% assumed (no historical weekly demand data)

Why should Honeywell have a safety inventory?

To mitigate risks associated with the volatility of supply chains, and to ensure a continuous flow of reliable security systems for its customers, Honeywell should keep a safety inventory for its Distributed Wireless system and its associated components.

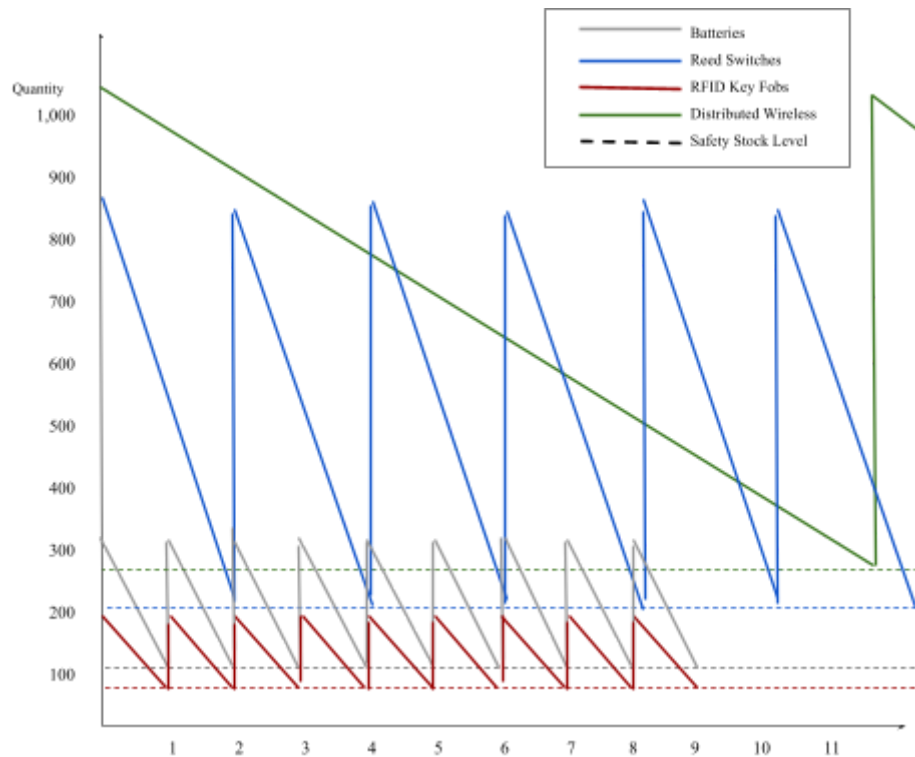
Calculations:

[Distributed Wireless Safety Inventory](#)

Results (in table form, for 2029)

Subsystem	Lead Time (weeks)	D_W	σ_W	$(D_L)_M$	σ_L	SS (units)	ROP (units)	Fill Rate (1-(ESC/Q*))
Batteries	22	29.8	14.9	655	69.8	115	770	99.48%
Reed Switches	44	37.2	18.6	1,637	123.4	204	1,841	99.63%
Key Fobs	44	14.9	7.4	655	49.4	81	736	99.26%
Dist. Wireless System	412	16.3	8.1	6,703	165.1	272	6,975	99.58%

**Note: The model uses a 95% CSL ($z=1.65$), which measures probability of NOT stocking out during a given replenishment cycle. However, FILL RATE is typically higher than CSL for the same ss level, and discovers the fraction of demand that is met immediately from stock, with zero delay assumed.*



Considering the figures found in the Cycle Inventory calculations, these results make sense. The system's high cycle inventory is the primary driver for the high ROP, and a consequence of the high lead time. The results are mathematically consistent.

4. Financial Modeling

The following section determines the base case profit according to the five year NPV analysis framework. Based on the demand estimates developed in the earlier phase this analysis allows us to determine if the project offers sufficient return.

4.1 Sales & Production Volume

Year	Quarter	Period (t)	Demand (units)
2026	1	1	351
	2	2	1054
	3	3	1230
	4	4	878
TOTAL			3513
2027	1	5	402
	2	6	1096
	3	7	1205
	4	8	950
TOTAL			3653
2028	1	9	393
	2	10	1121
	3	11	1269
	4	12	952

TOTAL			3735
2029	1	13	403
	2	14	1150
	3	15	1302
	4	16	977
TOTAL			3832

Following the estimated demand data projected for years 1-4 (using the estimation from Section 1 and the forecast from Section 2), the value of the sales and production volume is 3,513 (2026), 3,653 (2027), 3,735 (2028), and 3,832 (2029).

4.2 Base Case Input Justifications

Sales & Production Volume

Sales & Production Volume = Unit Demand sum of years 1-4
 $= Q_{2026} + Q_{2027} + Q_{2028} + Q_{2029}$
 $= 3513 + 3653 + 3735 + 3832$
 $= 14,733$
 $\Rightarrow$ In NPV table values are displayed as per quarter demand

Development Cost

Development Cost (D) = Revenue₂₀₂₆ (R) * development percentage allocation (d)

- Base Year Revenue(2026)

$$\begin{aligned}
 R_{2026} &= V_1 * P \\
 &= 3,513 * 500 \\
 &= 1,756,500 \\
 \Rightarrow \text{Base year (2026) revenue is } \$1,756,500
 \end{aligned}$$

- Development percentage allocation

$$r = 10\% = 0.1$$

$\Rightarrow$ Development Cost

$$\begin{aligned}
 D &= R_{2026} * r \\
 &= 1,756,500 * 0.1
 \end{aligned}$$

$$= 175,650$$

$$\Rightarrow \text{Development costs is } \$175,650$$

Development cost is estimated to be \$2,900 as 10% of the projected base-year (2026) revenue as similar products do not publicize detailed development costs.

Unit price

The unit price of the Distributed Wireless (\$500) is determined by market research on similar products to ensure competitive pricing:

Leading Competitor Products	Price (\$ USD)
SimpliSafe 21-Piece Security System	\$622.79 ~ 650
Vivint HomeProtect Pro	\$499
Ring Alarm Pro Security Kit	\$595

Market Price range determination

- Minimum: \$499
 - Maximum: \$650
- $$\Rightarrow (499 \leq P \leq 650)$$

$\Rightarrow$ Unit Price

$$P = 500$$

- Unit Price of Distributed Wireless = \$500

Based on the pricing of competitor home security products, setting the price at \$500 keeps the product in the expected margin over production ($\$499 \leq P \leq \650) while still offering affordability to the customers.

Unit Productions Cost

Unit Production Cost (C_{unit}) = Sum of Component Costs (C) *Overhead Costs (k * C)

- Sum of Component Costs (C)

$$C = \sum_{i=1}^n c_i$$

$$= \sum_{i=1}^{11} c_i$$

$$= 12.10 + 5.54 + 6 + 35.99 + 10 + 4.29 + 24.72 + 4.26 + 15.37 + 2 + 127.99$$

$$C = 248.26$$

⇒ sum of major components of Distributed Wireless is \$248.26

- Overhead Allocation ($k * C$)
where $k = 25\% = 0.25$ (percentage)

$$\begin{aligned} k * C &= 0.25 * C \\ &= 0.25 * 248.26 \\ &= 62.065 \end{aligned}$$

$$k * C = 62.07$$

⇒ allocate an overhead of \$62.07 to account for indirect manufacturing costs

⇒ Unit Productions Cost:

$$\begin{aligned} C_{\text{unit}} &= C + (k * C) \\ &= 248.26 + 62.07 \end{aligned}$$

$$C_{\text{unit}} = 310.33$$

⇒ Unit Productions Cost is \$310.33

Unit production costs are estimated at \$310.33 using a bottom-up component pricing of all major components including an added 25% allocation for indirect manufacturing costs to ensure realistic cost estimates for all production expenses.

Ramp-up cost

Ramp-up Cost (RC) = Revenue₂₀₂₆ (R) * initial percentage allocation (r)

- Initial percentage allocation
 $r = 5\% = 0.05$

- Base Year Revenue(2026)

$$\begin{aligned} R_{2026} &= V_1 * P \\ &= 3,153 * 500 \\ &= 1,756,500 \end{aligned}$$

⇒ Base year (2026) revenue is \$1,756,500

⇒ Ramp-Up Costs

$$\begin{aligned} RC &= R_{2026} * r \\ &= 1,756,500 * 0.05 \\ &= 87,825 \\ &\Rightarrow \text{ramp up costs is } \$87,825 \end{aligned}$$

Ramp-up cost is \$14,500 which was estimated as 5% of the expected revenue in the base year,

representing the initial setup investments.

Marketing & support cost

Marketing Cost (M) = Revenue₂₀₂₆ (R) * marketing and support percentage allocation (m)

- Marketing and Support percentage allocation (m)

$$m = 2\% = 0.02$$
- Base Year Revenue(2026)

$$\begin{aligned} R_{2026} &= V_1 * P \\ &= 3,513 * 500 \\ &= 1,756,500 \\ \Rightarrow \text{Base year (2026) revenue is } \$1,756,500 \end{aligned}$$

⇒ Marketing & support Cost

$$\begin{aligned} M &= R_{2026} * m \\ &= 1,756,500 * 0.02 \\ &= 35,130 \\ \Rightarrow \text{market and support costs is } \$35,130 \end{aligned}$$

Market and support costs is \$35,130 which was estimated using the percentage of sales method which takes 2% of the expected revenue value in the base year.

Annual Discount Factor

$$\begin{aligned} r &= 10\% \\ r &= 0.1 \end{aligned}$$

The annual discount factor is 10% which is a common value used for NPV analysis.

4.3 NPV Analysis

NPV analysis was used to determine whether launching the Distributed Wireless system would be financially beneficial over the project period.

- Using the demand estimates from the previous sections, projected revenue was calculated by multiplying the forecasted sales volume by the unit price of \$500.
- Production costs were estimated using the unit manufacturing cost of \$310.33 per system, along with additional costs such as development, ramp-up, and annual market and support expenses.
- These costs and revenues were organized into quarterly cash flows from 2025 to 2029. Each future cash flow was then discounted using a 10% annual discount rate to account for the time value of money. The discounted cash flows were added to calculate the project's NPV.

x Final Report - Financial Modeling & Sensitivity Analysis.xlsx

SCENARIO INPUT PARAMETERS	
Development Cost (total \$)	175,650
Unit price (\$/unit)	500
Unit Productions Cost (\$/unit)	310.33
Ramp-up cost (total \$)	87,825
Marketing & Support cost (\$/year)	35,130
Annual Discount Factor (%)	10

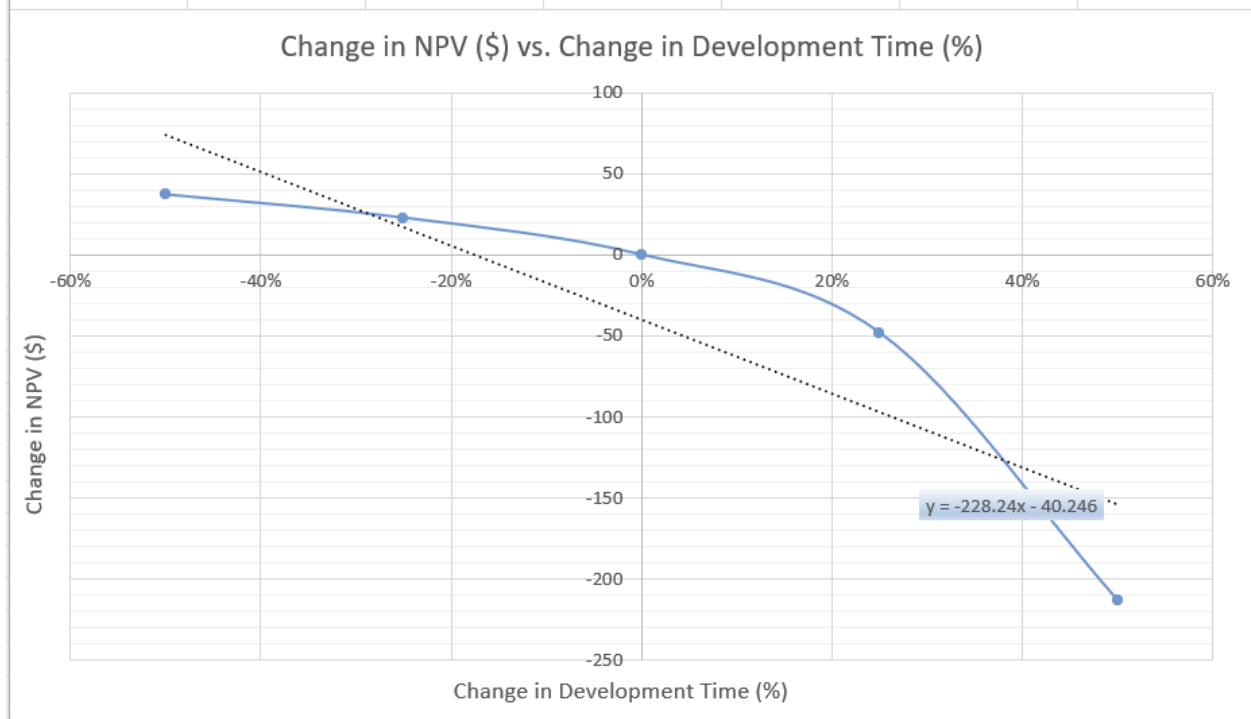
BaseCase	2025				2026				2027				2028				2029			
period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
(\$ values in thousands)	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Development Cost	-43.91	-43.91	-43.91	-43.91																
Ramp-up cost			-43.91	-43.91																
Marketing & support cost				-8.7825	-8.78	-8.78	-8.78	-8.78	-8.78	-8.78	-8.78	-8.78	-8.78	-8.78	-8.78	-8.78	-8.78	-8.78	-8.78	-8.78
Production cost					-108.93	-327.09	-381.71	-272.47	-124.75	-340.12	-373.95	-294.81	-121.96	-347.88	-393.81	-295.43	-125.06	-356.88	-404.05	-303.19
Production volume					351	1,054	1,230	878	402	1,096	1,205	950	393	1,121	1,269	952	403	1,150	1,302	977
Unit production cost					-0.31	-0.31	-0.31	-0.31	-0.31	-0.31	-0.31	-0.31	-0.31	-0.31	-0.31	-0.31	-0.31	-0.31	-0.31	-0.31
Sales Revenue					175.50	527.00	615.00	439.00	201.00	548.00	602.50	475.00	196.50	560.50	634.50	476.00	201.50	575.00	651.00	488.50
Sales volume					351.00	1,054.00	1,230.00	878.00	402.00	1,096.00	1,205.00	950.00	393.00	1,121.00	1,269.00	952.00	403.00	1,150.00	1,302.00	977.00
Unit price					0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
Period Cash Flow	-43.91	-43.91	-87.83	-96.61	57.79	191.13	224.51	157.75	67.46	199.10	219.77	171.40	65.76	203.84	231.91	171.78	67.65	209.34	238.17	176.53
PV year 1, r=10%	-43.91	-42.84	-83.59	-89.71	52.36	168.93	193.60	132.71	55.37	159.42	171.68	130.63	48.89	147.87	164.13	118.61	45.57	137.58	152.71	110.42
Project NPV, \$		1,730																		
all units in thousands, \$																				

The nominal profit of the Distributed Wireless project, determined by a 5-year NPV analysis with a discount factor of 10%, is expected to be \$1,730,000. Given the current conditions, the project is expected to be profitable.

4.4 Sensitivity Analyses

4.4.1 Development Time

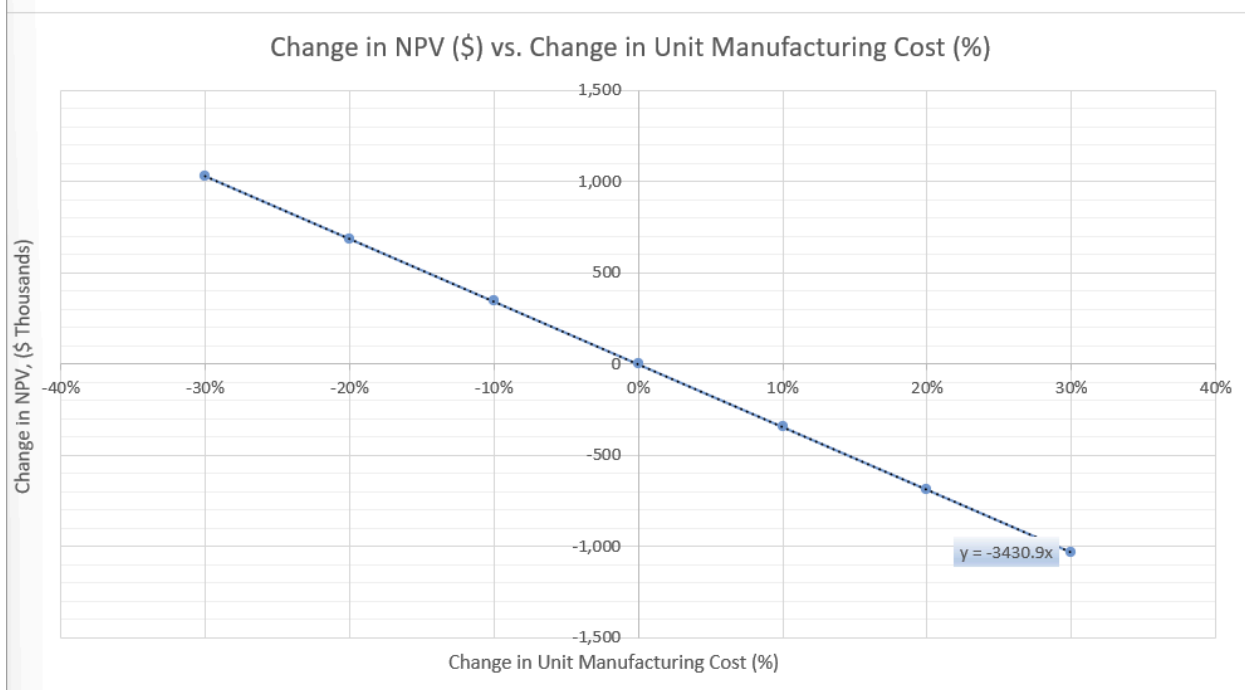
Base Development Time, Quarters	Change in Development Time, %	Development Time, Quarters	Change in Development Time, Quarters	NPV, \$ Thousands	Change in NPV, \$ Thousands	Change in NPV, %
4	50%	6	2	1,518	-213	-12%
4	25%	5	1	1,682	-48	-3%
4	0%	4	0	1,730	0	0%
4	-25%	3	-1	1,753	23	1%
4	-50%	2	-2	1,767	37	2%



Tradeoff Law: For each percentage increase in development time, the NPV changes by -\$2,282.40

4.4.2 Unit Manufacturing Cost

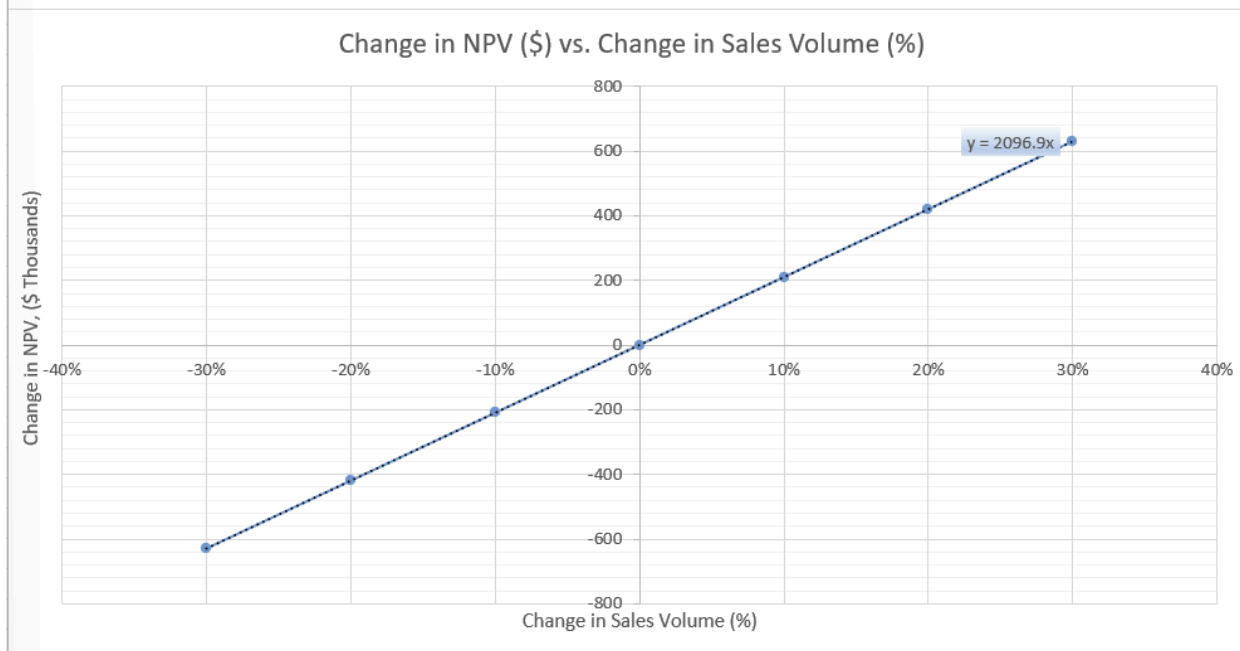
Base Unit Manufacturing Cost, \$	Change in Unit Manufacturing Cost, %	Unit Manufacturing Cost, \$	Change in Unit Manufacturing Cost, \$	NPV, \$ Thousands	Change in NPV, %	Change in NPV, \$ Thousands
310	30%	403	93	701	-59.48%	-1,029
310	20%	372	62	1,044	-39.65%	-686
310	10%	341	31	1,387	-19.83%	-343
310	0%	310	0	1,730	0.00%	0
310	-10%	279	-31	2,074	19.83%	343
310	-20%	248	-62	2,417	39.65%	686
310	-30%	217	-93	2,760	59.48%	1,029



Tradeoff Law: For each percentage increase in unit manufacturing cost, NPV changes by -\$34,309.00

4.4.3 Sales Volume

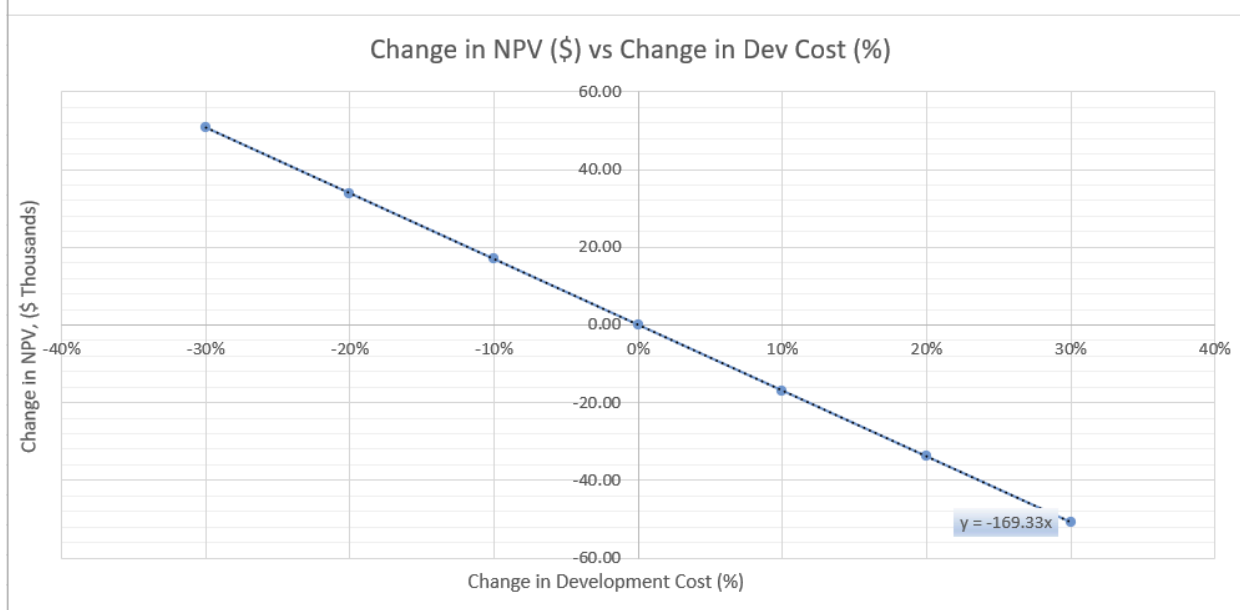
Base Sales Volume, Qty	Change in Sales Volume, %	Sales Volume, Qty	Change in Sales Volume, Qty	Change in NPV, %	NPV, \$ Thousands	Change in NPV, \$ Thousands
14,733	30%	19,153	4,420	36.35%	2,360	629
14,733	20%	17,680	2,947	24.24%	2,150	419
14,733	10%	16,206	1,473	12.12%	1,940	210
14,733	0%	14,733	0	0.00%	1,730	0
14,733	-10%	13,260	-1,473	-12.12%	1,521	-210
14,733	-20%	11,786	-2,947	-24.24%	1,311	-419
14,733	-30%	10,313	-4,420	-36.35%	1,101	-629



Tradeoff Law: For each percentage increase in sales volume, NPV changes by \$20,969.00

4.4.4 Development Cost

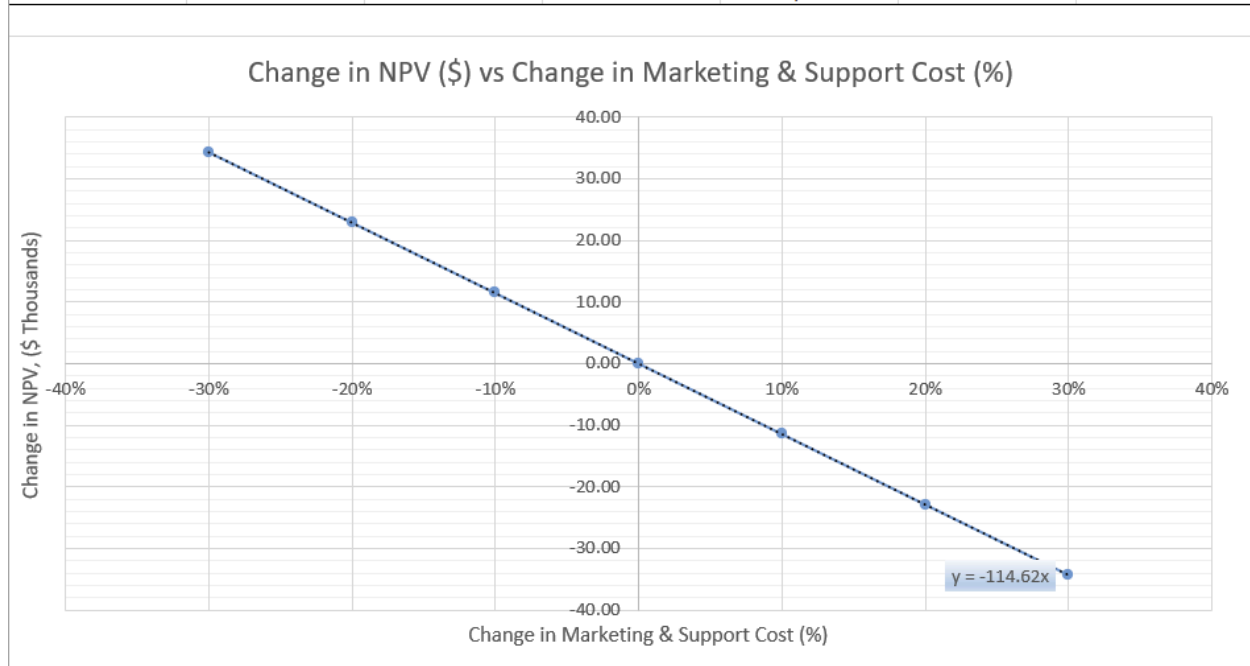
Base Dev Cost, \$ Thousands	Change in Development Cost, %	Development Cost, \$ Thousands	Change in Development Cost, \$ Thousands	NPV, \$ Thousands	Change in NPV, %	Change in NPV, \$ Thousands
176	30%	228.345	52.695	1,680	-2.94%	-50.80
176	20%	210.78	35.13	1,697	-1.96%	-33.87
176	10%	193.215	17.565	1,713	-0.98%	-16.93
176	0%	175.65	0	1,730	0.00%	0.00
176	-10%	158.085	-17.565	1,747	0.98%	16.93
176	-20%	140.52	-35.13	1,764	1.96%	33.87
176	-30%	122.955	-52.695	1,781	2.94%	50.80



Tradeoff Law: For each percentage increase in development cost, NPV changes by -\$1,693.30

4.4.5 Marketing & Support Cost

Base Marketing & Support Cost, \$ Thousands	Change in Marketing & Support Cost, %	Marketing & Support Cost, \$ Thousands	Change in Marketing & Support Cost, \$	NPV, \$ Thousands	Change in NPV, %	Change in NPV, \$ Thousands
35.13	30%	45.67	10.54	1,696	-1.99%	-34.39
35.13	20%	42.16	7.03	1,707	-1.32%	-22.92
35.13	10%	38.64	3.51	1,719	-0.66%	-11.46
35.13	0%	35.13	0.00	1,730	0.00%	0.00
35.13	-10%	31.62	-3.51	1,742	0.66%	11.46
35.13	-20%	28.10	-7.03	1,753	1.32%	22.92
35.13	-30%	24.59	-10.54	1,765	1.99%	34.39



Tradeoff Law: For each percentage increase in marketing & support cost, NPV changes by -\$1,146.20

5. Project Backlog

Phase	Deliverable	Comment	Status	Responsibility
Proposal	“Missing real dates for weeks and group meetings, and missing tentative project coordinators throughout Project Plan”	Added the real dates and times for the meetings (project coordinators have remained the same)	Complete	Hailey McElroy
	“Need to provide more bridging/explanatory text between Final Concept → FAST → Subsystems → Stage Representation → Network Representation.”	Added bridging text between each removing filler	Complete	Hailey McElroy
	“The contribution table should clearly state the name and include the dates for the following: (1) Classroom/Lecture attendance, (2) Project review attendance, (3) Group meeting attendance”	Format the contributions table with this format for each phase	Complete	Hailey McElroy
Phase 1	“Missing a meaningful executive summary that includes real results”	Include a clear executive summary that has the phases purpose/use and findings	Complete	Garrett Arcand/Abinaya Ramesh
	“Your SC business model includes too much filler, and you need to more explicitly align your SC business model (section 2) with your SC stage and network representations (section 1).”	- Get rid of filler - Connect section 2 with section 1	Complete	James Burch
	“Incorrect Competitive Strategy and incomplete SC Strategy; you must clearly show: Competitive strategy → SC strategy → SC Structure”	- Competitive strategy should be either Focus or Cost Leadership (more research needed to identify) - Put together a process that connects all three	Complete	Garrett Arcand
	“Too much filler within Demand Estimation (pages 18-23), and your demand estimation process needs to be mathematicized with symbols and equations.”	- Get rid of filler - mathematicize demand estimation process	Complete	Vivian Nguyen/Collin Yuen
	“Need to improve the quality of your integration and explanations throughout the document and significantly improve teamwork.”	- Improve quality of explanations - Too vague	Complete	N/A
Phase 2	“Your demand estimation has inconsistency with the number of units for demand (pages 10 & 14).”	Review demand estimation, see what is causing inconsistency	Complete	Vivian Nguyen
	In Demand Estimation and Demand Forecasting, you need to reduce filler and bulk (pages 5-16 & pages 17-34); you may want to use tables and/or an appendix to condense. (We expect to see an updated version of this in Task 1 for Phase 3.)	- use tables to reduce filler of chunks of text in demand estimation and forecasting	Complete	Abdul-Haadi
	“You need a clear mathematical, stepwise process to	- “mathematical” stepwise	Complete	Hailey

	develop a programmable equation within your Cycle Inventory Management section. (We expect to see an updated version of this in Task 2 for Phase 3)."	process for a "programmable equation" for Cycle Inventory Management		McElroy/ Garrett Arcand
	"You are missing a complete Network Representation of your SC in Cycle Inventory Management"	- missing network rep. of SC	Complete	Garrett Arcand
	You need to show multiple replenishment cycles in your CIM graphs (page 37, 39, 40), and include a more comprehensive conclusion.	- missing replenishment cycles in CIM graphs - conclusion is not "comprehensive" enough	Complete	Hailey McElroy
	"Within your Financial Modeling section, you need to show a clear mathematical, stepwise process for deriving financial model inputs. (We expect to see an updated version of this in Task 3 for Phase 3)."	- clear MATHEMATICAL stepwise process needed for financial model derivations	Complete	Vivian Nguyen
	"Your Project NPV ideally should not be negative, and your data is not properly aligned with its respective quarters within your financial model (page 44)."	- Make sure demand ends up positive - properly align data with respective quarters	Complete	James Burch
Phase 3	Mess with the smoothing constants in an excel spreadsheet and explain why we chose/got these constants.	- For the mess with smoothing constants and see how it changes the MAPE (graph)	Complete	Abdul-Haadi
	Complete cycle inventory management process for years 1-4 completely	Needs cycle inventory for all four years in an excel spreadsheet	Complete	Hailey McElroy
Phase 3 Post Meeting	Executive Summary: - Key results from each section should include a simple sentence of what we did and what we found out	Provided real results for each section's exec. summary	Complete	Garrett Arcand
	Demand Estimation: - Check equations - Take into account Honeywell's market share - Include the products life cycle	Took product lifecycle from last quarter's development into account	Complete	Vivian Nguyen
	Demand Forecasting: - Only include the absolutely necessary explanations/information in text - Include a table at the end summarizing the results(Can take from Phase 2 and edit)	Reduced filler text to only the necessary information	Complete	Abdul-Haadi
	Cycle inventory: - Finish diagram - Give structure to the equation	Corrected diagrams	Complete	Hailey McElroy
	NPV & Sensitivity analysis: - Correct NPV costs (marketing and support) - Make development 4 quarters of 2025	Corrected costs (NPV is now positive)	Complete	James Burch

Revisions Post Phase 3 Submission

Issue Identified	Changes Made	Team Members Responsible
TOC needs to be fixed with formatting/text issues	Reformatted the text and corrected some of the text	James Burch
Executive Summary is good, but you need to include more meaningful results (direct comparison of forecasting methods, ROP and fill rate from CSI, profit from your Financial Model, etc.).		Garrett Arcand
Demand Estimation, you must create a second method of estimating demand, to verify whether your original estimation is in the right ballpark or to accordingly scale up or down your estimated demand	Added a new method to estimating the demand Developed comparisons to the original estimation results Formed a new equation to rescale (up) the original estimation results Derived newly scaled demand estimation values for years 1 and 2	Vivian Nguyen Collin Yuen
Demand Forecasting for adaptive methods, you need to play around with more values for your smoothing constants. For Winter's method, your choice (value) for alpha, beta, gamma is essentially 0 and, therefore, the result for Winter's model and static model is similar.		Abdul-Haadi
Demand Forecasting conclusion needs to directly and explicitly tie with your results.		Abdul-Haadi
Cycle Safety Inventory section, your finished product replenishment cycle of 372 weeks is very large. Your Safety Inventory and ROP is also affected, so please ensure your calculations are correct.	Finished cycle/safety inventory double checking calculations	Hailey McElroy Garrett Arcand
Cycle Safety Inventory diagram (page 29) needs to include and explicitly define safety stock.	Added safety stock to graph	Hailey McElroy
need to explicitly mention the fill rate within your Safety Inventory calculations		Garrett Arcand
Within your Financial Model section, you need to use a table derived from your demand	Clarified the source of the data in the existing table	James Burch

estimation and demand forecasting results, for your financial model's Sales & Production Volume input (page 33).		
Within your Financial Model section, and all other references to it (TOC, Executive Summary), you need to clearly understand that NPV is only a tool, in this case used to calculate the Expected Profit. Therefore, you need to clearly include a statement to the effect: Our nominal (base case) profit based on a 5-year NPV analysis with a discount factor of x% is \$P; with meaningful conclusions.	Added statements clarifying that NPV is only a tool	James Burch
explicitly state your profit in your conclusion. Additionally, your profit of \$357k for 5 years is low.	With the new demand data provided by Vivian and Collin, the 5-NPV is significantly higher	James Burch
significantly improve the quality of your integration, reduce the amount of filler, and improve explanations throughout the document; as well as significantly improving your group's teamwork.		Abinaya Ramesh

Member Contributions Table (Pre-Review)

Name	Classroom/ Lecture Attendance		Project Review Attendance	Group Meeting Attendance		Initial Project Report Contributions
	Thurs 2/26	Tues 3/3	Weds 3/4	Mon 3/2	Tues 3/3	
James Burch	☒	☒	☒	☒	☒	- NPV Base Case - NPV Sensitivity Analyses
Hailey McElroy	☒	☒	☒	☒	☒	- Project Proposal edits - Formatted cycle inventory - Created cycle inventory spreadsheet - Added a third subsystem for cycle inventory - Added to Project Backlog
Vivian Nguyen	☒	☒	☒	☒	☒	- Financial modeling Base Case justification & mathematicization - NPV Base Case
Garrett Arcand	☒	☒	☒	☒	☒	- Safety Inventory Calculations and table formatting - Wrote executive summary with real results
Mohammed Abdul-Haadi Ansari	☐	☐	☒	☒	☒	Task 1 - Demand forecasting for years 3 and 4
Collin Yuen	☒	☒	☒	☒	☒	- Create a comprehensive backlog in a table - Identify deliverable and areas of improvement
Abinaya Ramesh	☒	☒	☐	☐	☐	

Post-meeting Action Table

Issue	Assigned Member	Timeline
Demand Estimation	Vivian Nguyen	☒ Update numbers based on

○ Ensure all equations work (add explanatory text) ○ Account for product lifecycle ○ Research Honeywell's actual market share of home security market and scale ours accordingly	Collin Yuen	product lifecycle (3/10) ☒ Implement market share of Honeywell (3/10) ☒ Re-estimate demand values for years 1 and 2 (3/10)
● Demand Forecasting ○ Insert connecting text ○ Format equation formulations into tables ○ Insert a table at the end summarizing results ○ Rename “Avg MAPE” to “Final MAPE”	Garrett Arcand Hailey McElroy	☒ Reincorporate new demand estimations from above section (3/10) ☒ Re-forecast demand and make new selection for best method (3/10)
● Cycle Inventory ○ Finish diagram ○ Conclusion in table form ○ More detail in backlog	Hailey McElroy Garrett Arcand	☒ Update diagram (3/10) ☒ Convert results to table form (3/10)
● NPV & Sensitivity analysis ○ NPV should not be negative ○ Marketing & support costs are too high ○ Development should take place over all 4 quarters of 2025 ○ Write explanatory text	James Burch	☒ Correct development time (3/4) ☒ Correct negative NPV (3/5) ☒ Reassess & lower marketing & support cost (3/9) ☒ Write explanatory text (3/9)

Member Contributions Table (Post-Review)

Name	Classroom/ Lecture Attendance		Group Meeting Attendance		Initial Project Report Contributions
	Thurs 3/5	Tues 3/10	Wed 3/4 (in-person)	Mon 3/9	
James Burch	☒	☒	☒	☒	● Corrected NPV analysis according to changes in prior sections ● Corrected errors in initial NPV estimate (NPV is no longer negative)

					• Reassessed marketing and support costs • Justified everything with explanatory text
Hailey McElroy	☒	☒	☒	☒	• Reforecasted demand for Static, Moving Avg., S.E.S., and Holt's models • Completed cycle inventory re calculations • Attached all forecasting and cycle inventory charts
Vivian Nguyen	☒	☒	☒	☒	• Market share research and re-calculation of share percentage • Revision of annual unit demand formula and unification with variables • Re-establishment of product life-cycle calculations • Adjusted validation criteria and final checks in accordance to changes • Reformatting of Demand Estimation
Garrett Arcand	☒	☒	☒	☒	• Re-forecasted demand for Winter's method, made final selection • Calculated demand per-subsystem for cycle inventory • Re-calculated safety inventory using new demand info • Wrote executive summary using real results
Mohammed Abdul-Haadi Ansari	☒	☒	☒	☒	• NPV check
Collin Yuen	☐	☐	☐	☐	• Updated numbers based on product lifecycle • Implemented market share of Honeywell to estimate demand • Input demand graph based on demand table

Abinaya Ramesh	☒	☒	☐	☒	• Explanation of Life-Cycle weights table • Explanation of Validation Criteria • Added explanatory text to NPV Analysis
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Conclusions and Next Steps

Conclusions

The Honeywell *Distributed Wireless* home security system offered a comprehensive opportunity to design a supply chain for a new consumer product in the home products industry. Through iterative phases of demand estimation, inventory modeling, and financial analysis, we developed several key insights which shaped our final design and understanding of supply chain strategy for emergent products.

Demand for the Distributed Wireless system was estimated at 3,513 units in Year 1 (2026) and forecasted to grow steadily to 3,832 units by Year 4 (2029), driven by seasonal peaks in Q2 and Q3 consistent with home sales and property crime patterns. Winter's Method was selected as the most appropriate forecasting approach given its ability to capture both trend and seasonality, achieving a MAPE of 2.90%. Cycle inventory analysis yielded optimal lot sizes and replenishment cycles that were reasonable for all subsystems, with batteries and RFID key fobs turning over every 7 days and the completed Distributed Wireless system replenishing on an 11–12 week cycle. Safety inventory calculations, targeting a 95% cycle service level, produced corrected safety stock requirements of 115, 204, 81, and 272 units for batteries, reed switches, key fobs, and the DW system respectively, with corresponding fill rates between 99.26% and 99.63%, confirming that the chosen inventory policy will meet nearly all demand without delay. Financially, the project yields a nominal base-case profit of \$1,730,000 over a 5-year NPV analysis at a 10% discount rate, with unit manufacturing cost and sales volume identified as the most sensitive drivers of profitability.

Next Steps

Looking ahead, several areas warrant further refinement as the product moves toward launch. First, the 50% demand coefficient of variation assumed throughout the safety inventory analysis should be replaced with empirically observed demand data as early sales become available, which will allow for more precise safety stock calibration. Second, the Distributed Wireless system's replenishment cycle of approximately 412 weeks and corresponding ROP of nearly 6,975 units merit a reassessment of the finished-goods ordering strategy, a smaller, more frequent ordering approach would better align with the product's annual demand of 600–850 units and reduce excess inventory risk. Third, as the product enters its growth and maturity phases, Winter's Method smoothing constants should be updated with real sales data to ensure the forecast continues to track actual demand patterns. Finally, continued monitoring of the two primary NPV sensitivities, unit manufacturing cost and sales volume, will be critical to ensuring the project remains profitable, particularly if component costs or market adoption deviate meaningfully from current projections.

Individual Contributions

Name	Project Proposal	Project Phase I	Project Phase II	Project Phase III
James Burch			- Mathematically clarified and developed the stepwise process to deriving input values - Corrected unit conversion error in demand data - Added statements clarifying that NPV is only a tool	- Added statements clarifying that NPV is only a tool
Hailey McElroy	- Added real times and dates to the tables - Bridging and explanations for sections	- Removed Filler - Corrected the competitive strategy - Removed filler to focus on explaining key information	- Corrected cycle inventory formulas when recalculating the cycle inventory - Redid the cycle inventory graph - Added safety inventory to graph - Double checked work against plan and formulas developed in class	- Added multiple cycles to the graph - Revisited the calculations and development of sections process and execution
Vivian Nguyen	- general explanatory content bridging	- Reduced explanatory filler of Demand estimation - Revised Demand estimation process to be more clearly mathematicized.	- Reworked the Demand Estimation Section to remove filler and remove inconsistencies - Mathematically clarified and developed the stepwise process to deriving input values - clear mathematical stepwise process needed for financial model derivations - Took product lifecycle from last quarter's development into account to develop unified equations -	- Added a new method to estimating the demand - Developed comparisons to the original estimation results - Formed a new equation to rescale (up) the original estimation results - Derived newly scaled demand estimation values for years 1 and 2 - Created final quarterly demand visualization graph
Garrett Arcand			- Reinstated network representation diagram to its entirety, and added individual condensed diagrams	- Revised Safety Inventory - Completed Conclusions and next steps (report)

				-Completed assessment section (report)
Abinaya Ramesh	- Updated the stages in the Supply chain network representation with more detailed explanations - Created an executive summary for all phases for final report	- Added a meaningful executive summary that includes real results - Forecasted demand data for distributed wireless system using various methods from TIM172B	- Forecasting & Error Analysis Spreadsheet	- Added a project reflection by identifying key lessons learned - Explanation of Life- Cycle weights table - Explanation of Validation Criteria - Added explanatory text to NPV Analysis
Mohammed Abdul-Haadi Ansari	• Supply chain stage and network representation • Use-Case diagram • FAST diagram contributions • Member contributions writeup (was flagged as incomplete at one point)	• Forecasted demand data for distributed wireless system using various methods from TIM172B	• Built the DW_Forecasting.xlsx workbook (all 5 methods — Static, MA, SES, Holt's, Winter's) • Wrote up forecasting sections including connecting text between subsections • Worked on the method comparison summary table • Verified MAPE/MAD values across methods	Expanded SES sensitivity trials (1 → 5 trials, $\alpha = 0.01$ through 0.50) Expanded Winter's sensitivity trials (3 → 6 trials, adding higher α/β values) Altered Executive Summary for accuracy Altered Forecasting Method Selection (Section 2.6) to resolve the SES vs. Winter's contradiction Identified and flagged the Quarterly Demand Total column error in Section 2.7 Identified Section 4.1 pulling from wrong demand column (~17% Y4 overstatement)
Collin Yuen	• Created the Supply Chain Network Representation diagram	• Found similar products demand data to improve the demand	• Reorganized the Estimating Demand section to match	• Create a comprehensive backlog in a table

	• Worked on both SC Stage Representation diagram and SC Network Representation diagram	estimation section • Name suppliers in our SC diagrams • Research suppliers for this type of product	expected layout • Simplified and mathematicized the Estimating Demand section • Calculated yearly revenue, sales volume, price in a mathematical numerical process	• Updated numbers based on product lifecycle • Implemented market share of Honeywell to estimate demand • Implemented a new method to rescale original estimation for a higher NPV • Formulated a chart with demand estimation data from all three methods
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5. Time-Phase Project Plan

Week	Phase	Member Assigned	Meeting Schedule
1	Project Proposal for SCM of Distributed Wireless Security System	See below	See below
2-3	Proposal: SCM Strategy/Design	<i>Garrett Arcand (Project Coordinator):</i> Time-Phase plan <i>Hailey McElroy (Co-Coordinator):</i> Conceptual design review/FAST diagram <i>James Burch:</i> Preliminary infrastructure (Stage rep) <i>Vivian Nguyen:</i> Preliminary infrastructure (Network rep) <i>Abdul-Haadi:</i> Post-processing <i>Collin Yuen:</i> Network representation <i>Abinaya Ramesh:</i> Post-processing	Thursday 1/15 post-lecture Sunday/Monday (1/18-19) 4:00PM-5:00PM
3	Proposal Review/Report Due	N/A	Sunday 1/25 7-8:00PM Monday 1/26 4-5:00PM
4-5	Phase 1: SC Modeling/Planning (demand forecasting)	<i>Garrett Arcand (Project Coordinator):</i> Assign work and fix time-phase plan errors, fix contribution table, add connecting text (1) <i>Hailey McElroy (Co-Coordinator):</i> Business Model Development (2) <i>James Burch:</i> Supply Chain Strategy Development (3) <i>Vivian Nguyen:</i> Data estimation process (4) <i>Abdul-Haadi:</i> Forecast demand (5) <i>Collin Yuen:</i> Work with Vivian to estimate data (4) <i>Abinaya Ramesh:</i> Work with Abdul-Haadi to forecast demand (5)	Sunday 2/1: 5-6:00PM Monday 2/2: TBD Tuesday 2/3: 1-2:20PM (post-lecture)
5	Phase 1 Review/Report Due	N/A	TBD
6-7	Phase 2: SC Cycle/Safety Inventory Mgmt.	<i>Garrett Arcand (Project Coordinator):</i> Coordinate with Hailey (3), update time-phase plan accordingly <i>Hailey McElroy (Co-Coordinator):</i> Cycle inventory management (3)	Monday 2/16: 5-6:00PM Tuesday 2/17: 1-2:20PM (post-lecture)

		<i>James Burch:</i> Determine base case profit (4) <i>Vivian Nguyen:</i> Re-examine demand data (1) and work with James to determine base case profit (4) <i>Abdul-Haadi:</i> Coordinate with Abinaya (2) <i>Collin Yuen:</i> Coordinate with Vivian for (1) <i>Abinaya Ramesh:</i> static/adaptive forecasting (2)	
7	Phase 2 Review/Report Due	N/A	Friday 2/20: 7:30-8:30PM Sunday 2/22: 5-6:00PM Tuesday 2/24: 3-4:00PM
8-9	Phase 3: SC Facilities/Transportation Network Design	<i>Garrett Arcand (Project Coordinator):</i> Cycle/Safety Inventory (2) <i>Hailey McElroy (Co-Coordinator):</i> Cycle/Safety Inventory (2) <i>James Burch:</i> Financial Modeling (3) <i>Vivian Nguyen:</i> Financial Modeling (3) <i>Abdul-Haadi:</i> Demand Forecasting (1) <i>Collin Yuen:</i> Backlog Structuring (4) <i>Abinaya Ramesh:</i> Demand Forecasting (1)	Monday 3/2: 7:30-8:30PM Tuesday 3/3: 4-5:00PM
9	Phase 3 Review/Report Due	N/A	“”
10	Integration using SCM and Network Design from 172B	“”	“”

**Each segment of the week-by-week project plan is subject to change*

This table outlines a ten-week project timeline for the SCM of the proposed Distributed Wireless security system. The project will be structured into distinct phases, beginning with a proposal, building into the four core SCM phases, and ending with final integration.

Assessment

Throughout both quarters of the project, our group was, according to the professor, “one of the better groups.” What worked particularly well for us was our ability to adjust assignments, meetings, and workflows on the fly, as every member of the group had a constantly changing schedule. Something that could have been improved upon was the frequency of in-person group meetings, as our constantly changing schedule conflicts caused the scheduling of such meetings to frequently fall apart.